

Machine Learning

CSE 4621

Topic: Linear Regression

Source: [DeepLearning.AI](https://deeplearning.ai)

Presented by

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Assistant Professor



Supervised Learning



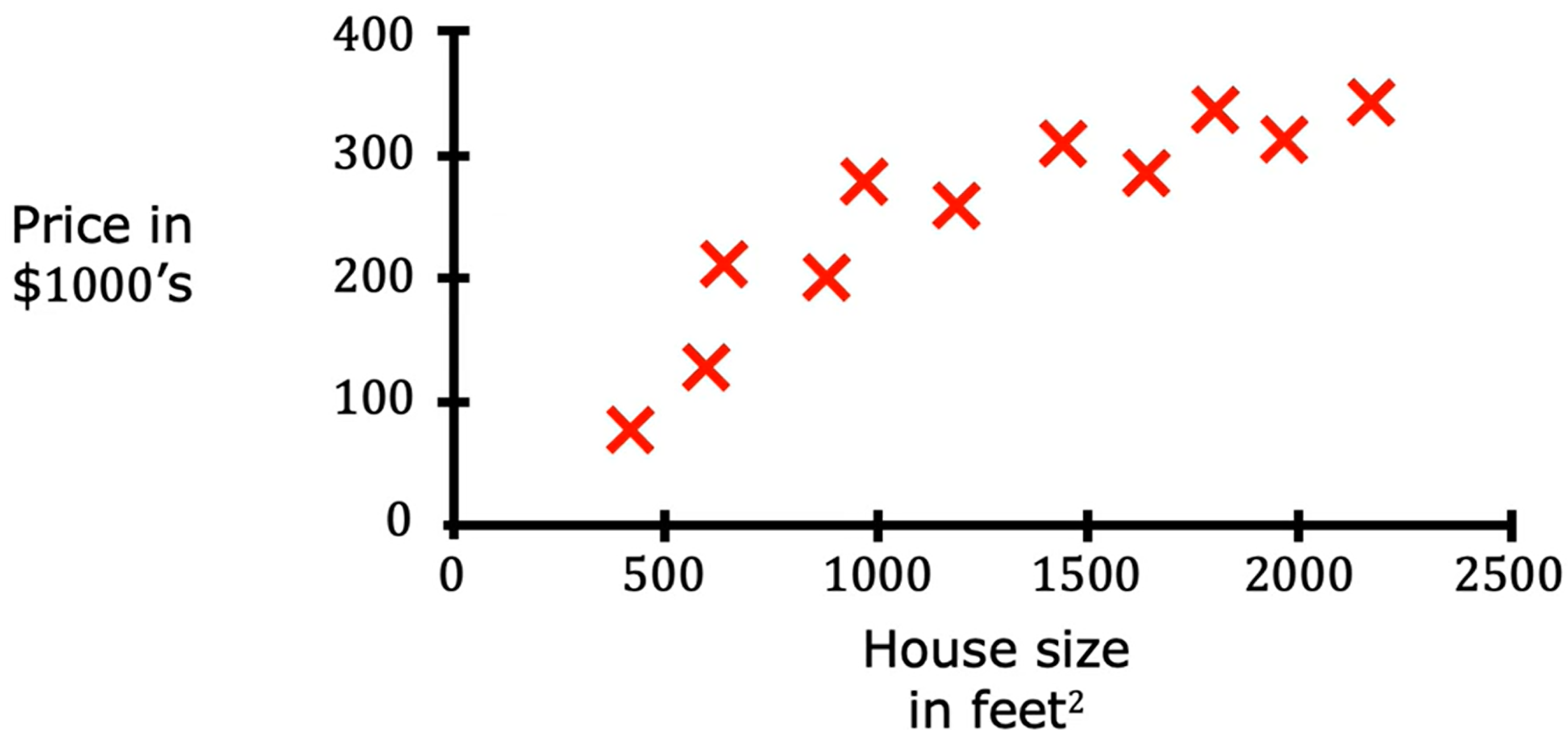
Learns from being given “right answers”

Supervised Learning

Input (X)		Output (Y)	Application
email	→	spam? (0/1)	spam filtering
audio	→	text transcripts	speech recognition
English	→	Spanish	machine translation
ad, user info	→	click? (0/1)	online advertising
image, radar info	→	position of other cars	self-driving car
image of phone	→	defect? (0/1)	visual inspection

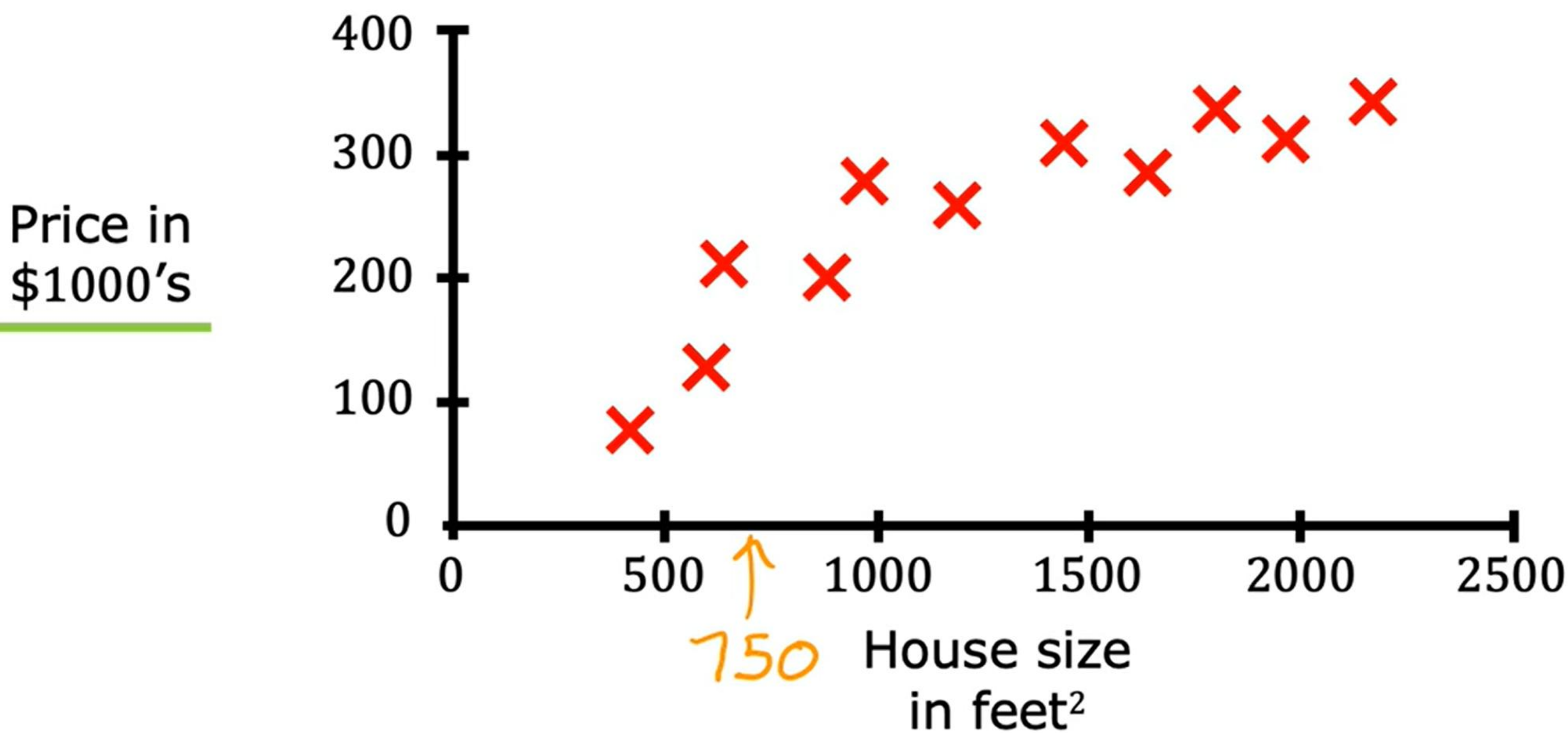
Supervised Learning

Regression: Housing price prediction



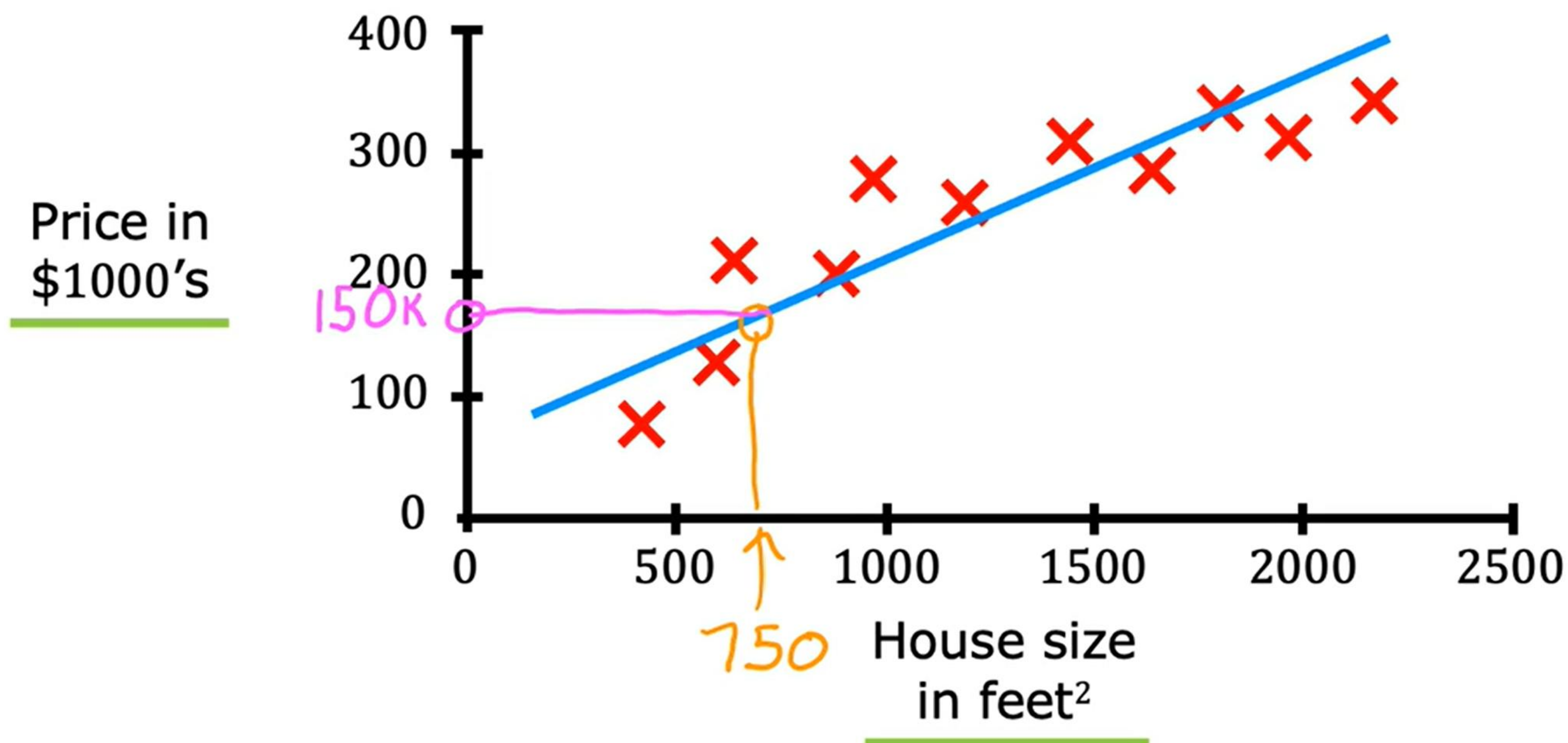
Supervised Learning

Regression: Housing price prediction



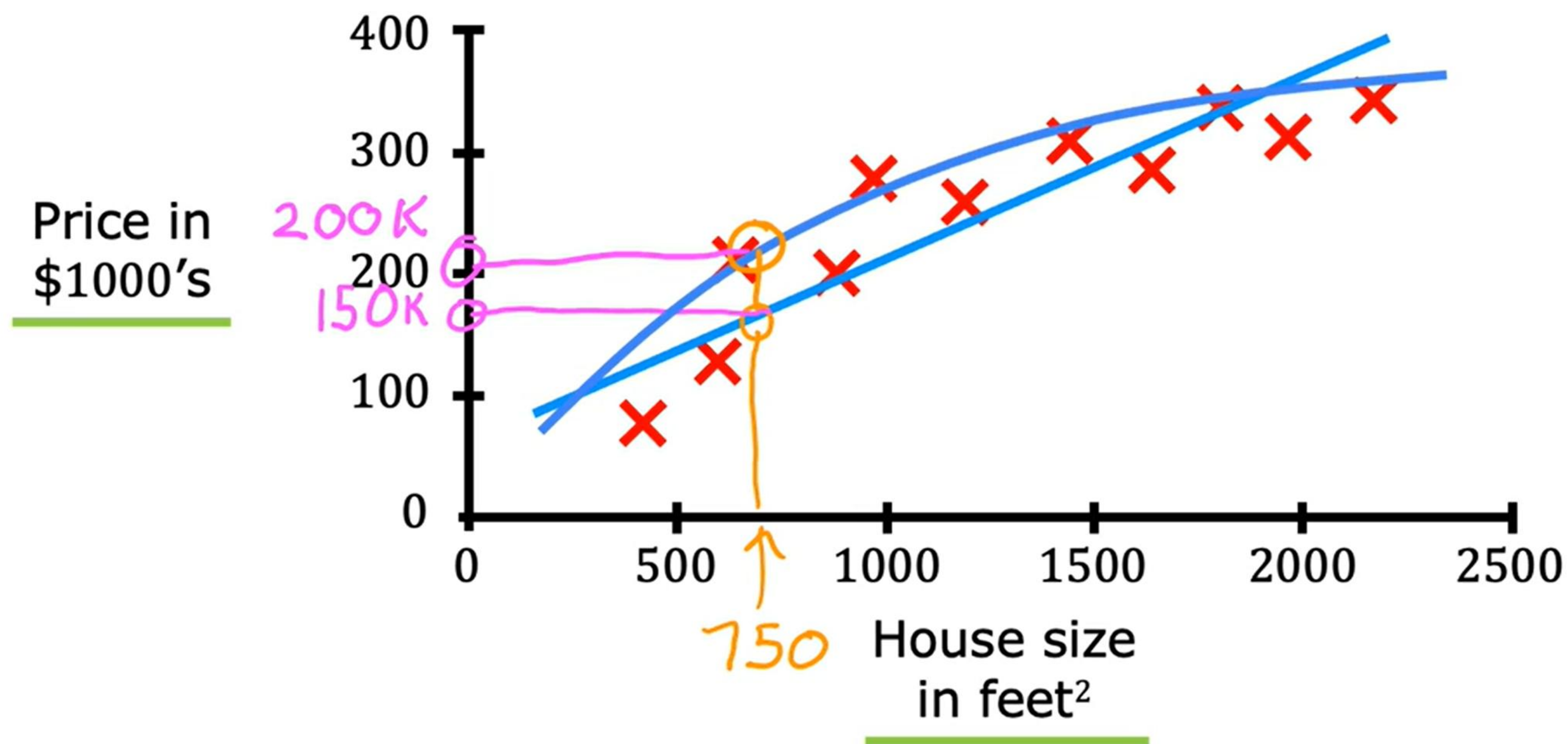
Supervised Learning

Regression: Housing price prediction



Supervised Learning

Regression: Housing price prediction



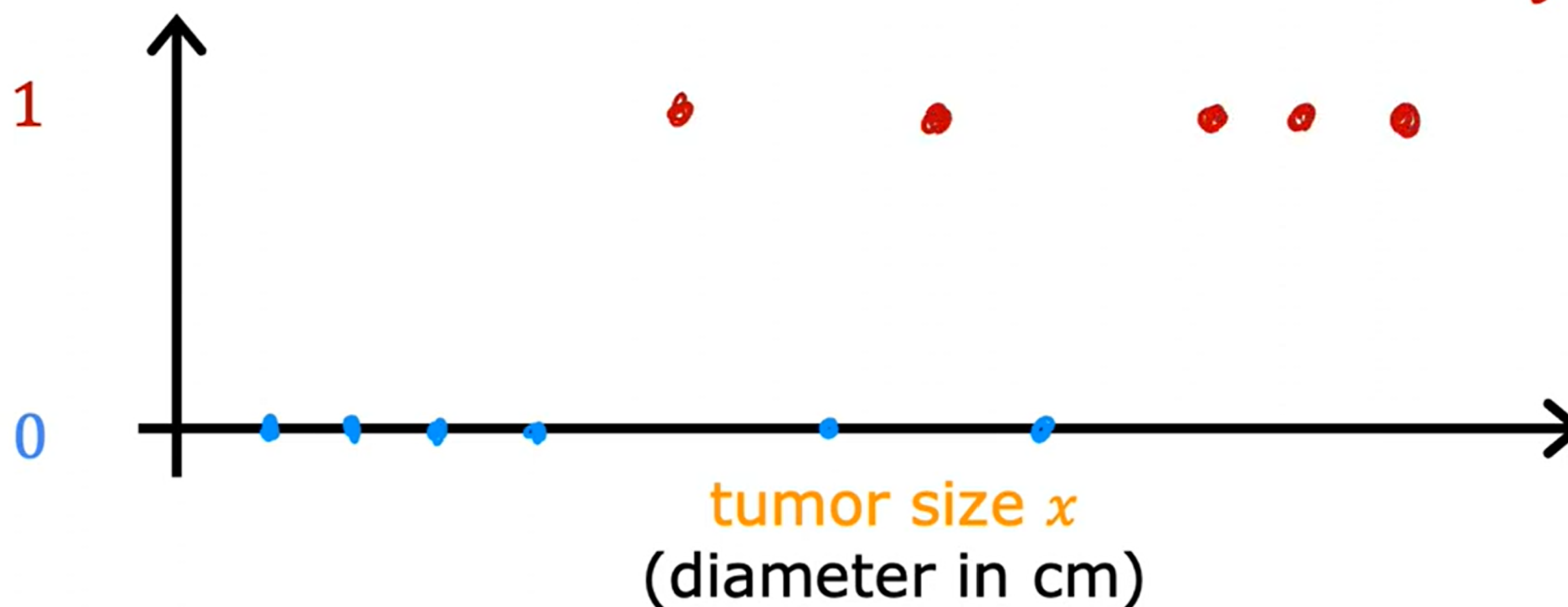
Supervised Learning

Classification:

cancer detection



malignant benign



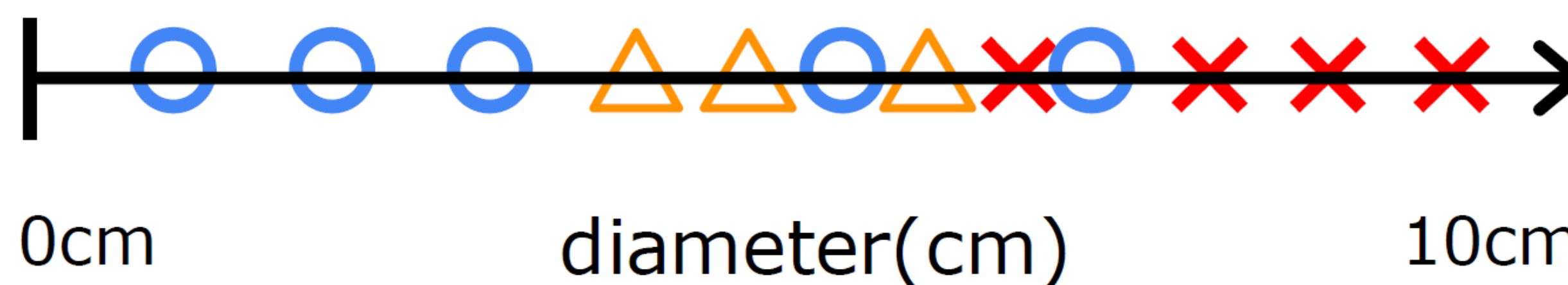
size	diagnosis
2	0
5	1
1	0
7	1
⋮	

Supervised Learning

Classification:

- benign
- ✗ malignant *type 1*
- △ malignant type 2

cancer detection



class category

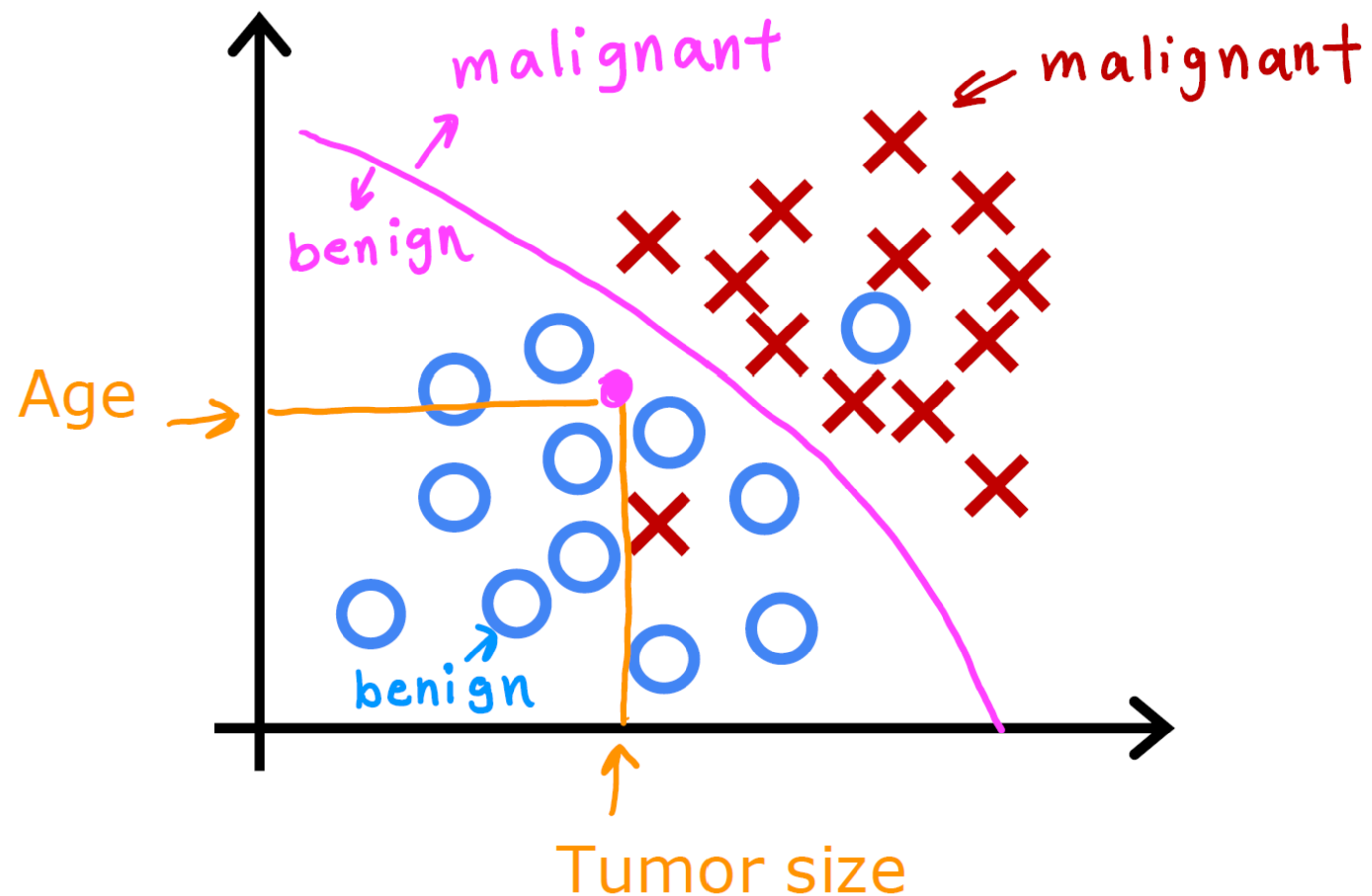
Classification

predict *categories* *cat dog* benign malignant 0, 1, 2

small number of possible outputs

Supervised Learning

Two or more inputs



Supervised Learning

Learns from being given “right answers”

Regression

Predict a number

infinitely many possible outputs

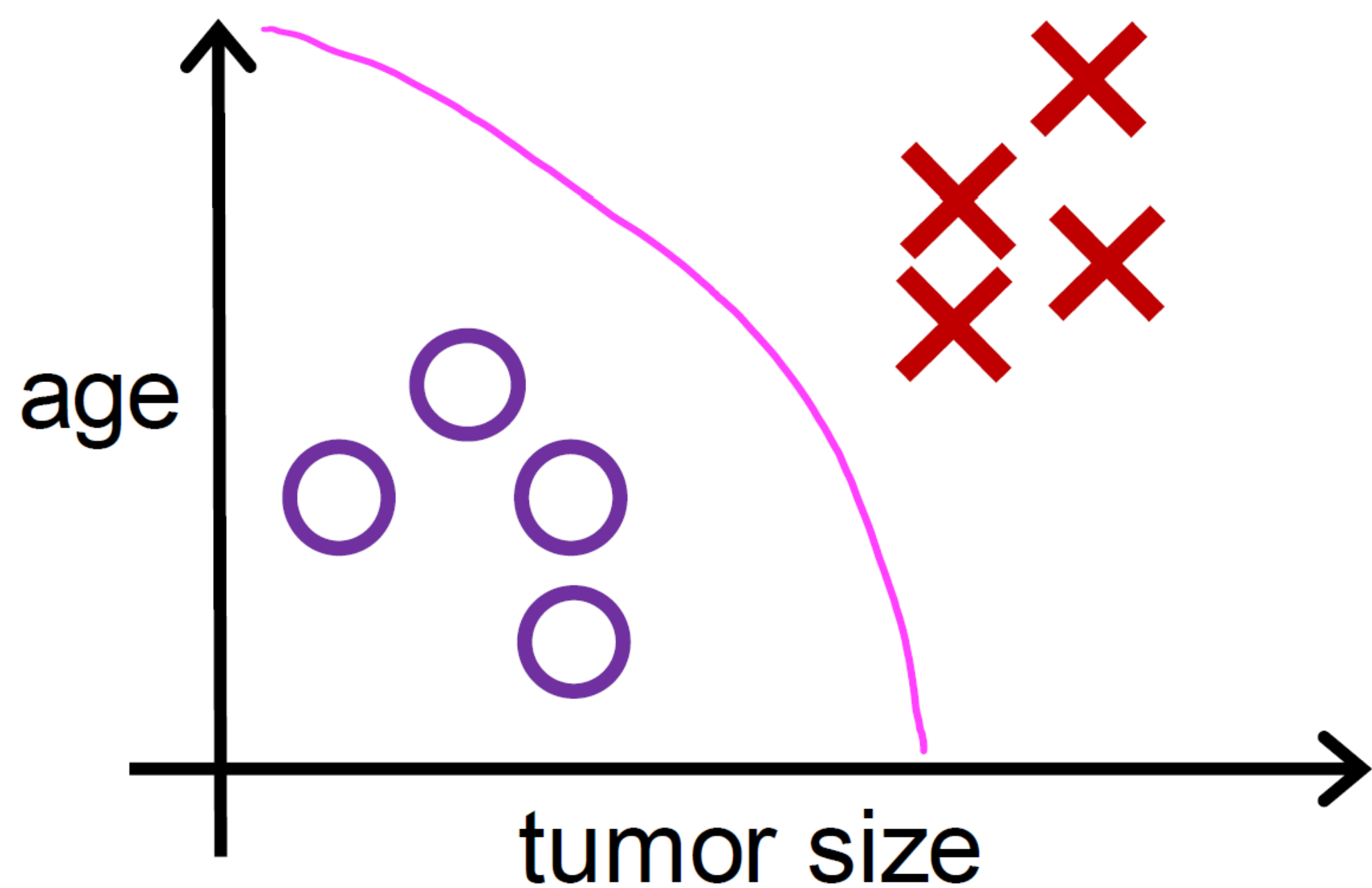
Classification

predict categories

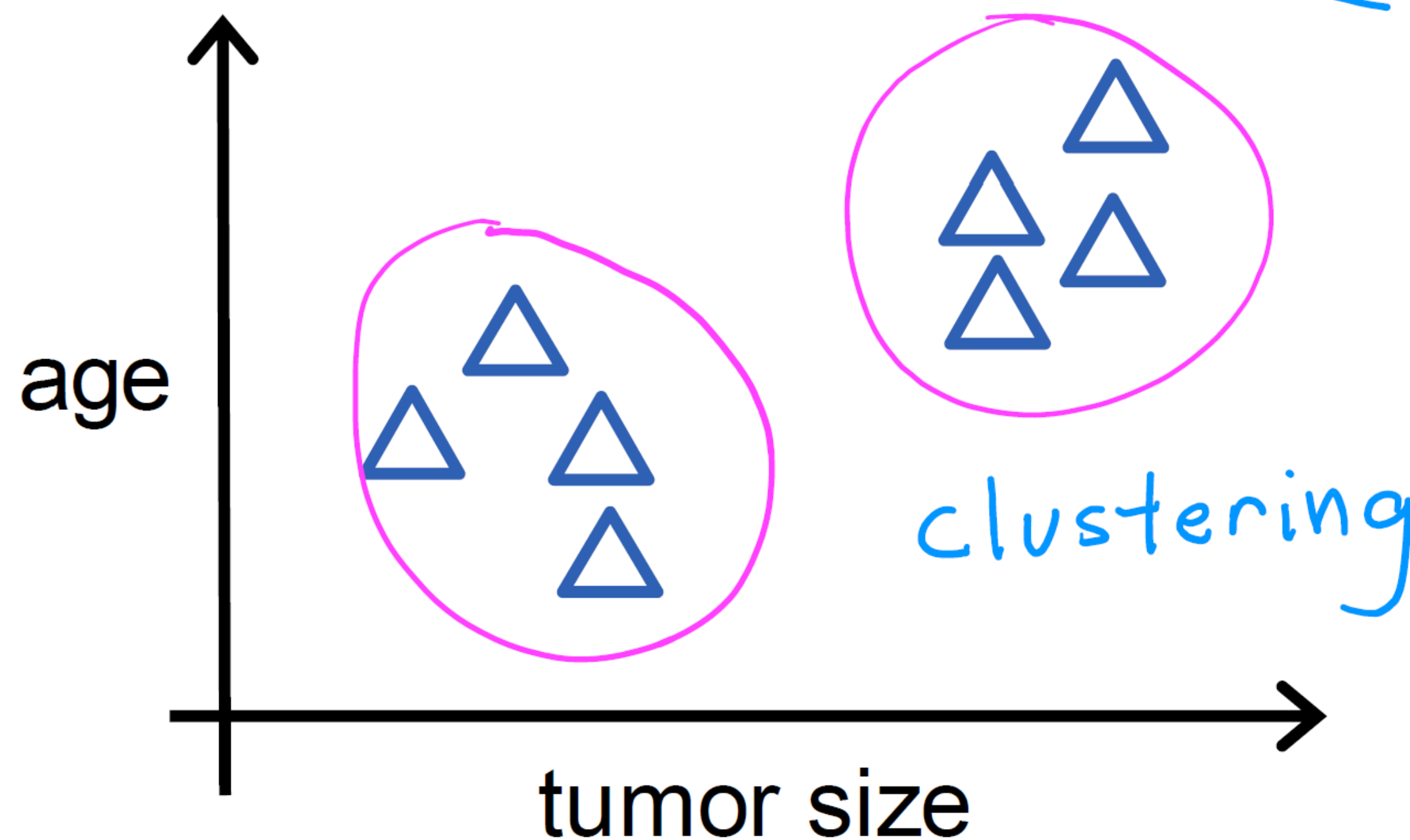
small number of possible outputs

Unsupervised Learning

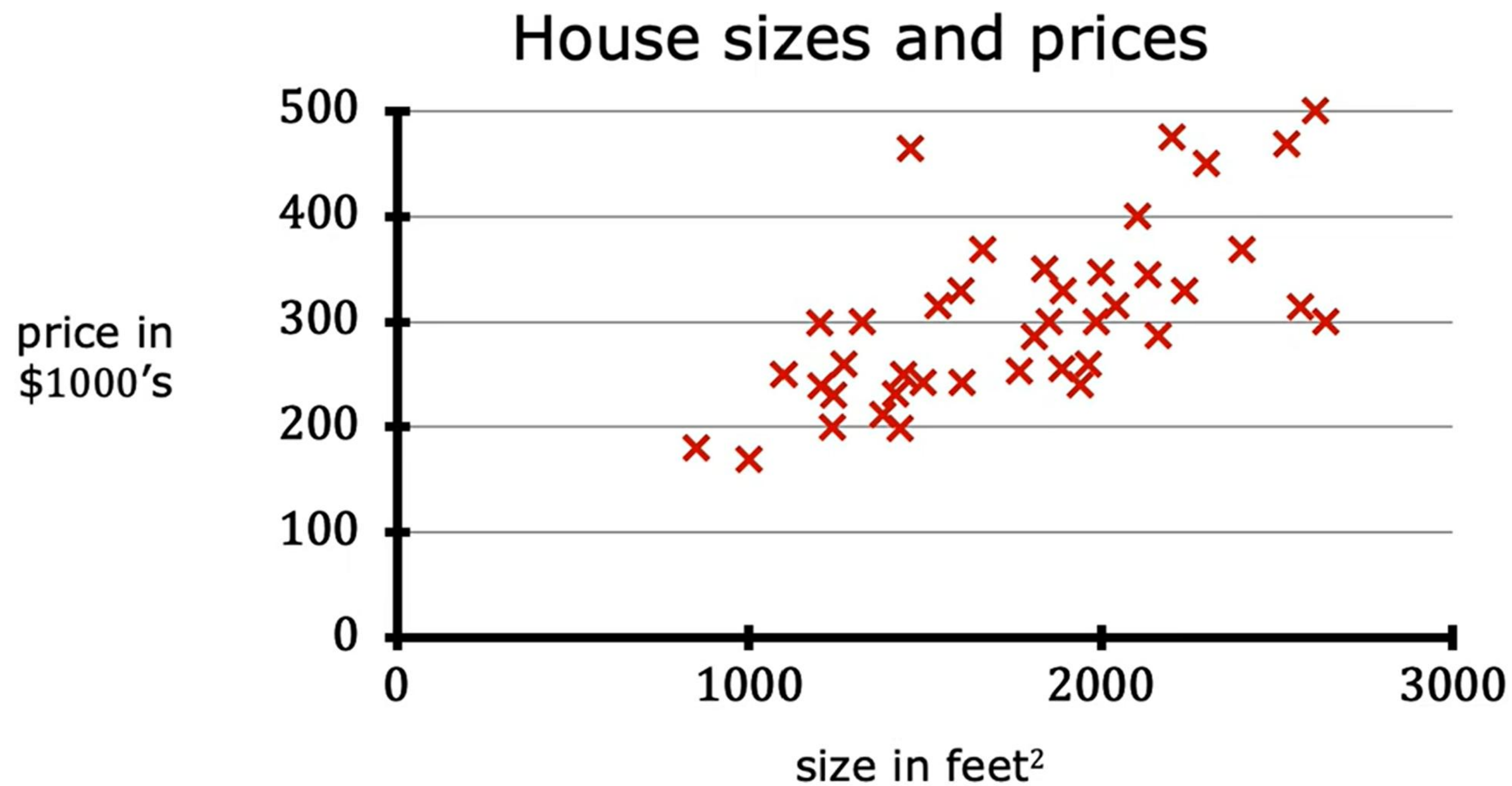
Supervised learning
Learn from data **labeled**
with the "right answers"



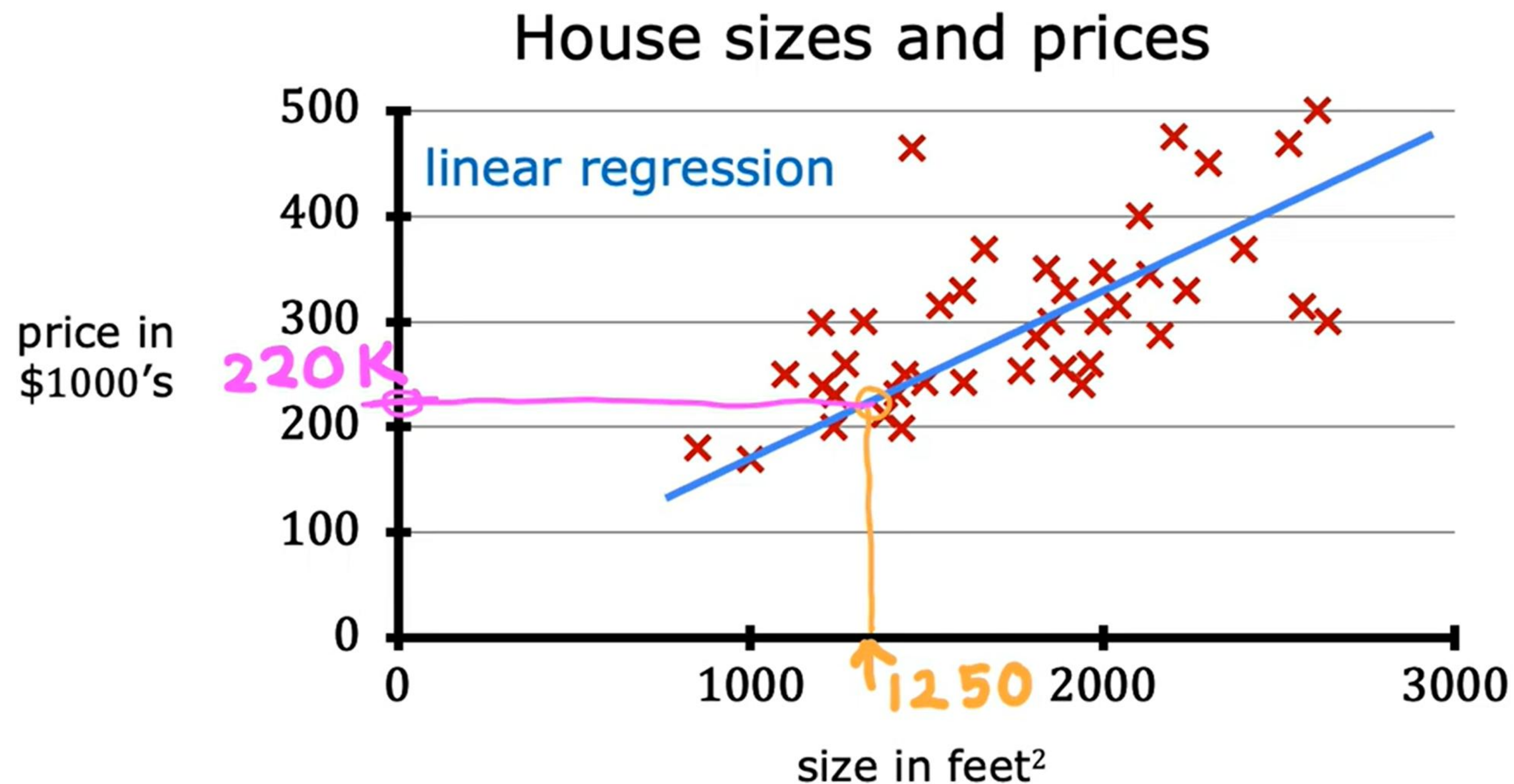
Unsupervised learning
Find something interesting
in **unlabeled** data.



Linear Regression



Linear Regression



Linear Regression

House sizes and prices



Data table

size in feet ²	price in \$1000's
2104	400
1416	232
1534	315
852	178
...	...
3210	870

Linear Regression

Terminology

Training set: Data used to train the model

	x size in feet ²	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

Notation:

x = "input" variable
feature

y = "output" variable
"target" variable

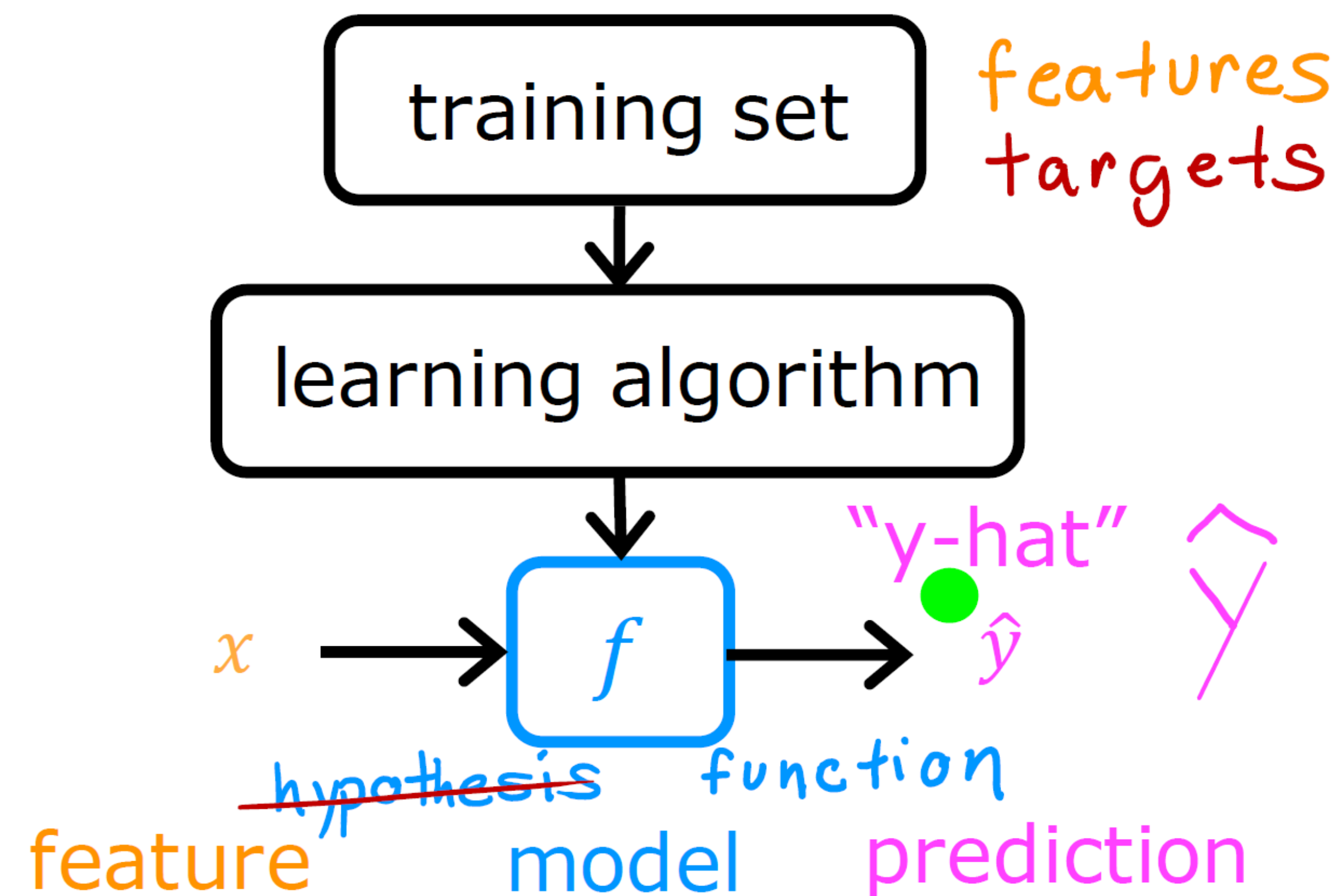
m = number of training examples

$$(x^{(i)}, y^{(i)})$$

$(x^{(i)}, y^{(i)})$ = i^{th} training example

(1st, 2nd, 3rd ...)

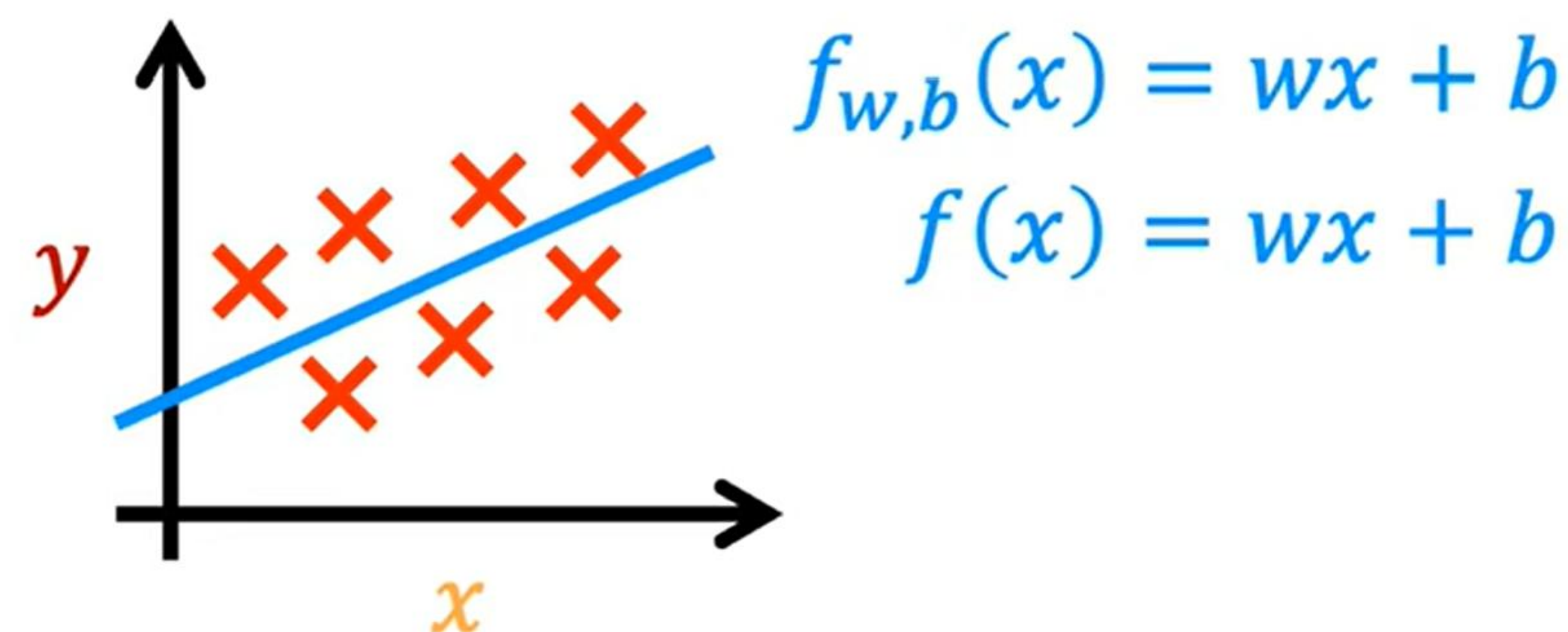
Linear Regression



How to represent f ?

$$f_{w,b}(x) = wx + b$$

$f(x)$



Linear regression with **one** variable.

Univariate linear regression.
one variable

Linear Regression (Cost function)

Training set

<i>features</i> size in feet ² (x)	<i>targets</i> price \$1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

Model: $f_{w,b}(x) = wx + b$

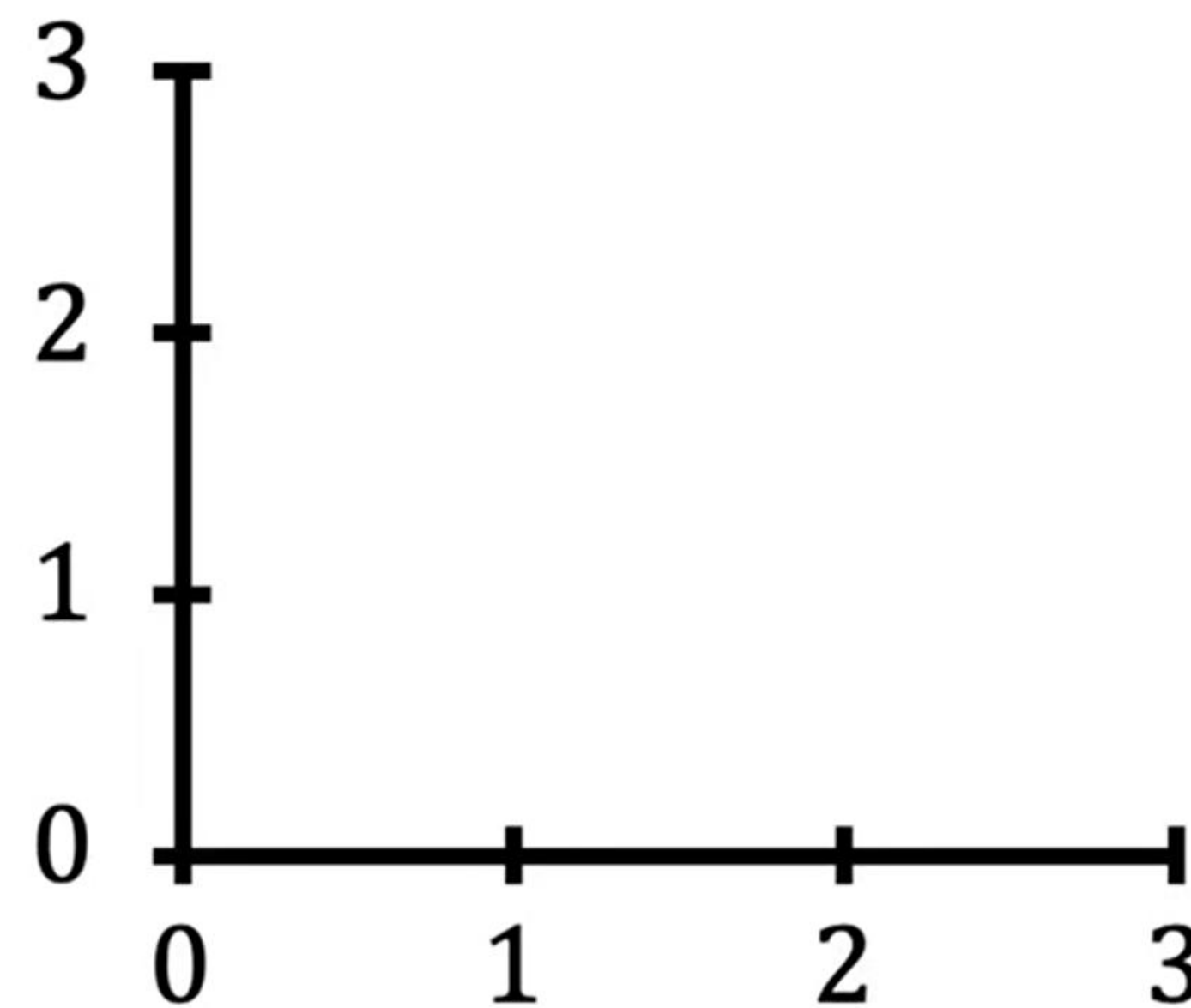
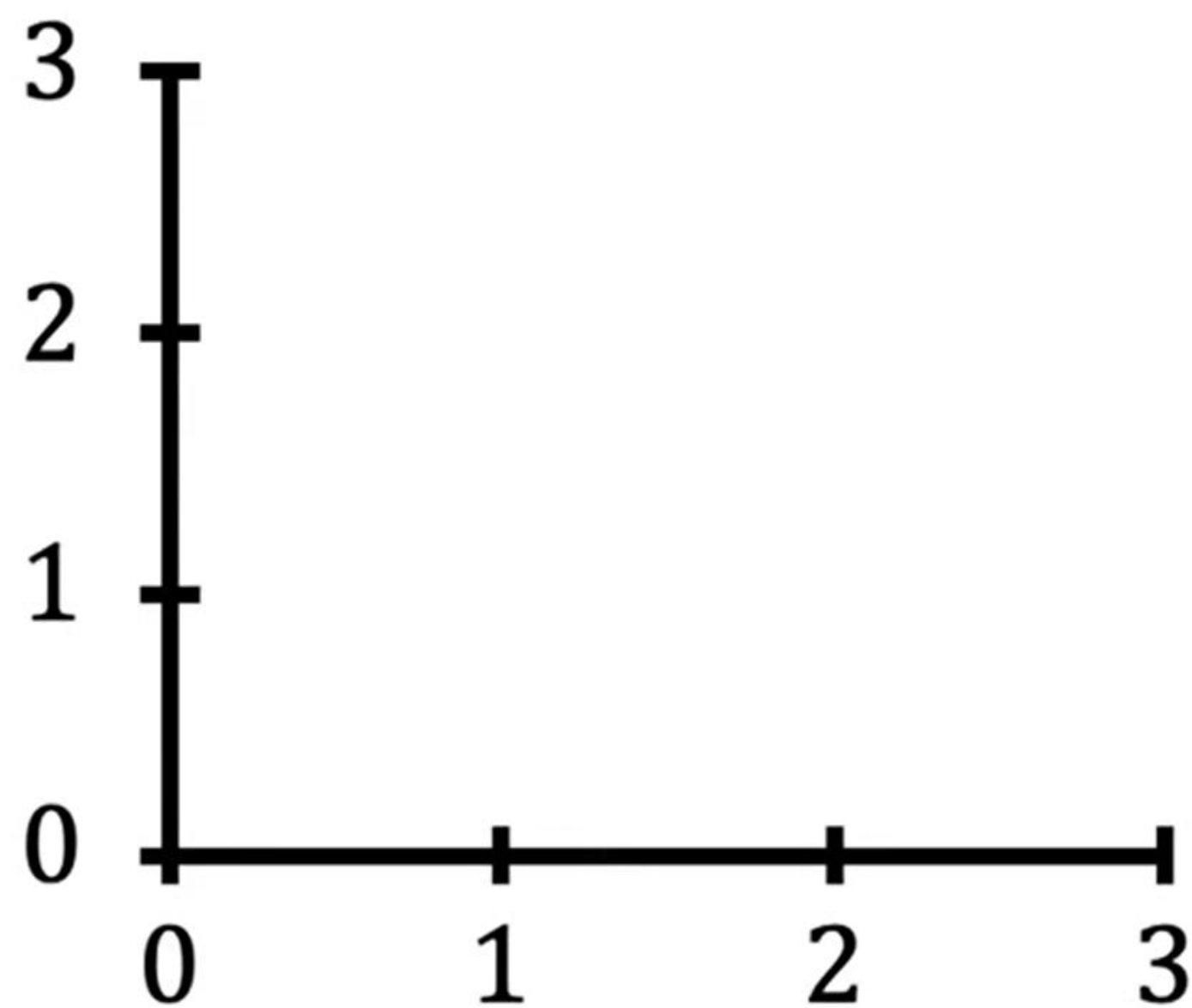
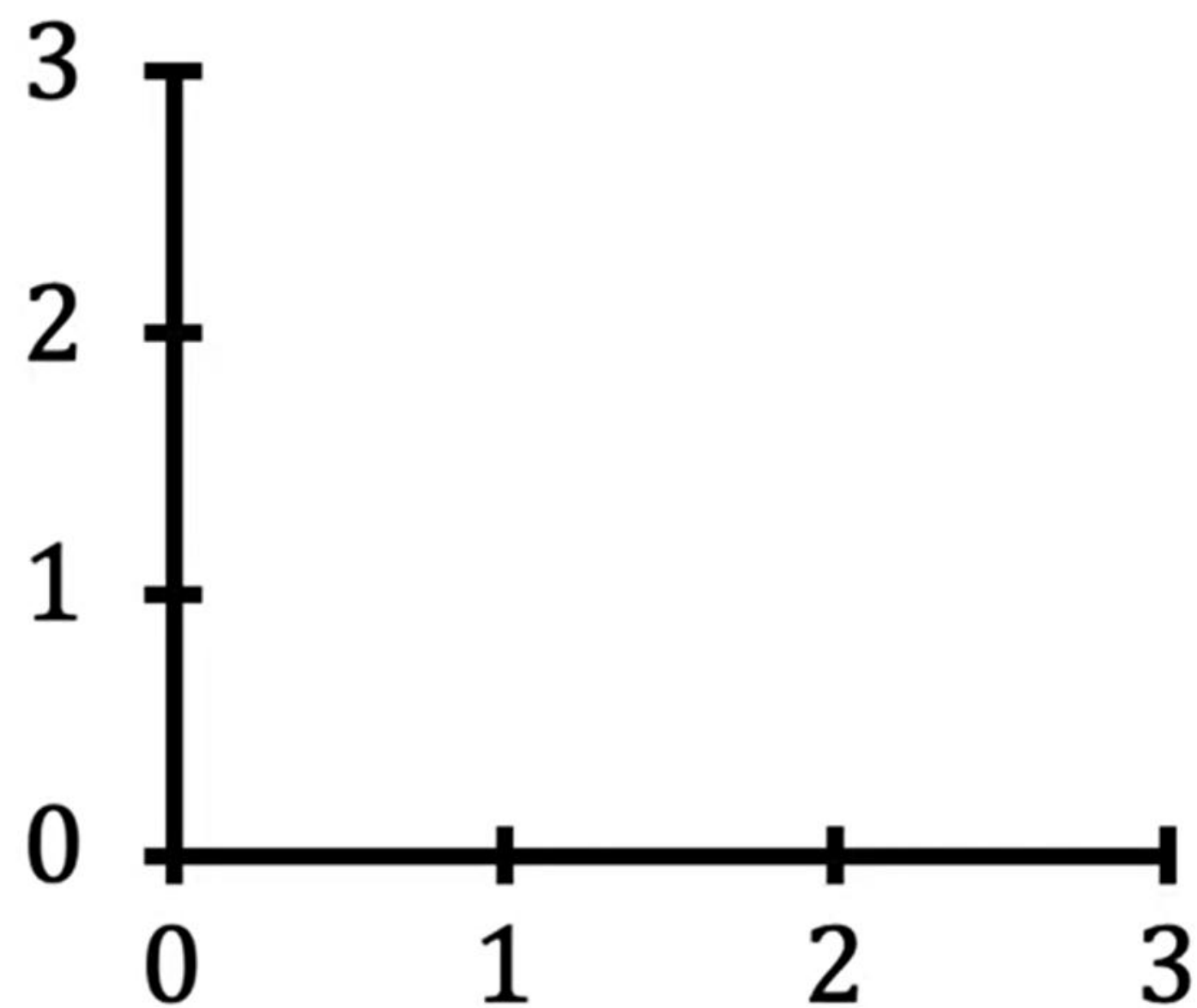
w, b : parameters
coefficients
weights

What do w, b do?

Linear Regression (Cost function)

$$\underline{f_{w,b}}(x) = wx + b$$

$f(x)$



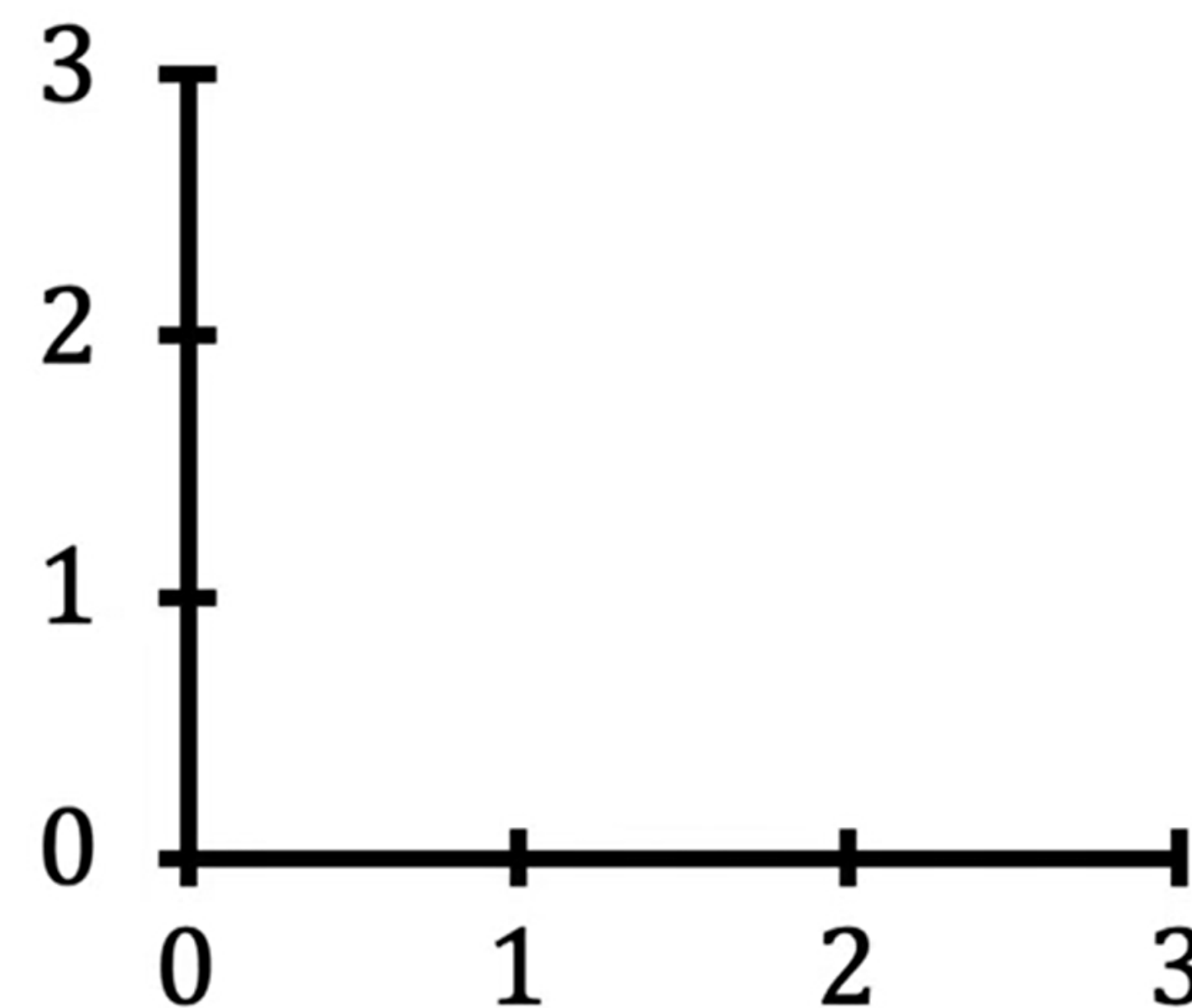
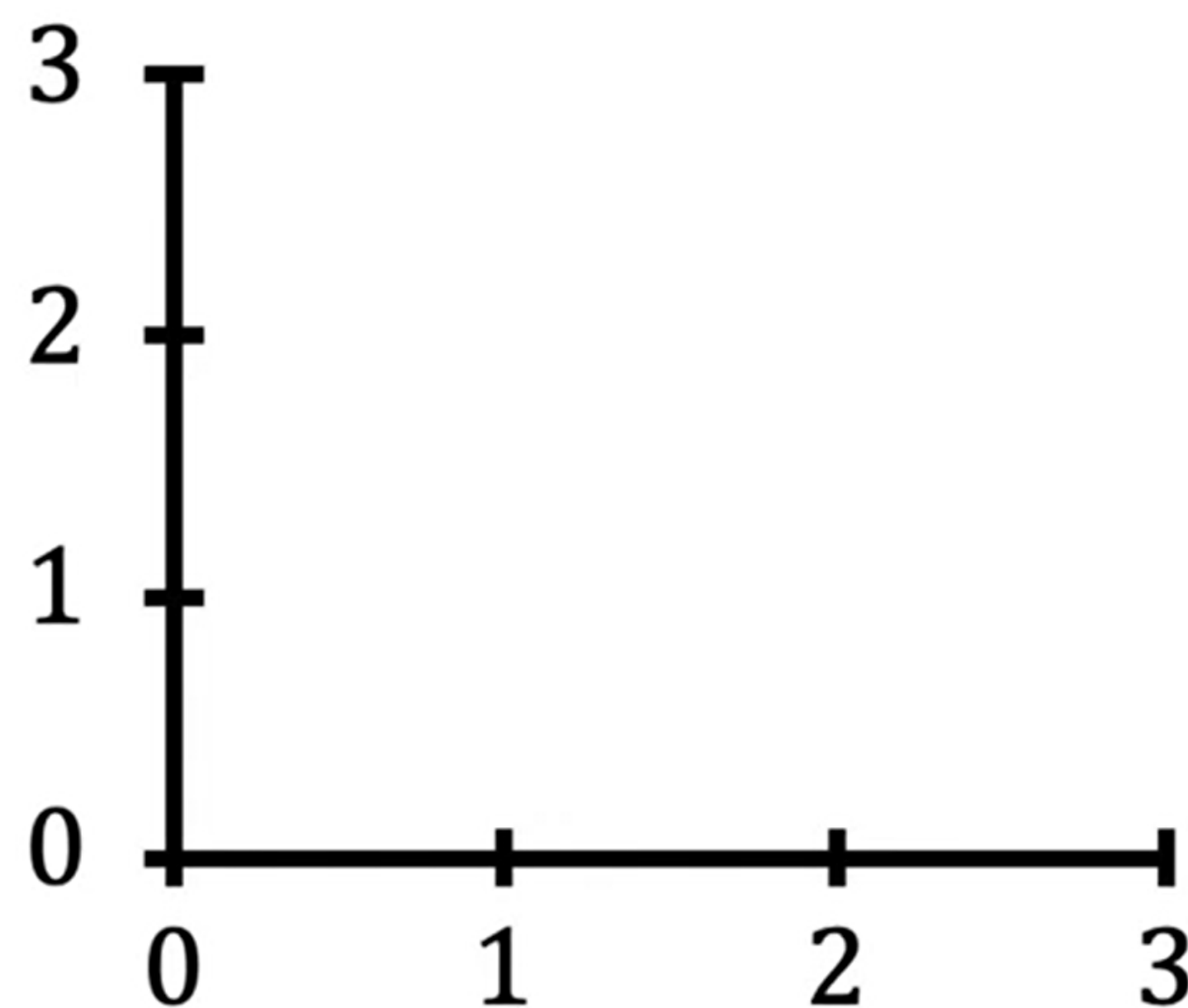
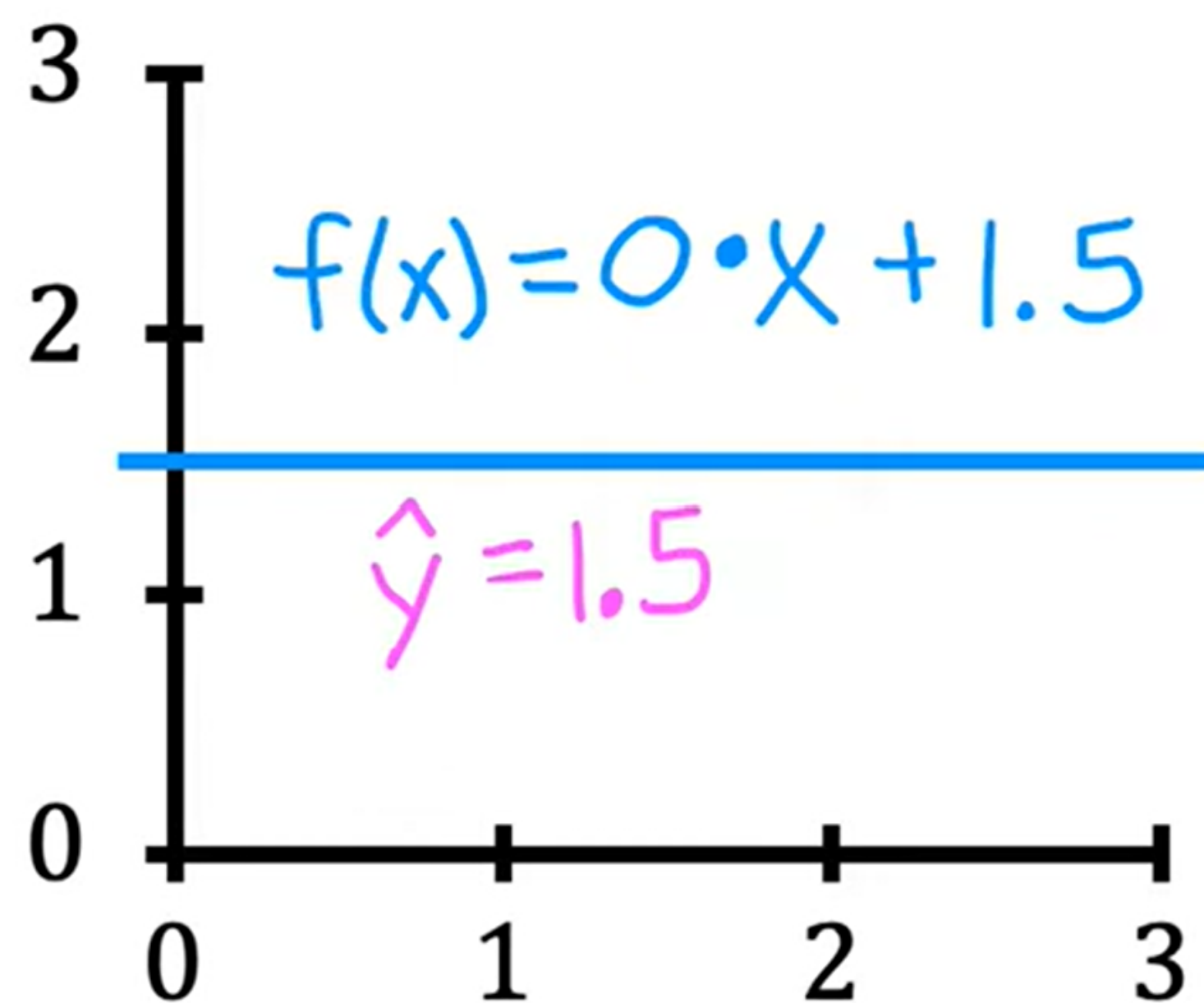
→ $w = 0$

→ $b = 1.5$

Linear Regression (Cost function)

$$\underline{f_{w,b}}(x) = wx + b$$

$f(x)$



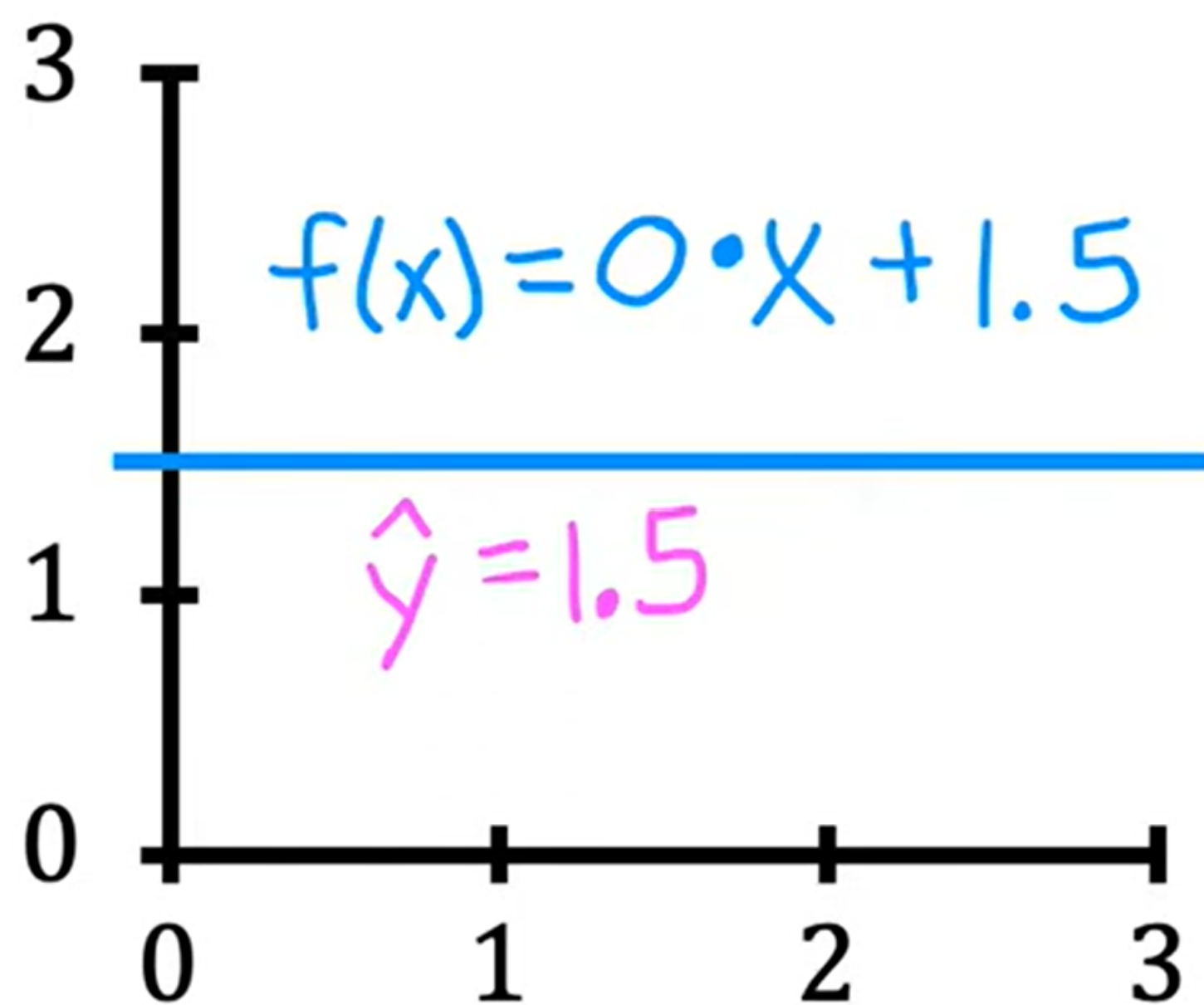
→ $w = 0$

→ $b = 1.5$

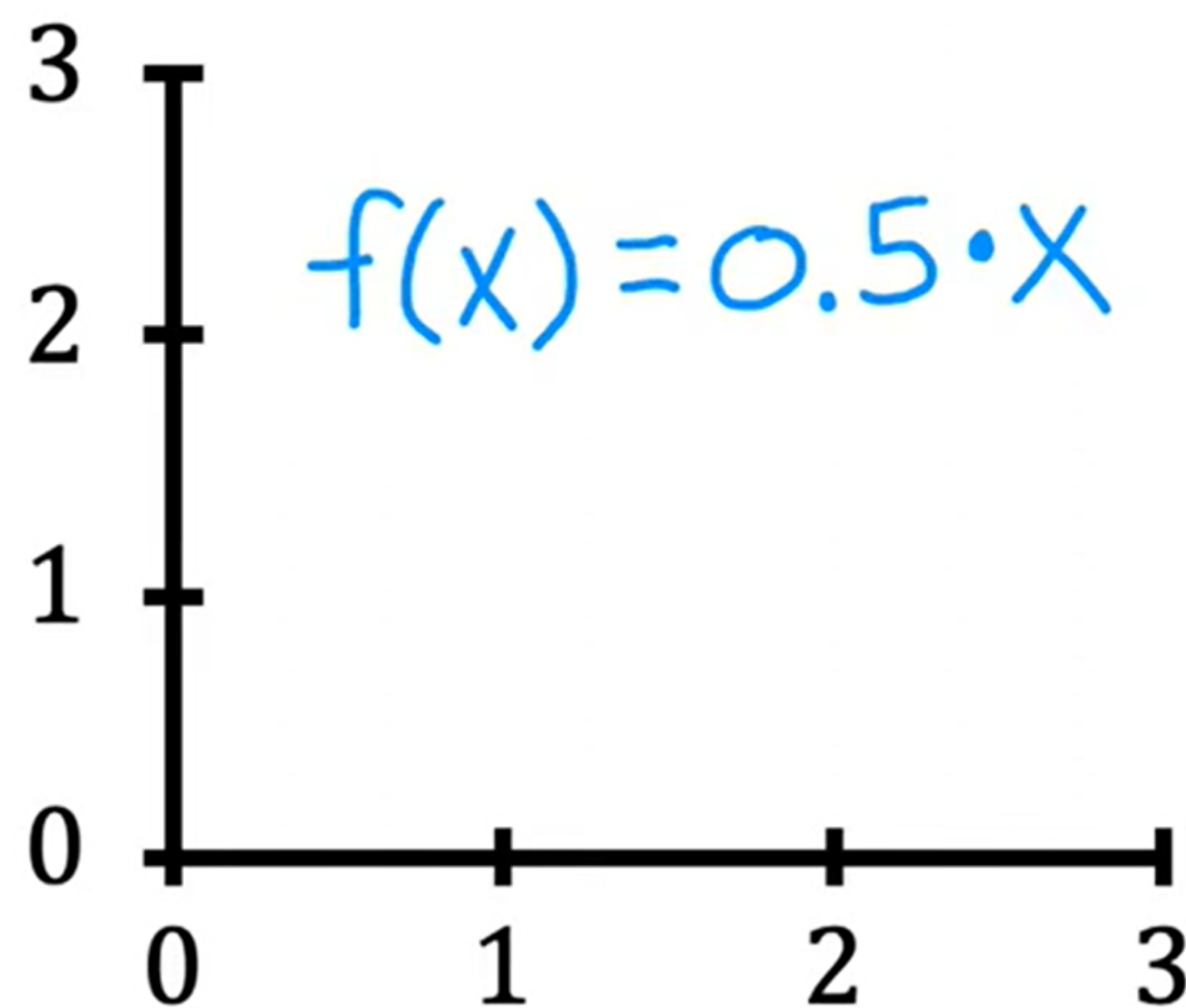
Linear Regression (Cost function)

$$\underline{f_{w,b}}(x) = wx + b$$

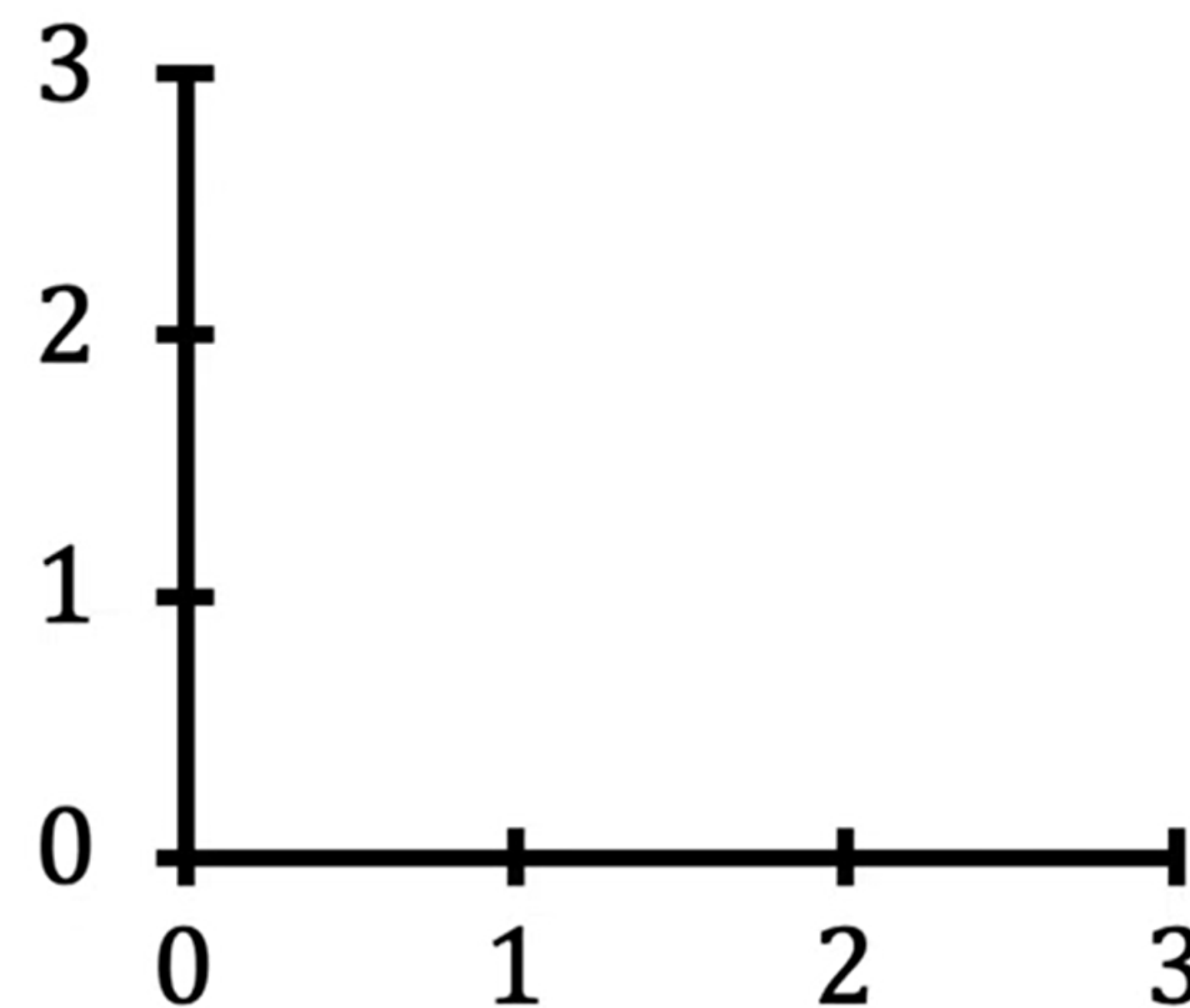
$f(x)$



→ $w = 0$
→ $b = 1.5$



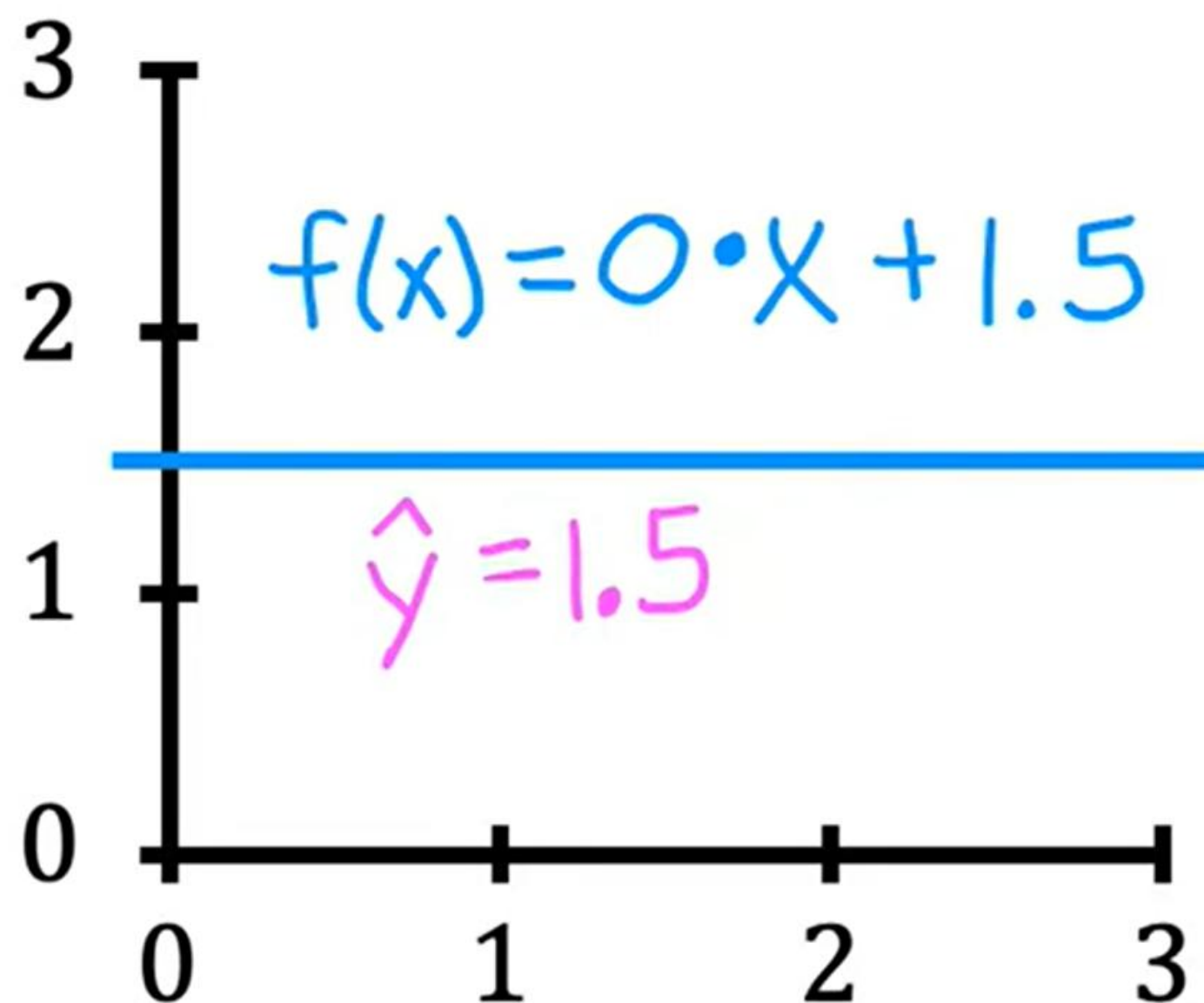
→ $w = 0.5$
→ $b = 0$



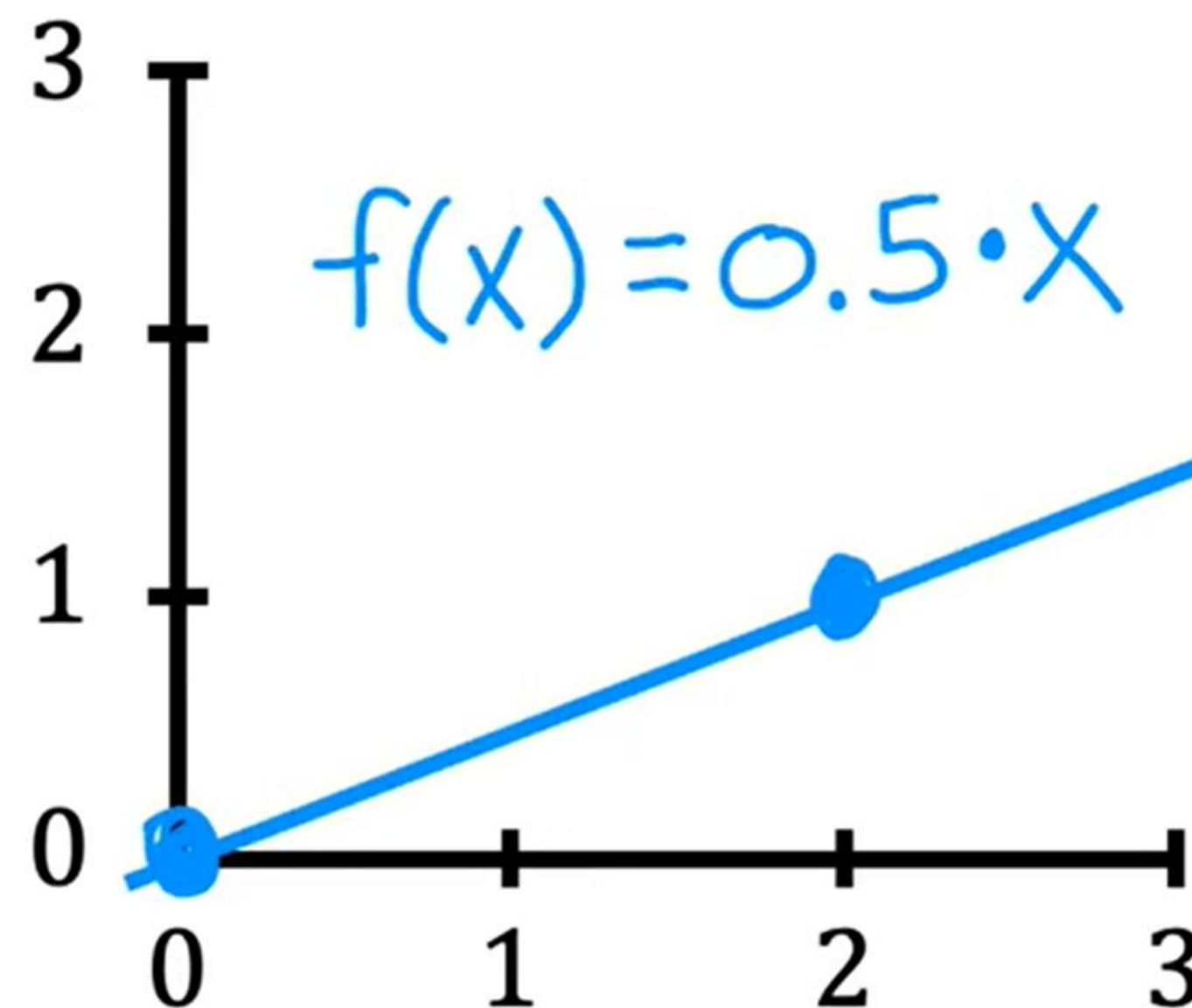
Linear Regression (Cost function)

$$\underline{f_{w,b}}(x) = wx + b$$

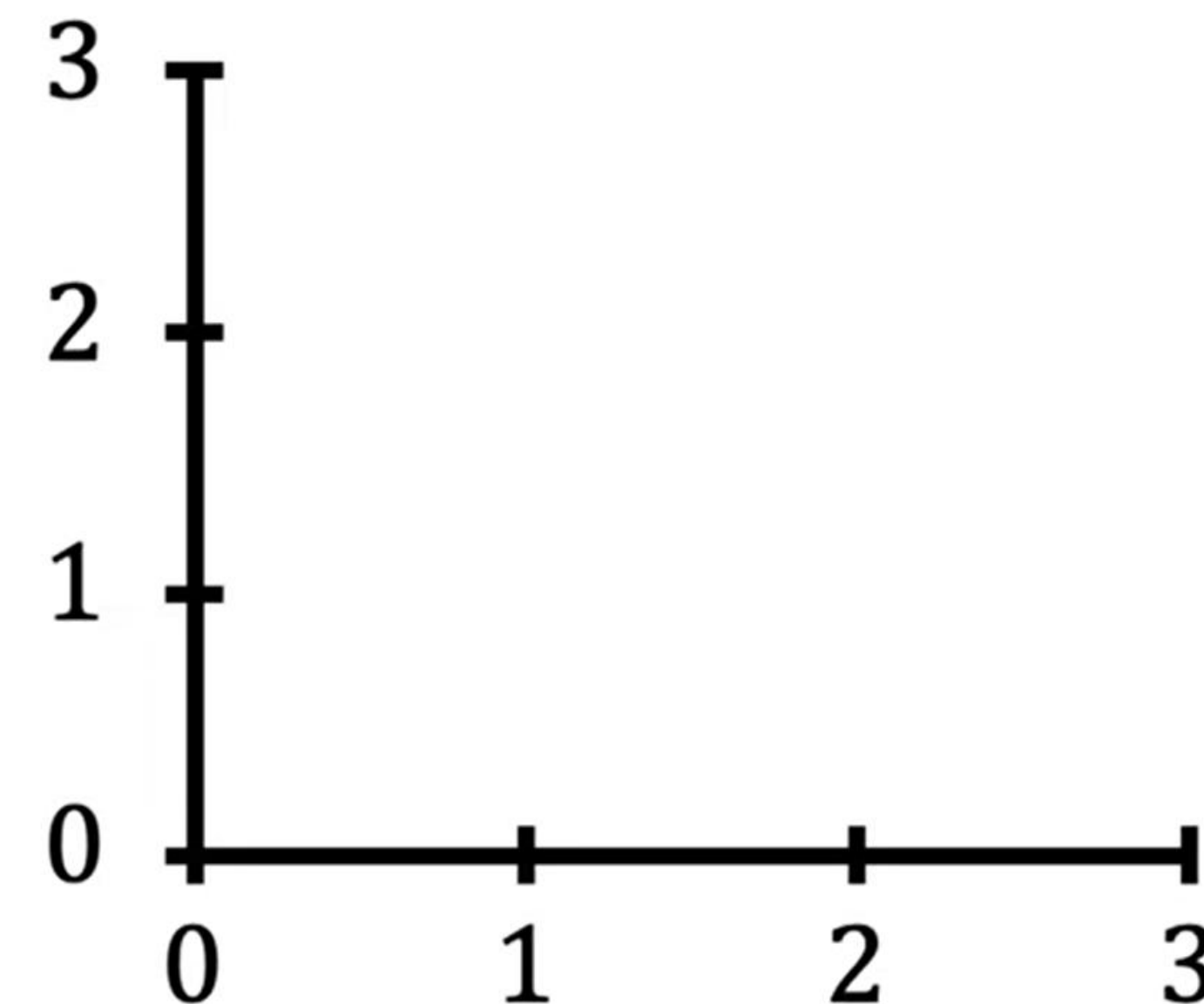
$f(x)$



→ $w = 0$
→ $b = 1.5$



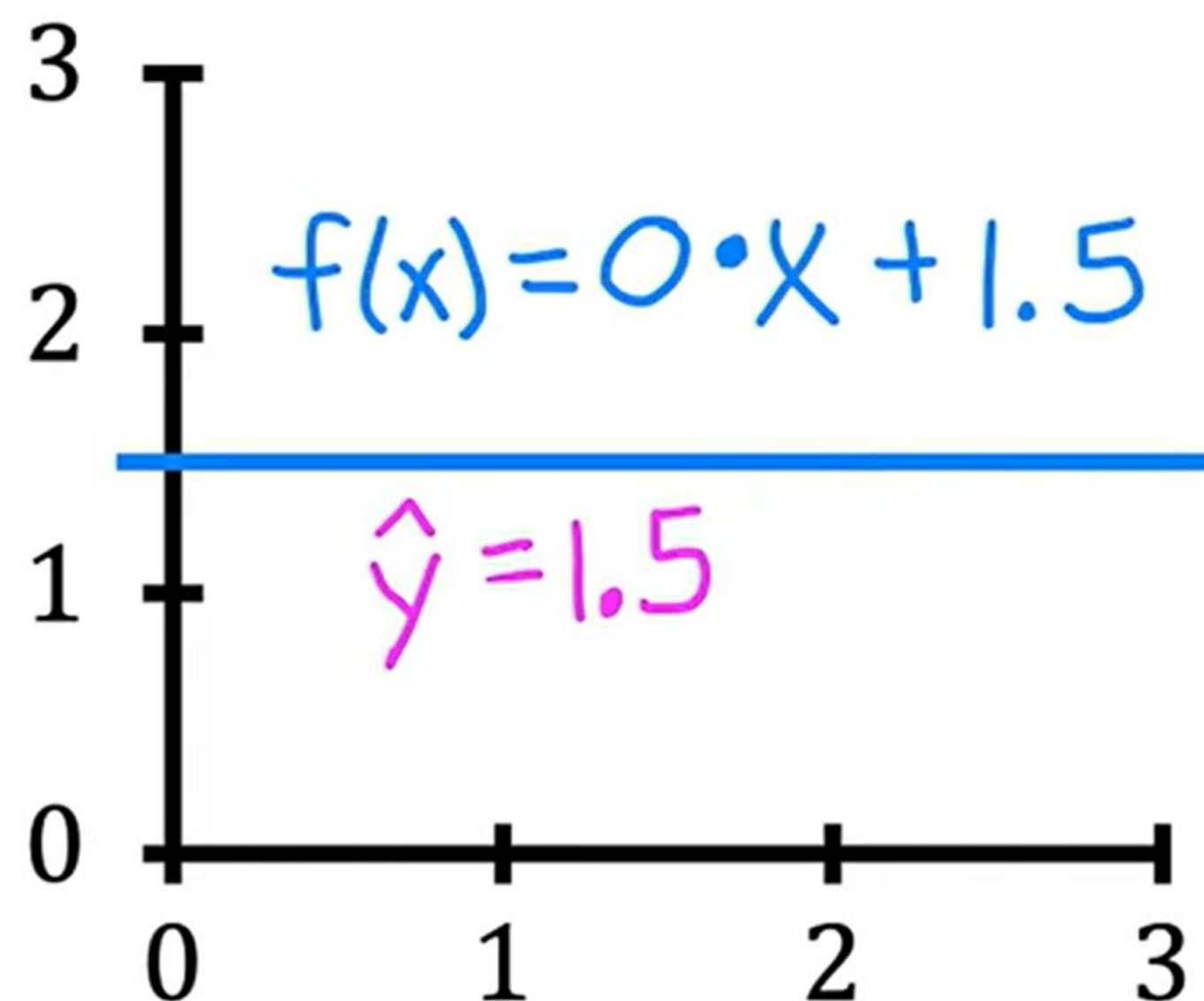
→ $w = 0.5$
→ $b = 0$



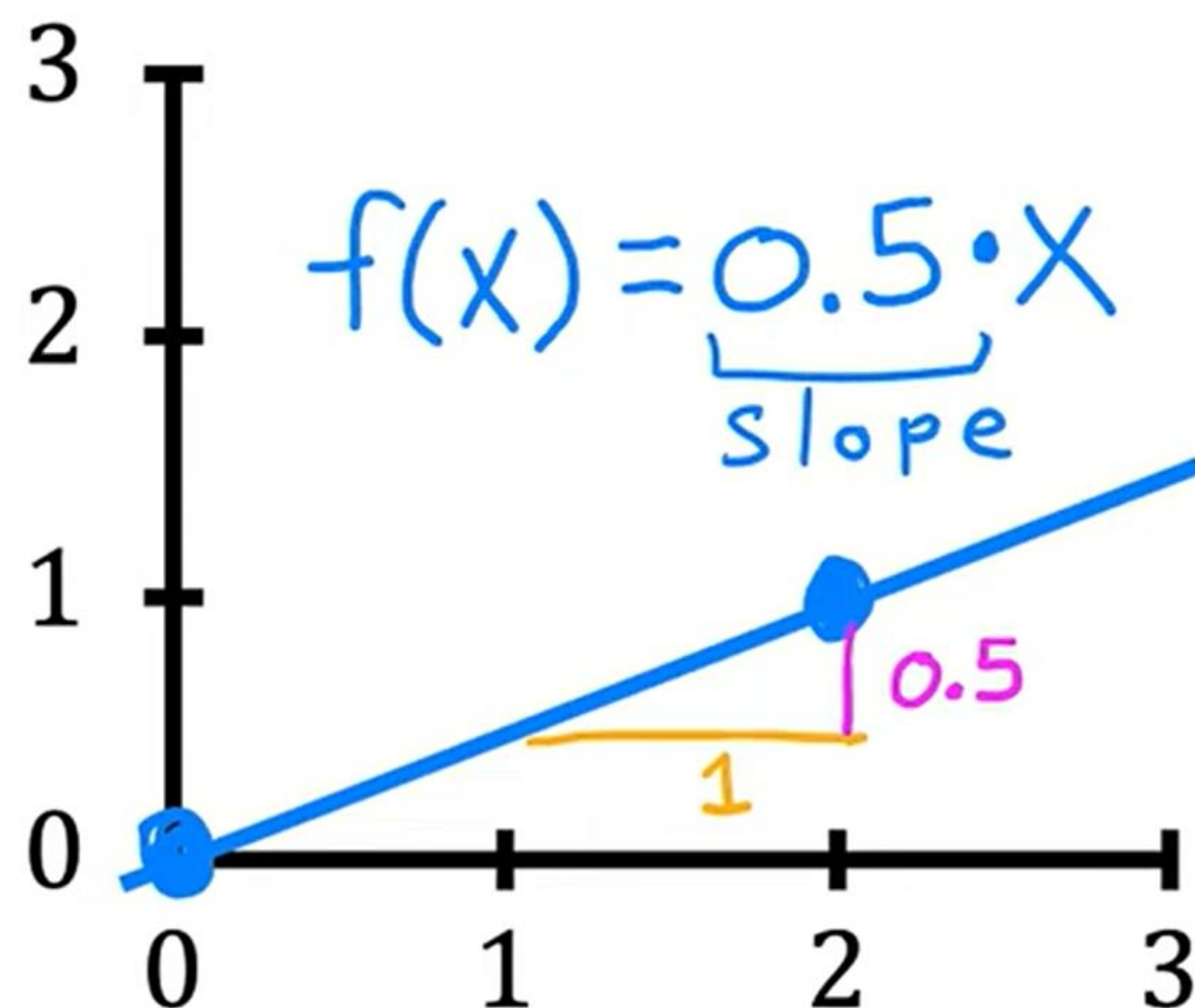
Linear Regression (Cost function)

$$\underline{f_{w,b}}(x) = wx + b$$

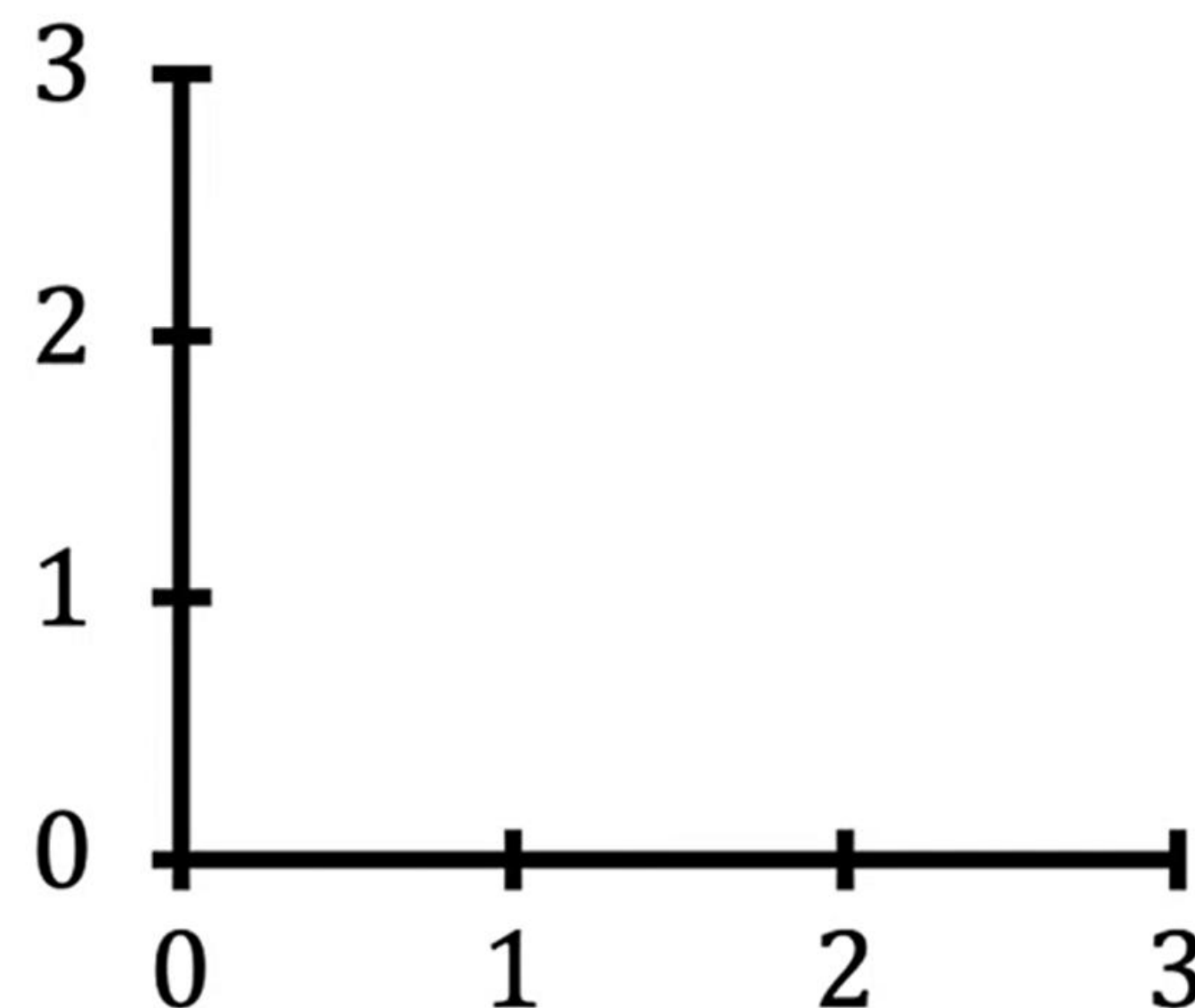
$f(x)$



→ $w = 0$
→ $b = 1.5$



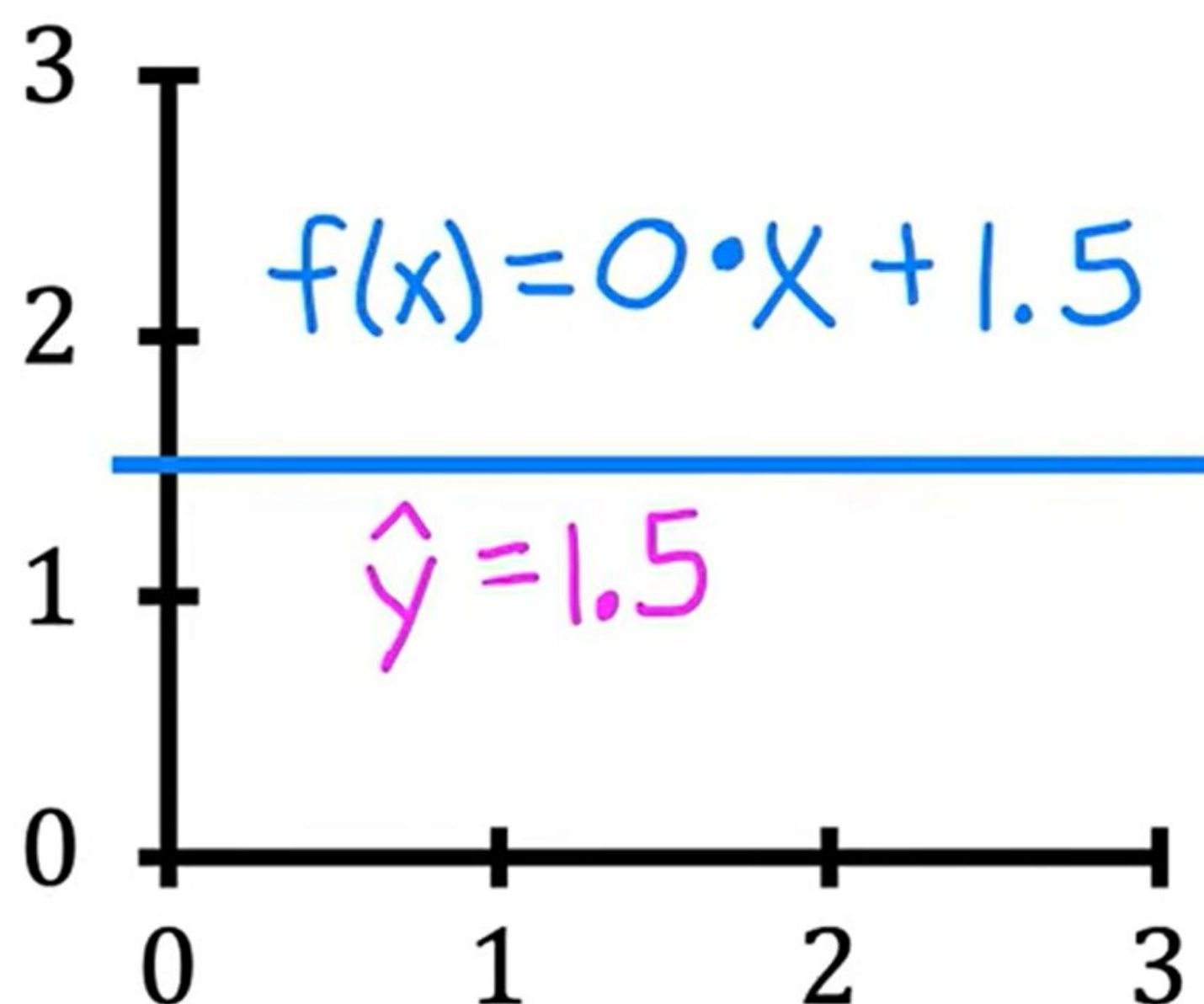
→ $w = 0.5$
→ $b = 0$



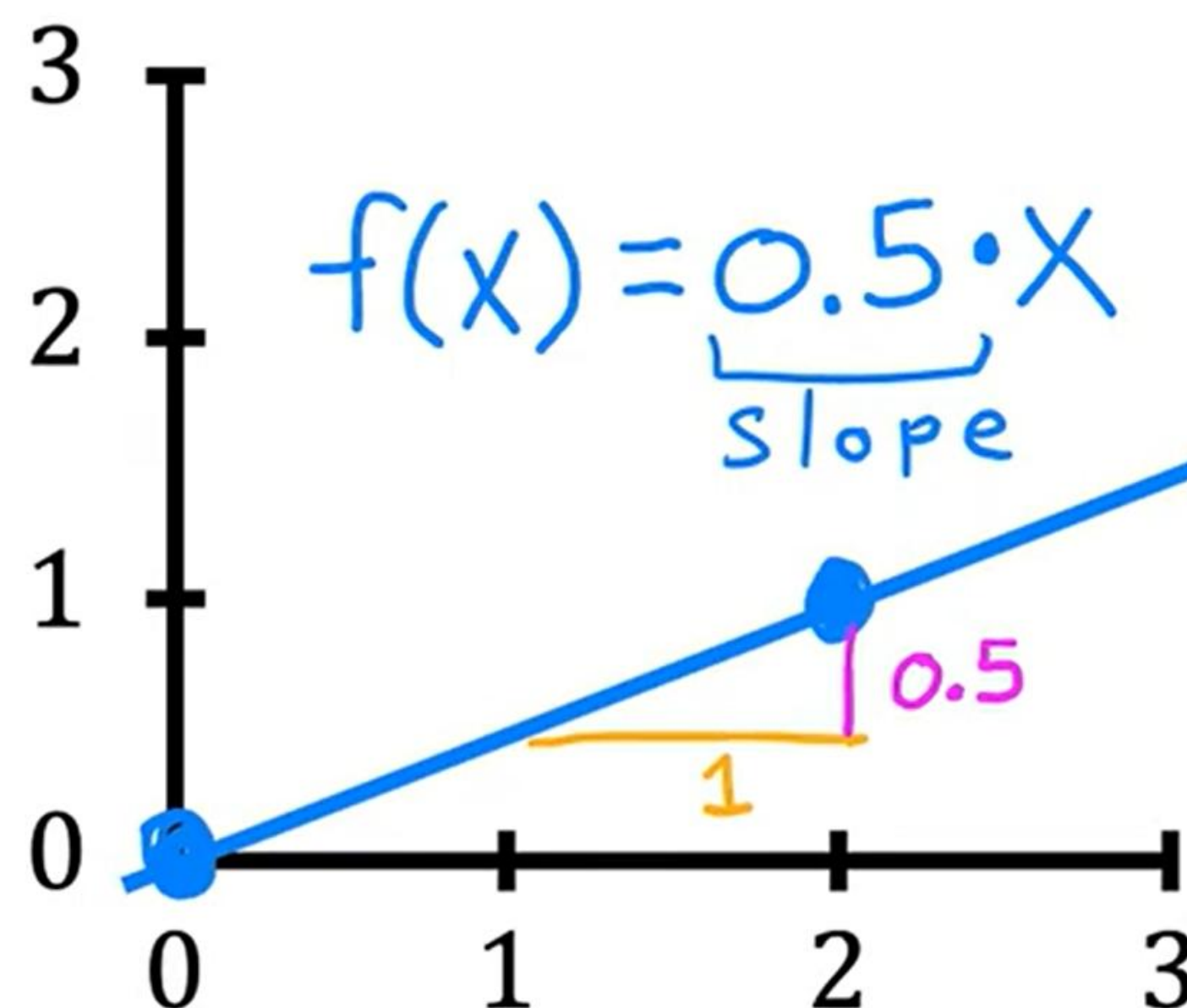
Linear Regression (Cost function)

$$\underline{f_{w,b}}(x) = wx + b$$

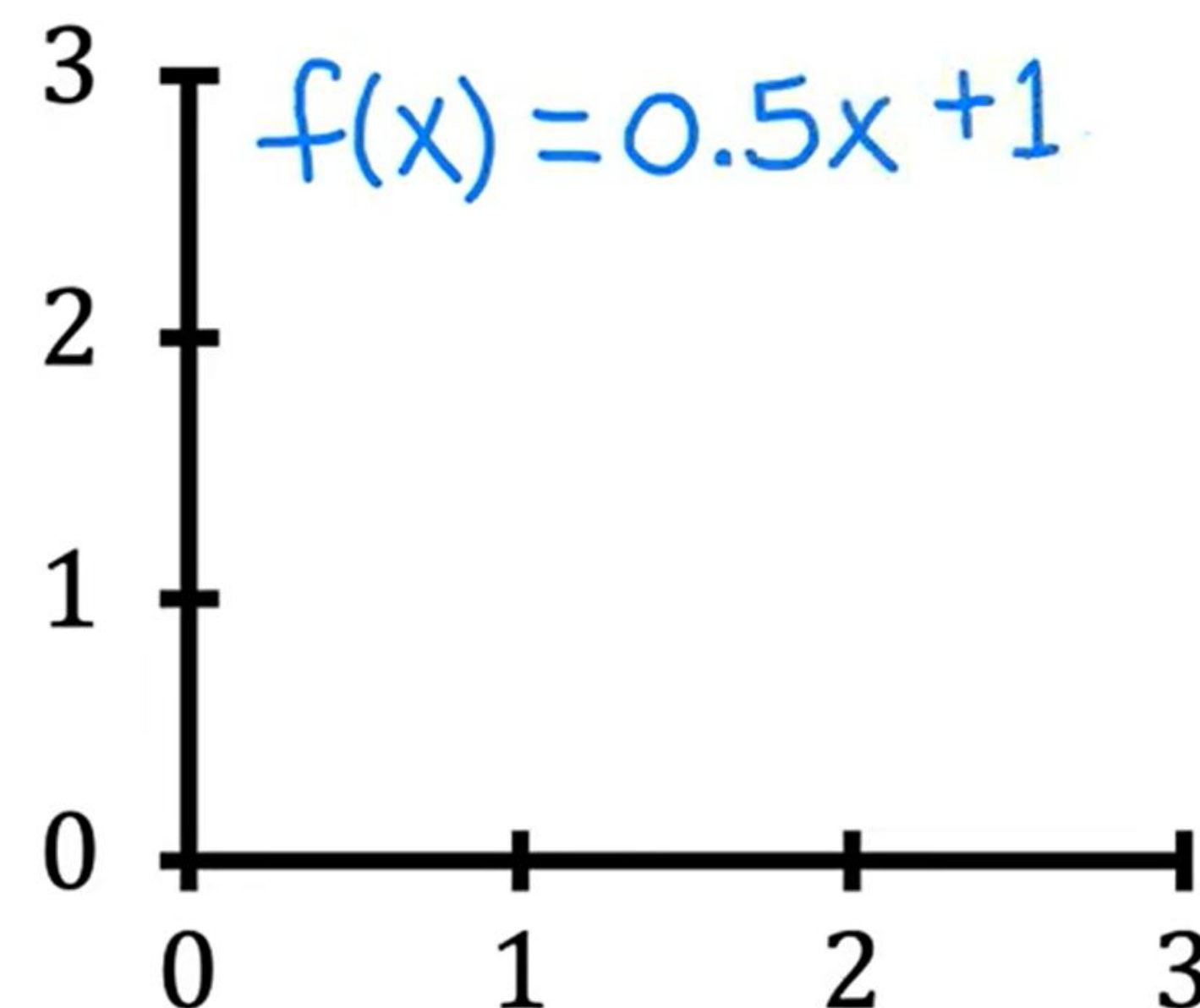
$f(x)$



→ $w = 0$
→ $b = 1.5$



→ $w = 0.5$
→ $b = 0$

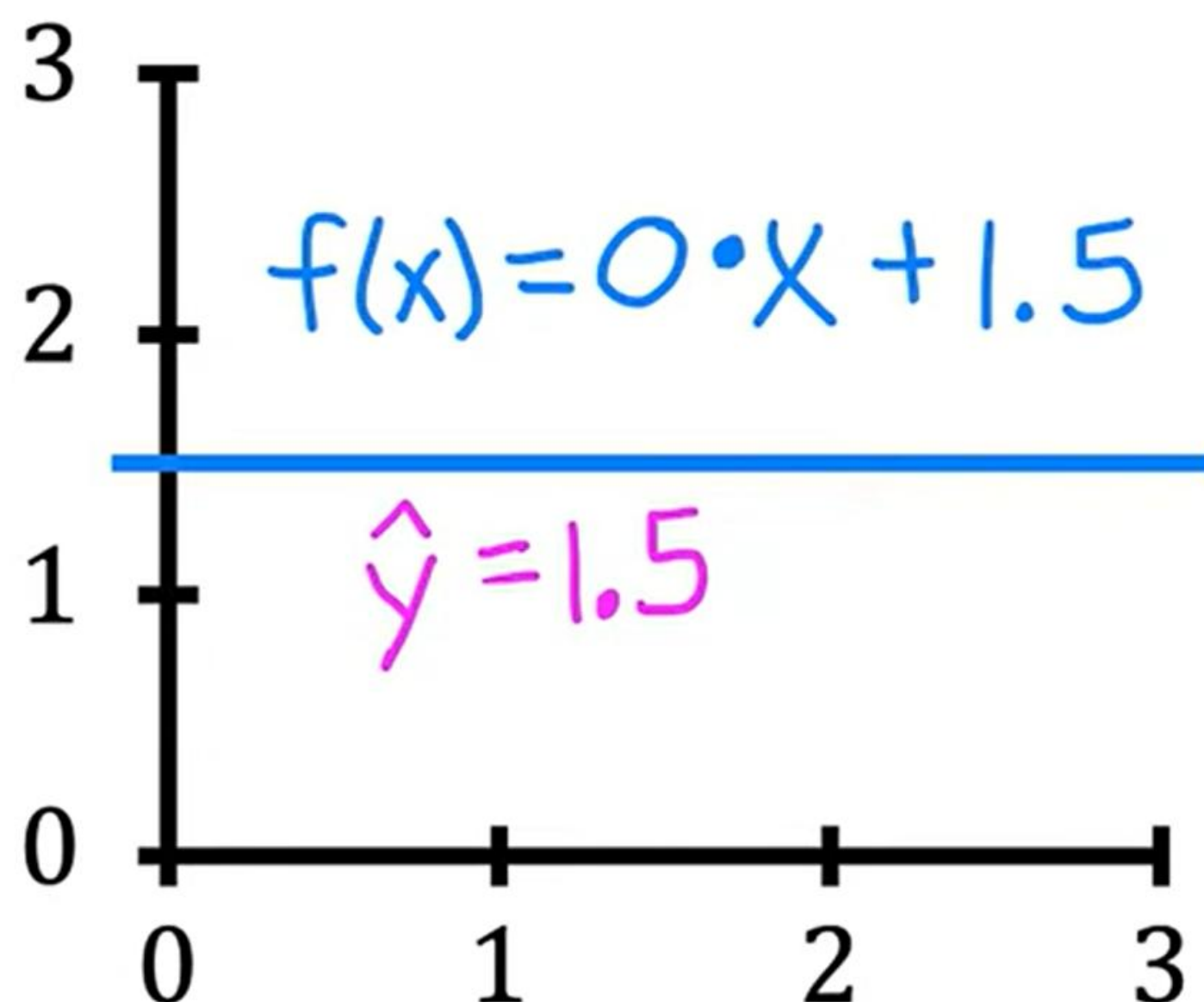


→ $w = 0.5$
→ $b = 1$

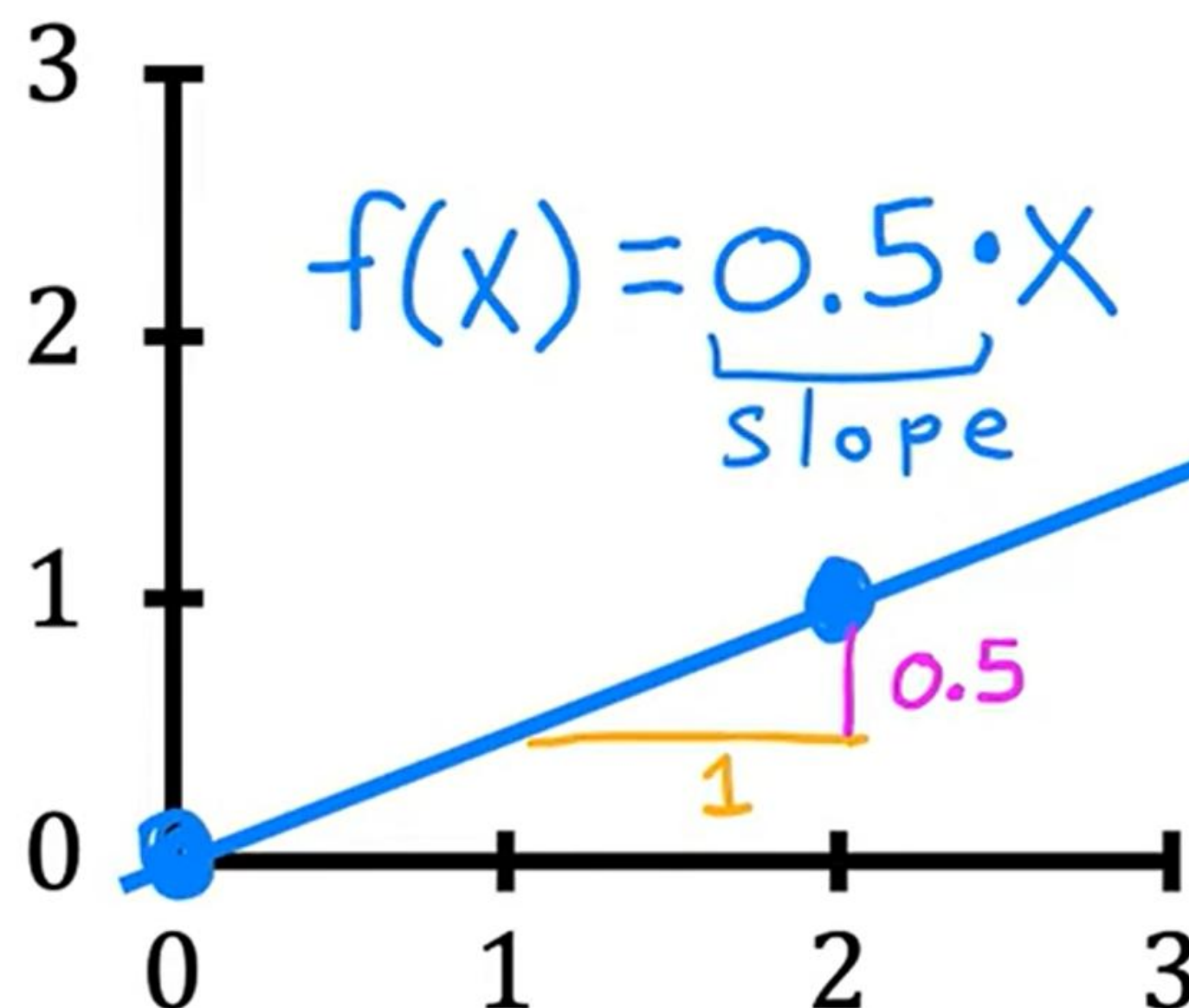
Linear Regression (Cost function)

$$\underline{f_{w,b}}(x) = wx + b$$

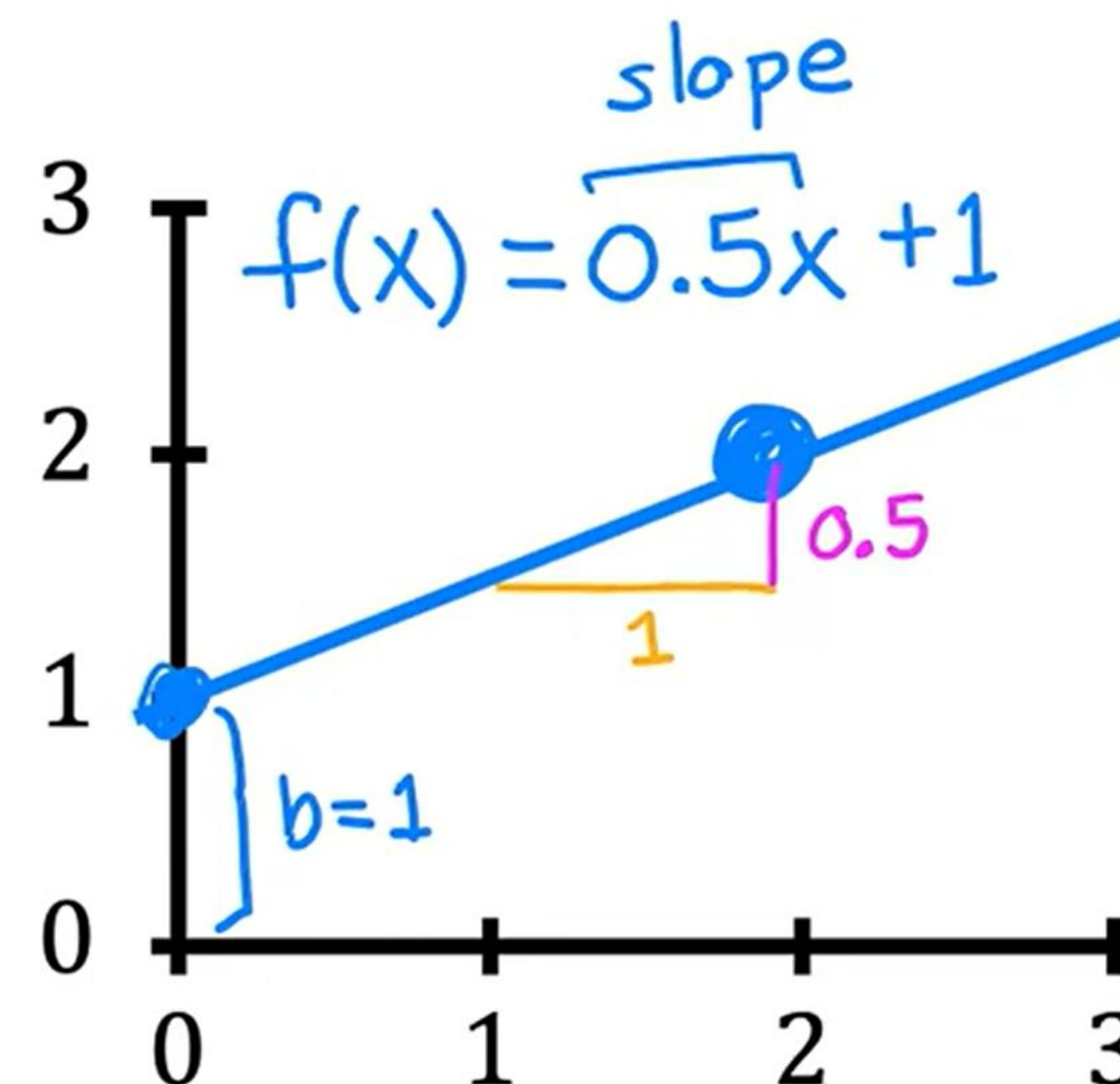
$f(x)$



→ $w = 0$
→ $b = 1.5$

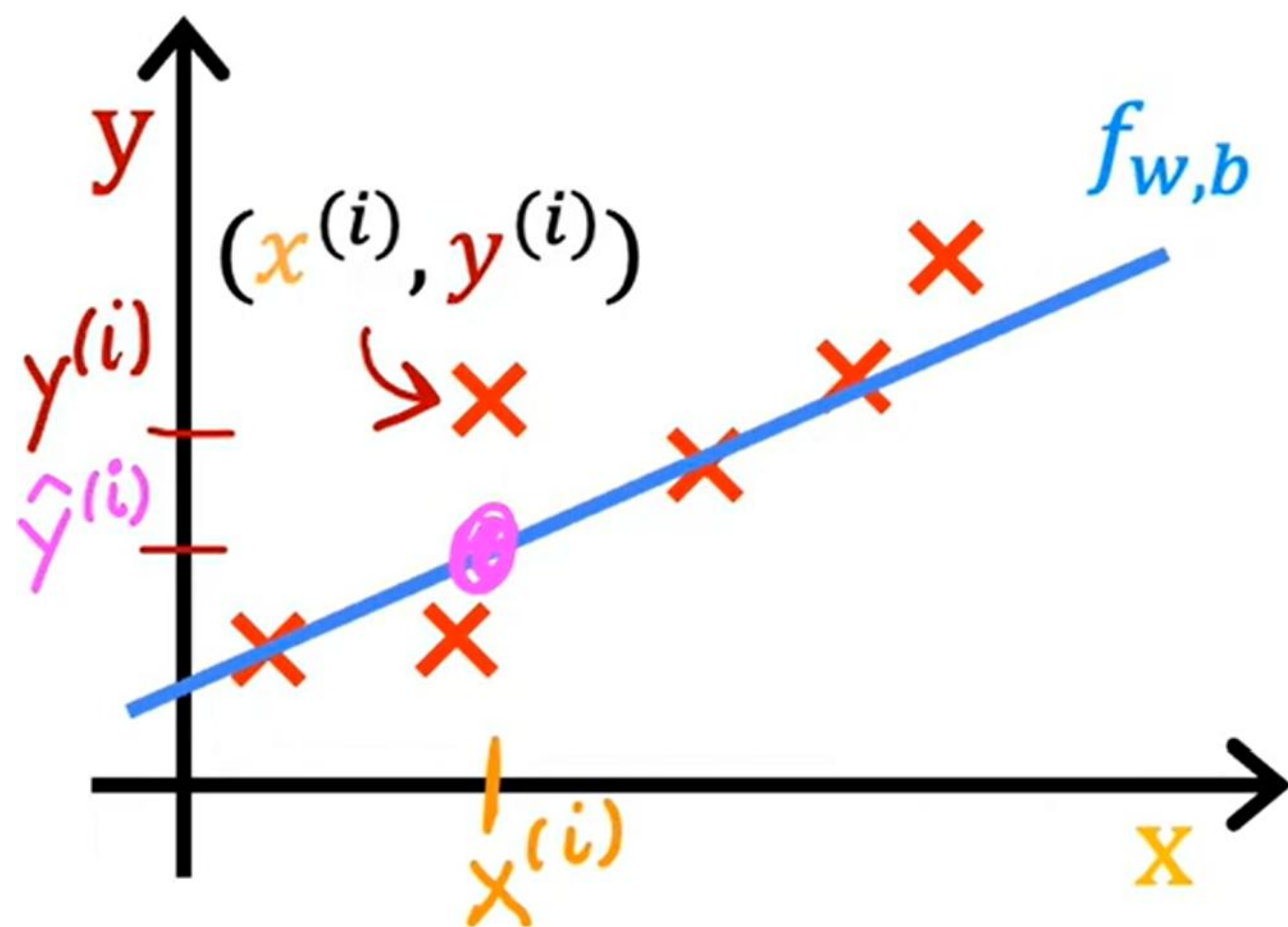


→ $w = 0.5$
→ $b = 0$



→ $w = 0.5$
→ $b = 1$

Linear Regression (Cost function)

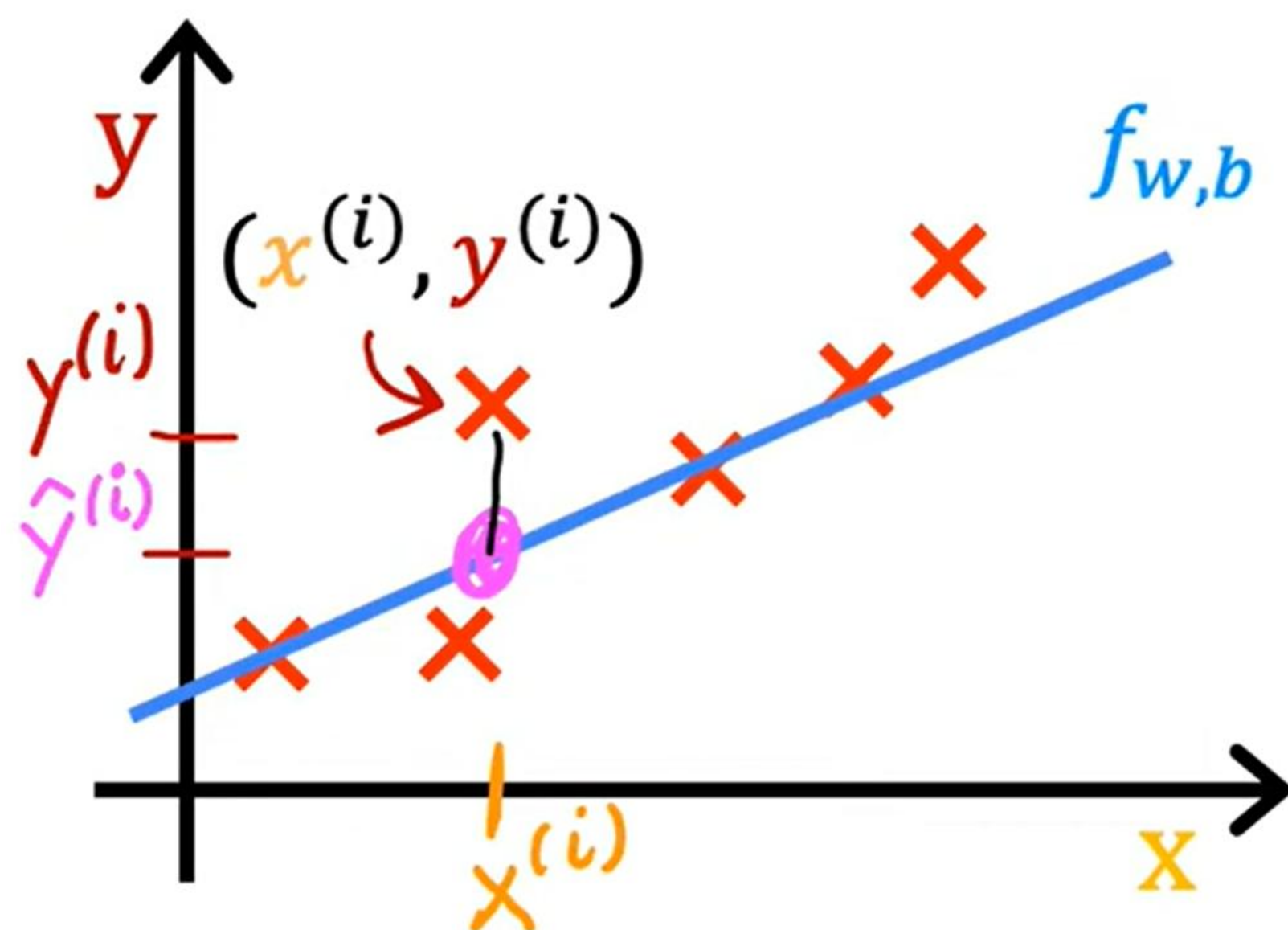


Find w, b :
 $\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.

$$\hat{y}^{(i)} = f_{w,b}(x^{(i)})$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

Linear Regression (Cost function)



$$\hat{y}^{(i)} = f_{w,b}(x^{(i)})$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

Cost function

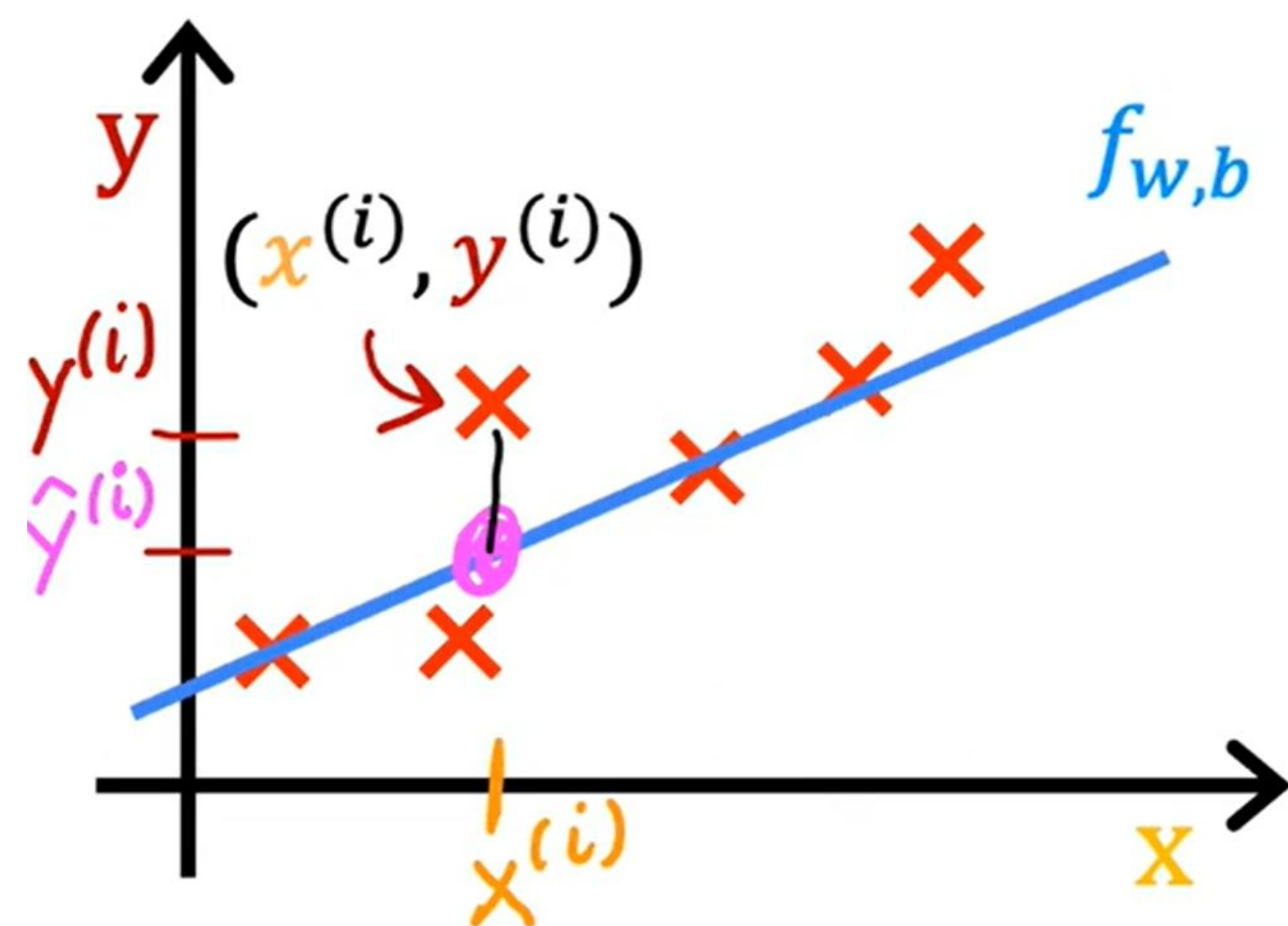
$$(\hat{y} - y)^2$$

error

Find w, b :

$\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.

Linear Regression (Cost function)



$$\hat{y}^{(i)} = f_{w,b}(x^{(i)})$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

Cost function

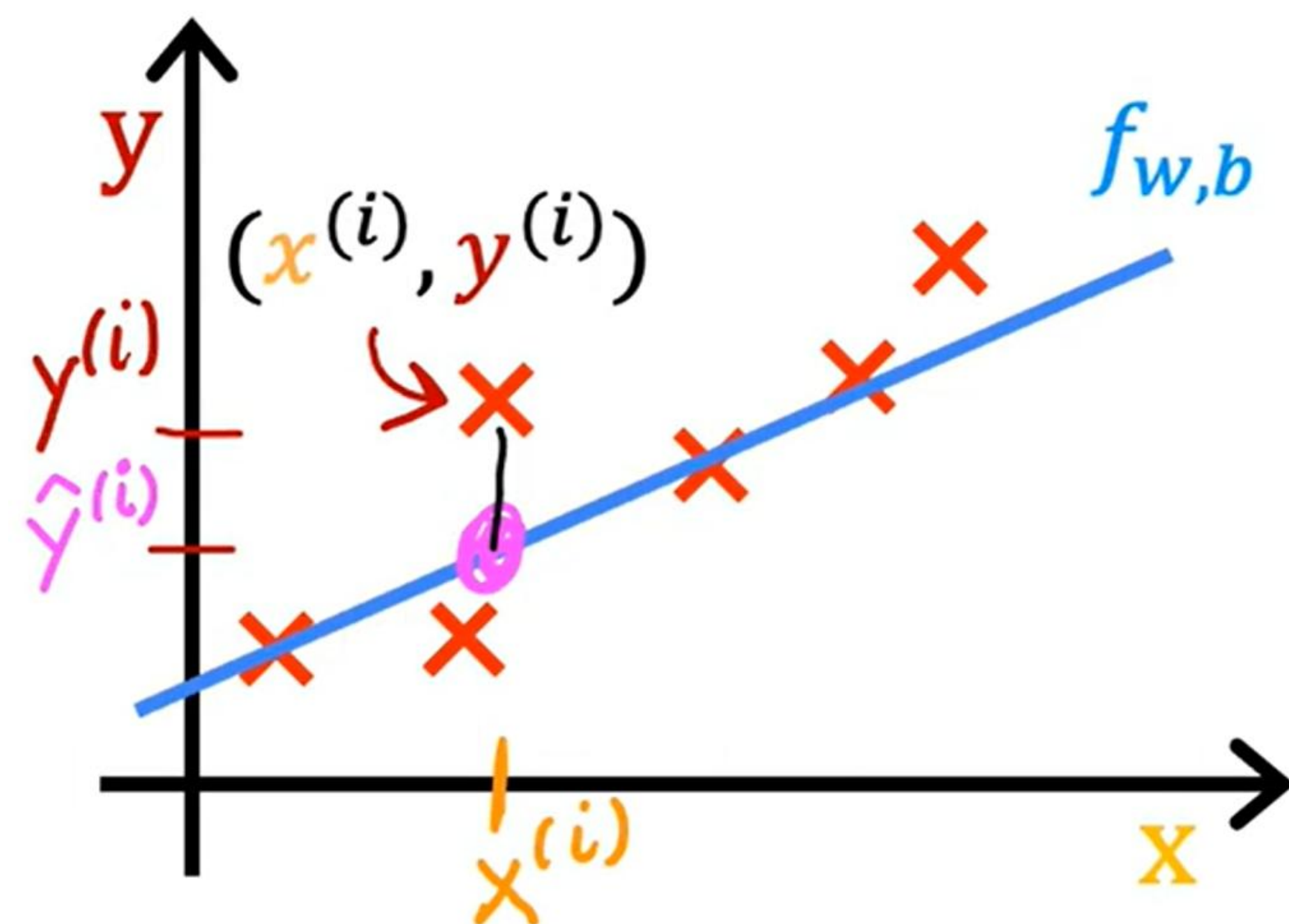
$$\sum_{i=1}^m \left(\underset{\text{error}}{\hat{y}^{(i)}} - y^{(i)} \right)^2$$

m = number of training examples

Find w, b :

$\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.

Linear Regression (Cost function)



$$\hat{y}^{(i)} = f_{w,b}(x^{(i)})$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

Cost function Squared error cost function

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m \left(\underset{\text{error}}{\hat{y}^{(i)}} - y^{(i)} \right)^2$$

m = number of training examples

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m \left(f_{w,b}(x^{(i)}) - y^{(i)} \right)^2$$

Find w, b :

$\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.

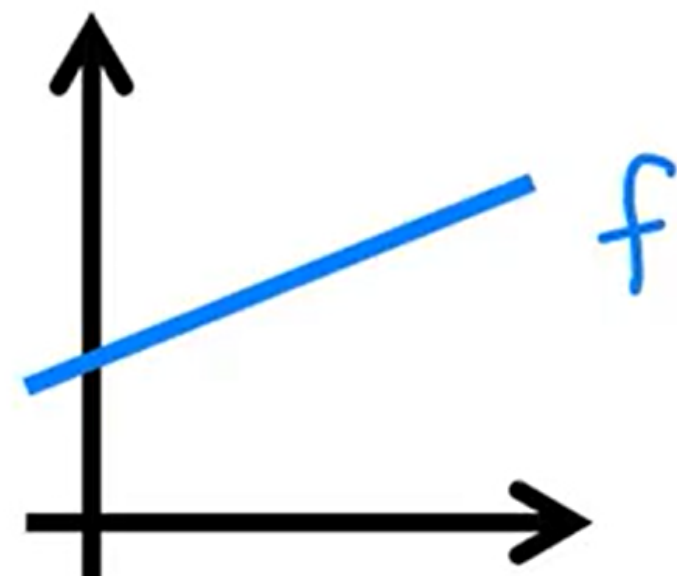
Linear Regression (Cost function Intuition)

model:

$$\underline{f_{w,b}(x) = wx + b}$$

parameters:

$$\underline{w, b}$$



cost function:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

goal:

$$\underset{w, b}{\text{minimize}} J(w, b)$$

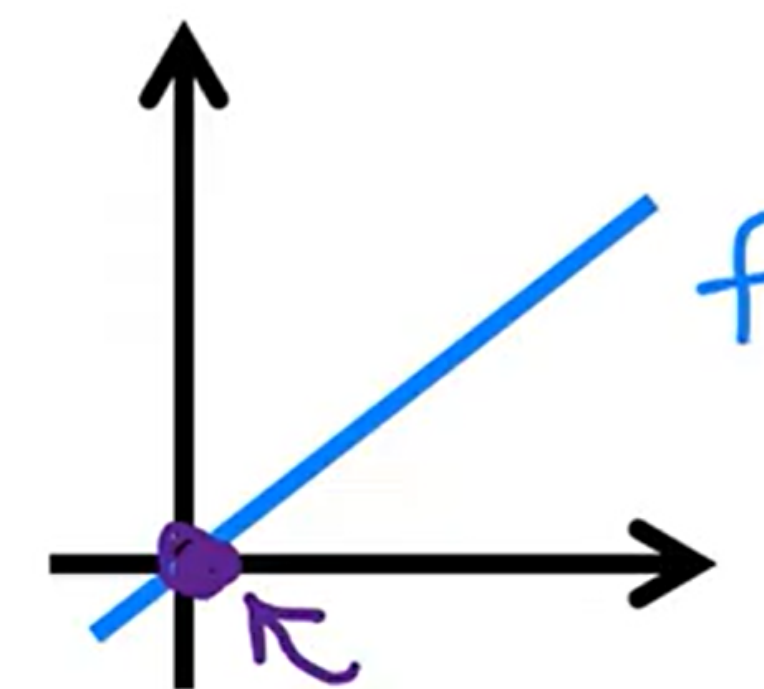
simplified

$$f_w(x) = \underline{wx}$$

$$b = \emptyset$$

w

w

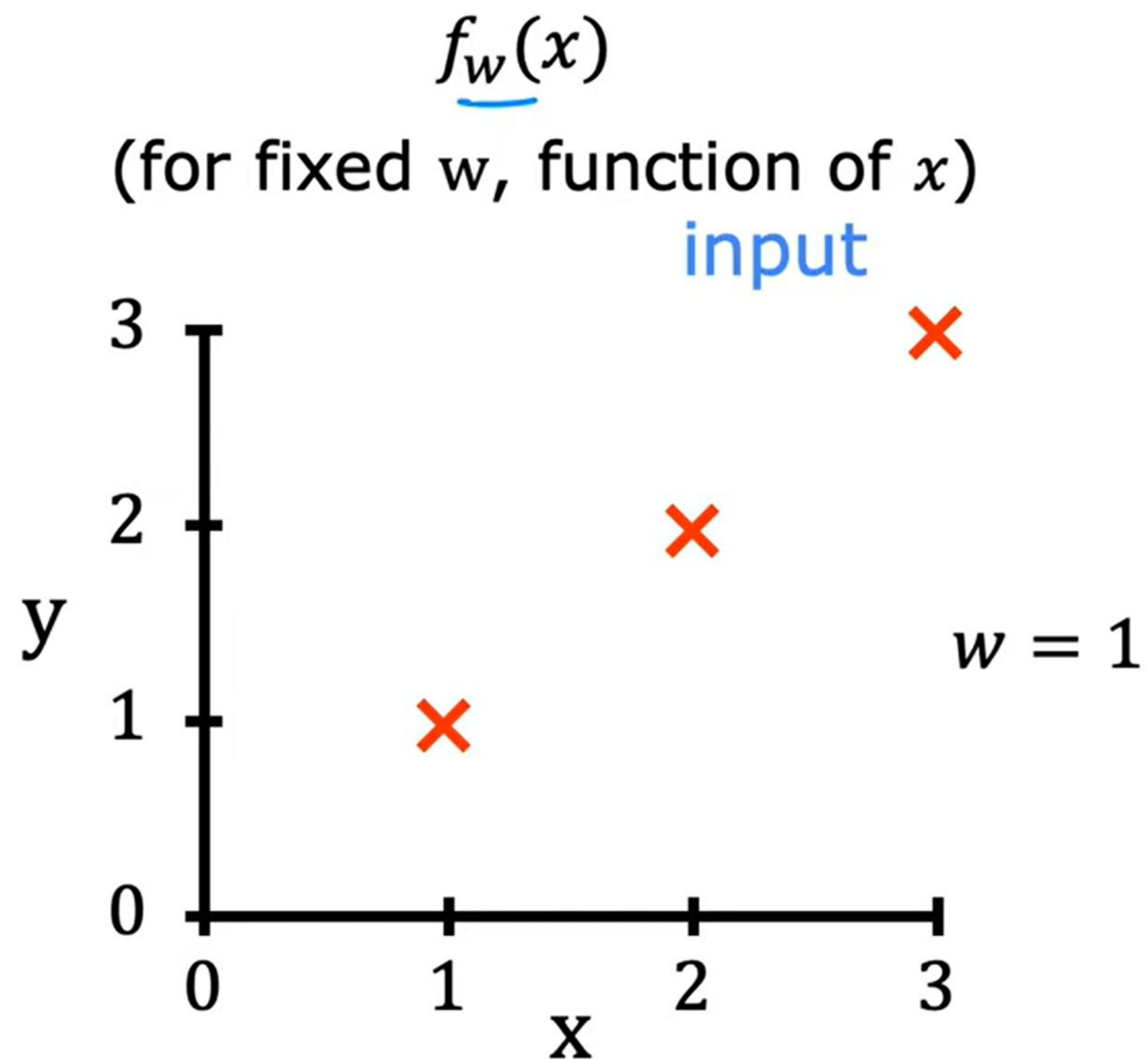


$$\underline{J(w)} = \frac{1}{2m} \sum_{i=1}^m (\underline{f_w(x^{(i)})} - y^{(i)})^2$$

$w x^{(i)}$

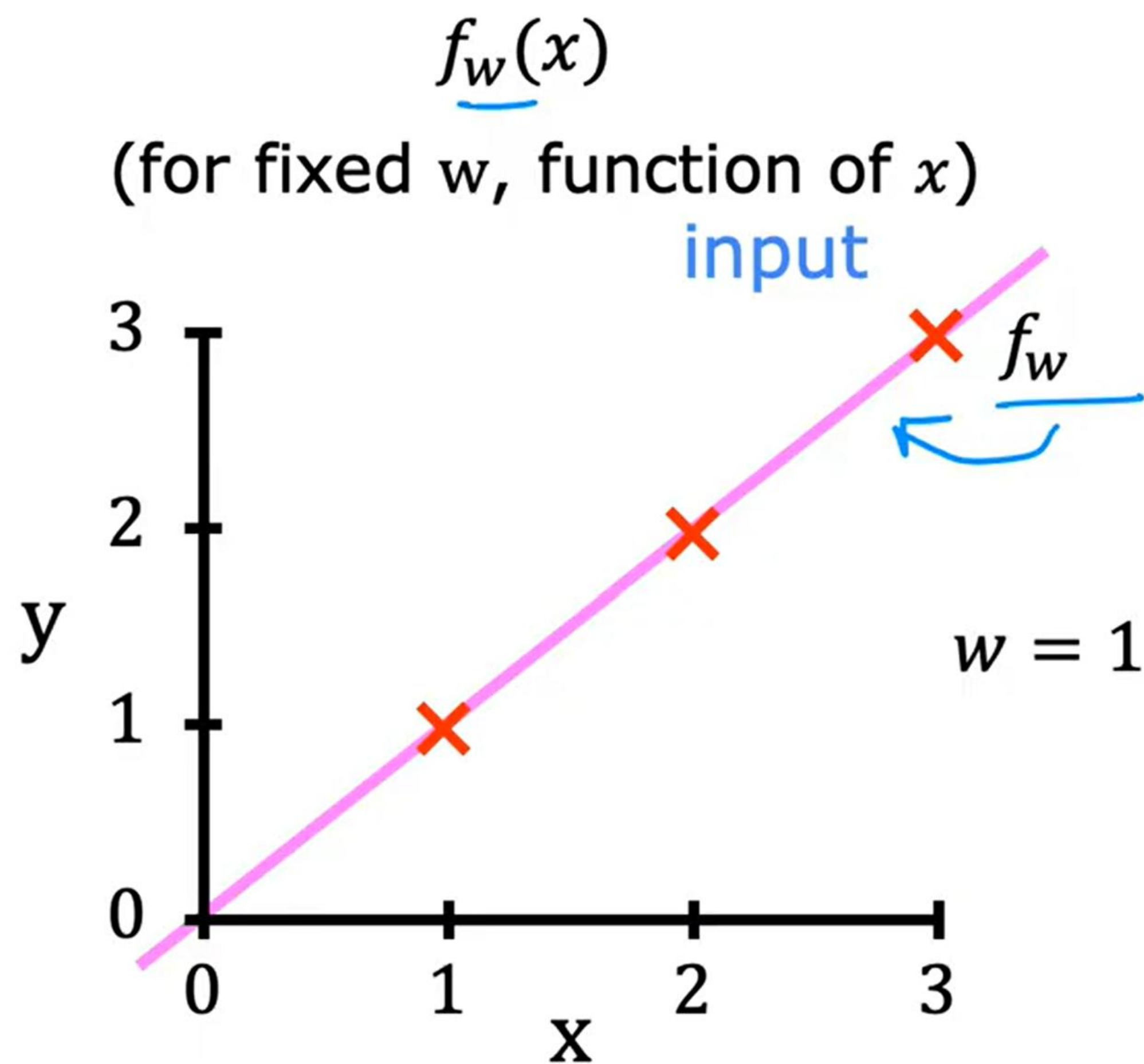
$$\underset{\underline{w}}{\text{minimize}} \underline{J(w)}$$

Linear Regression (Cost function Intuition)



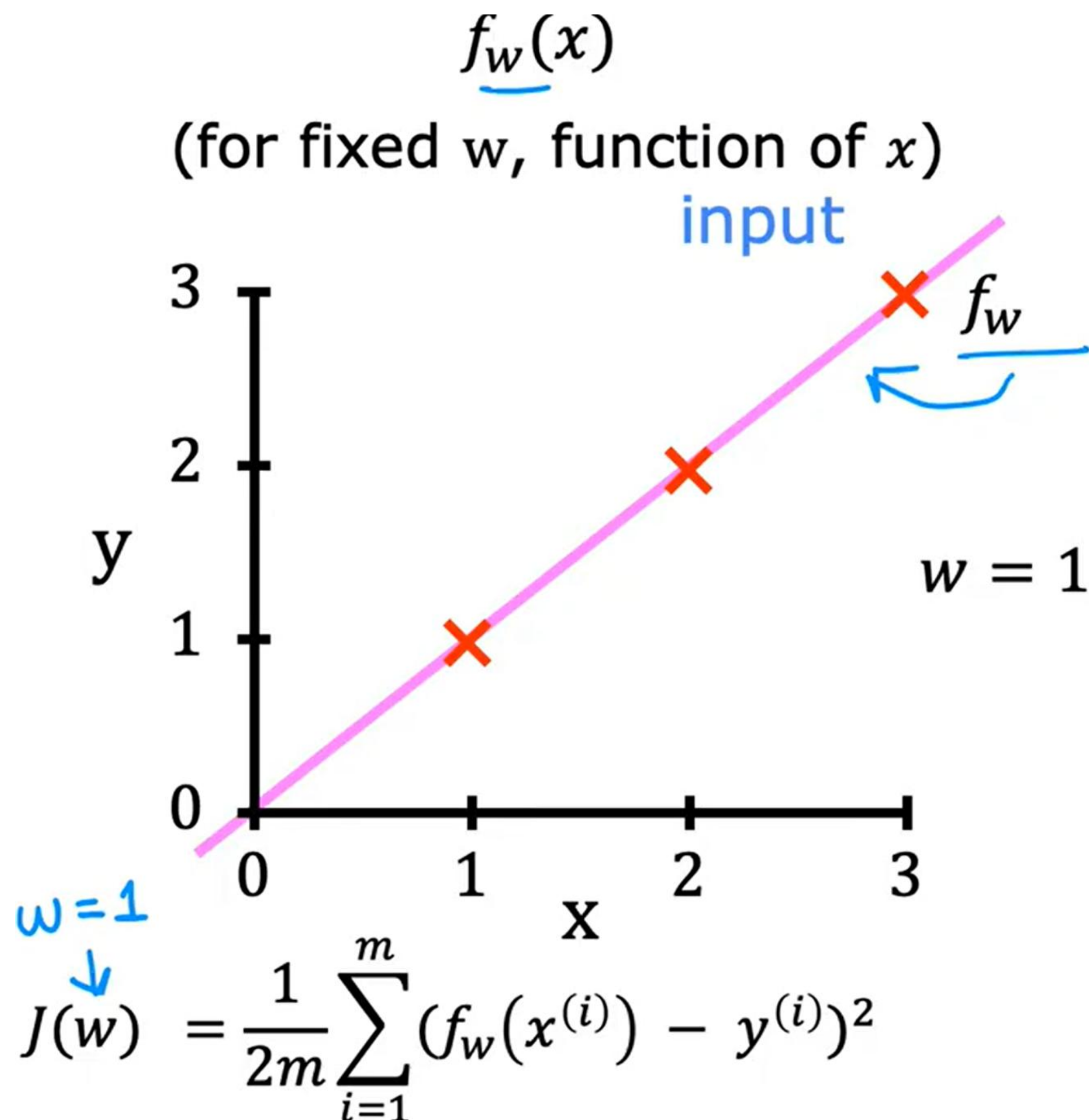
$J(w)$
(function of w)
parameter

Linear Regression (Cost function Intuition)



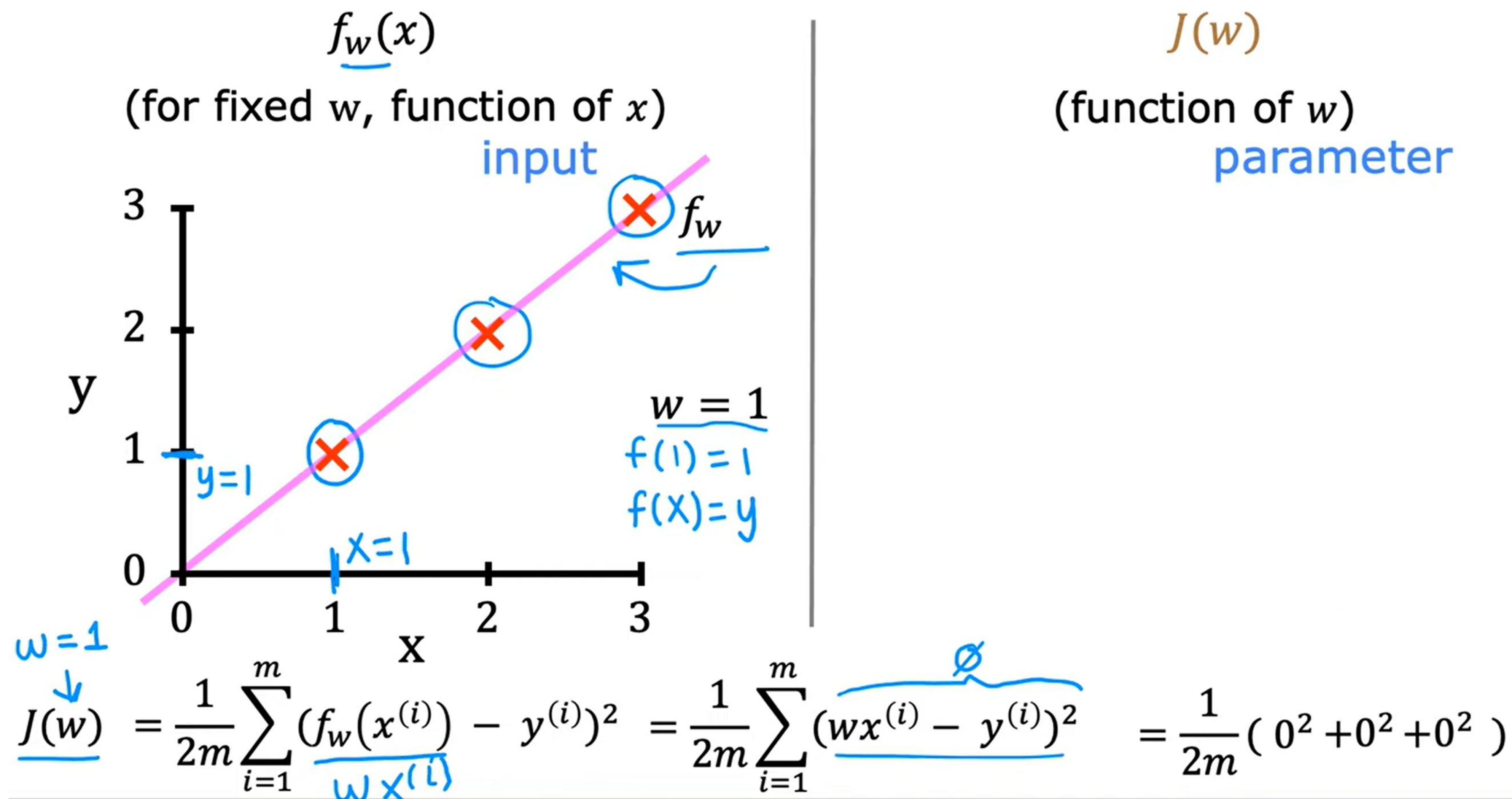
$J(w)$
(function of w)
parameter

Linear Regression (Cost function Intuition)

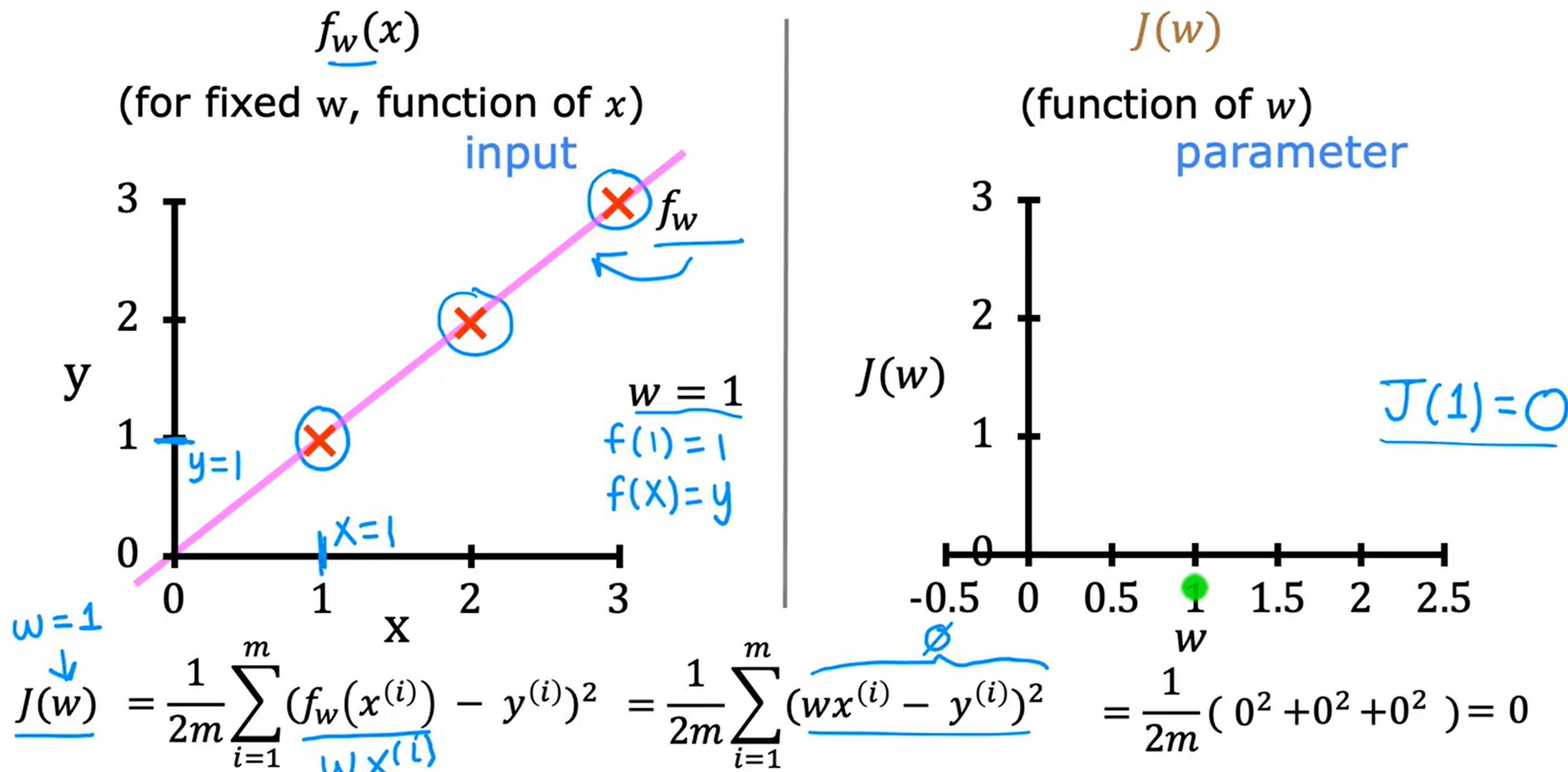


$J(w)$
(function of w)
parameter

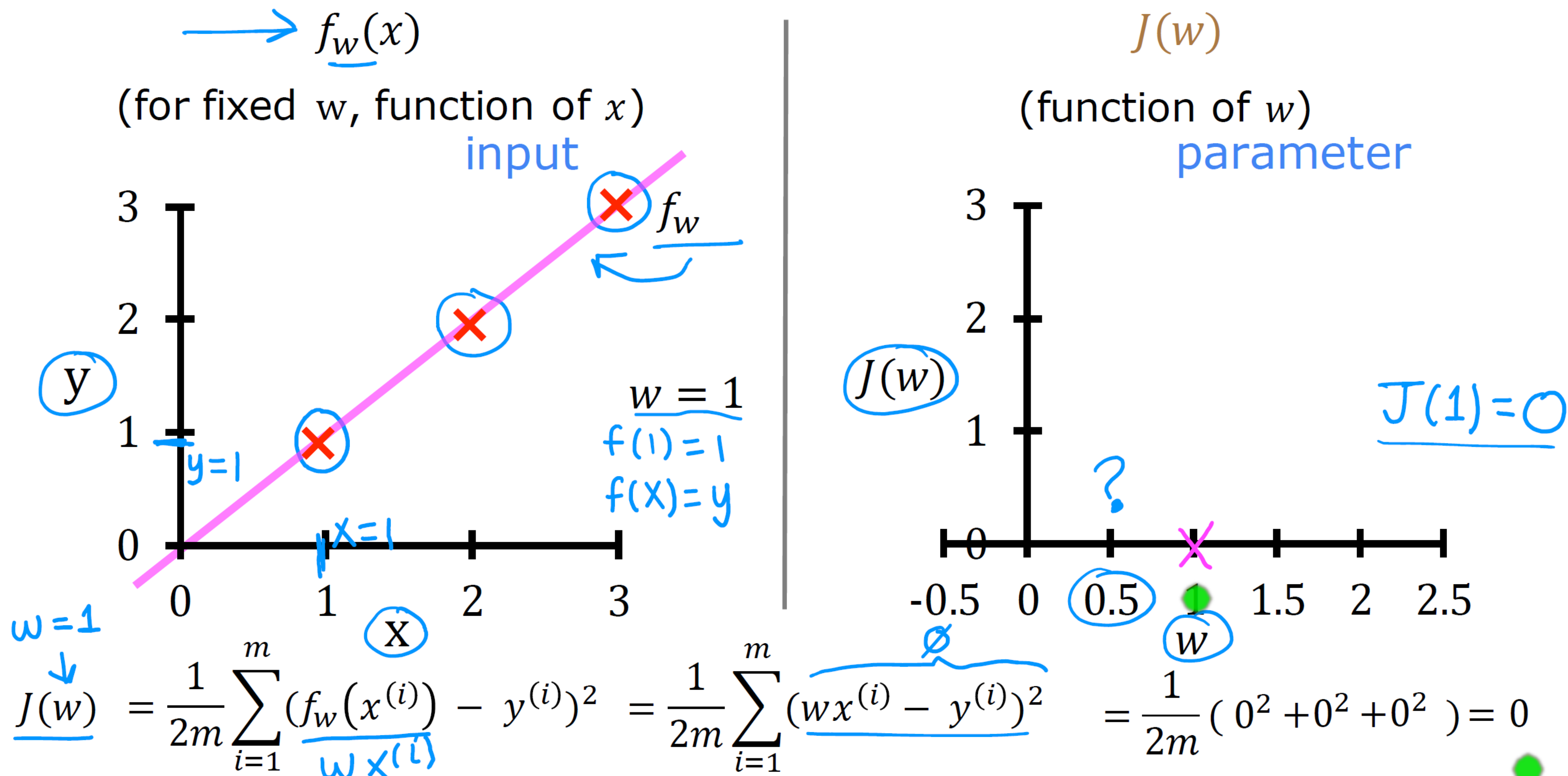
Linear Regression (Cost function Intuition)



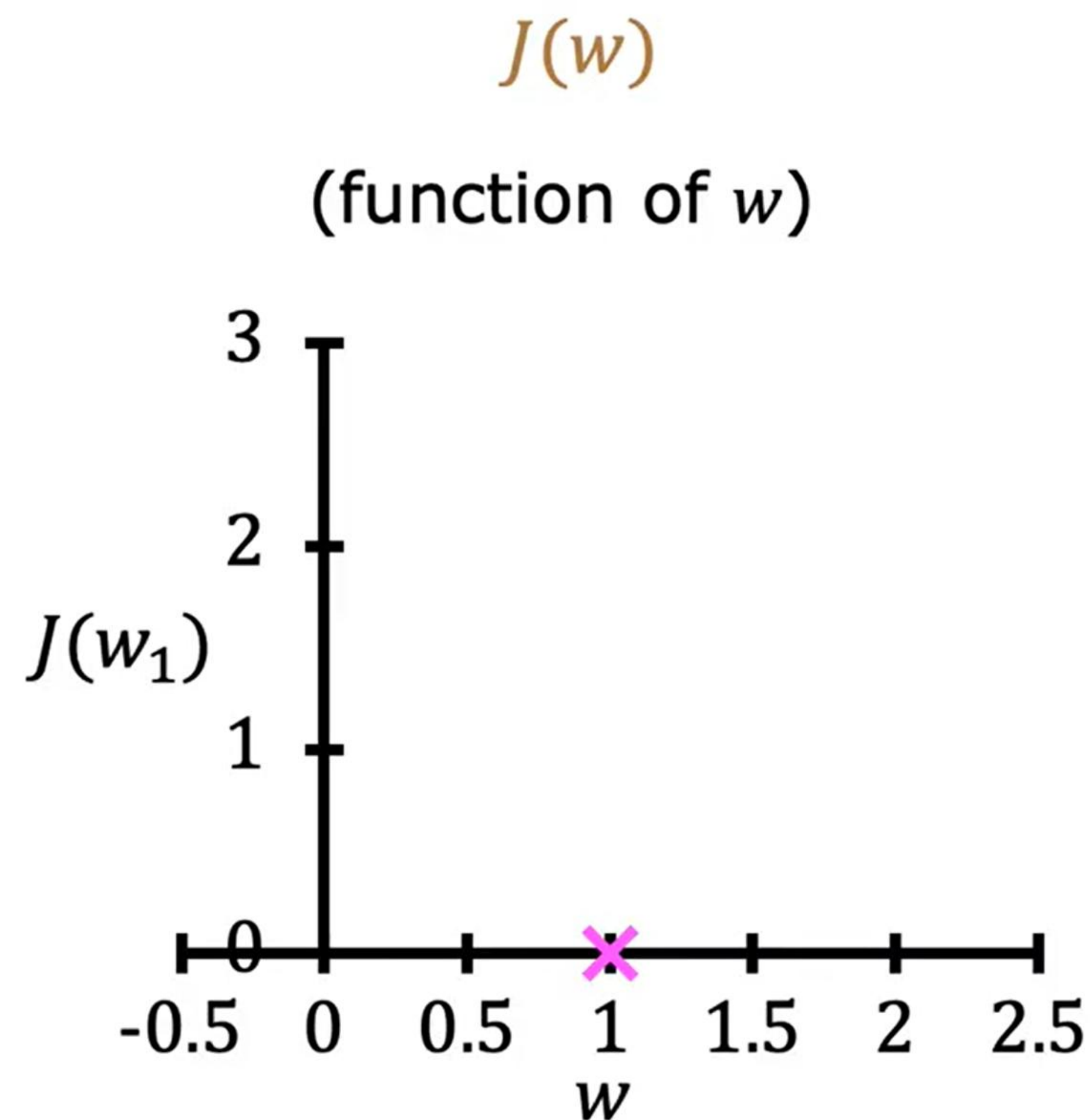
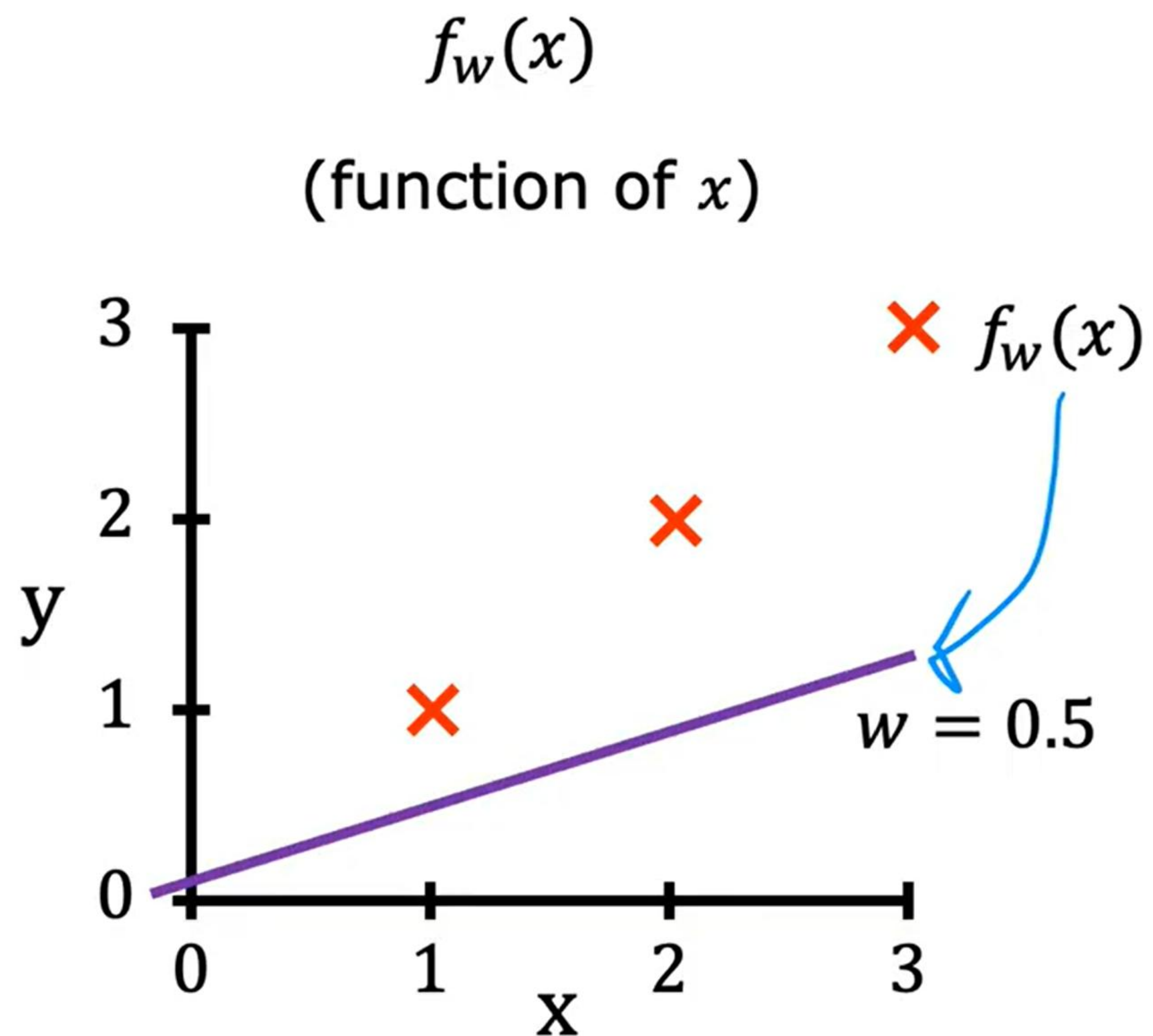
Linear Regression (Cost function Intuition)



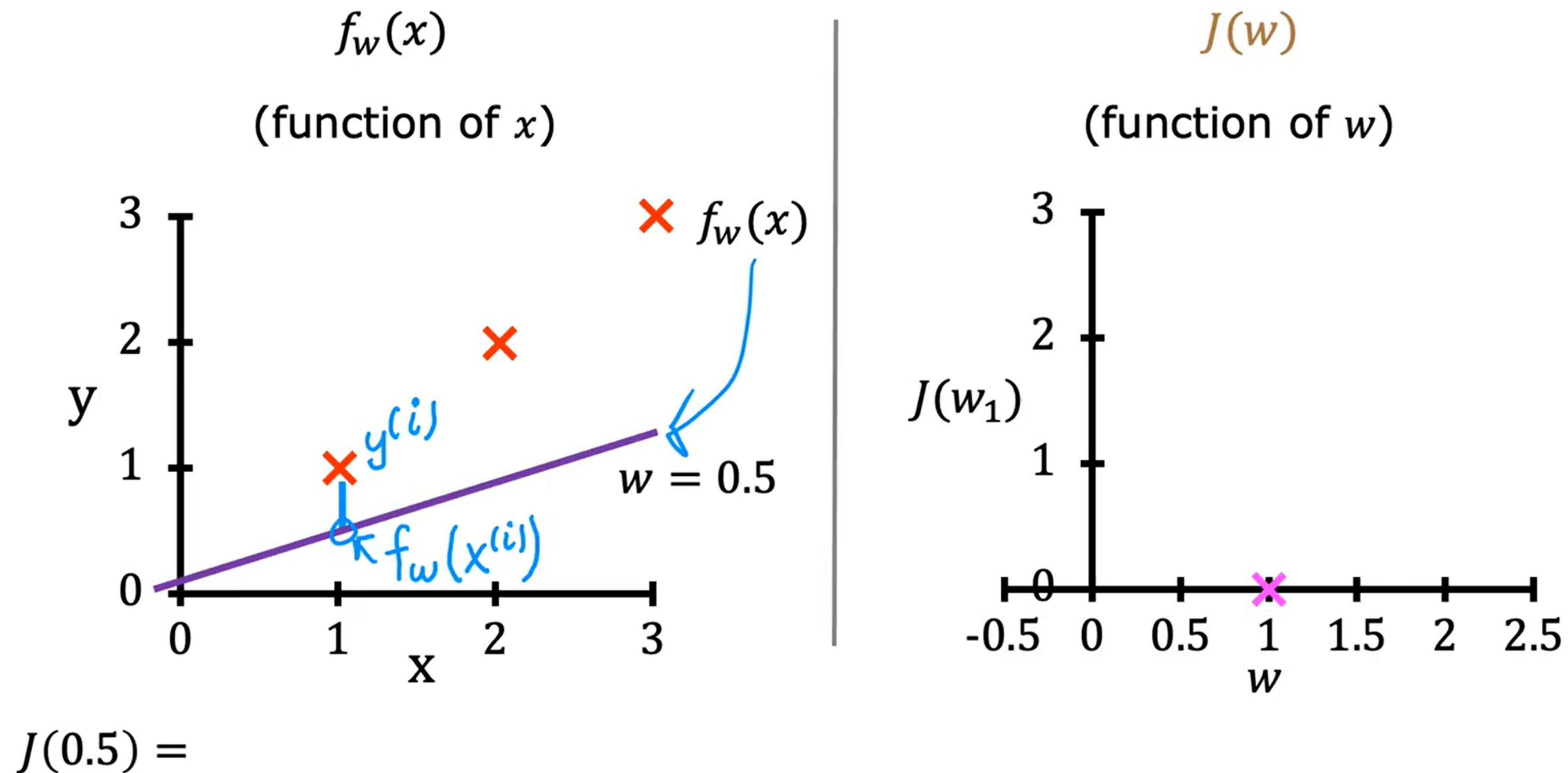
Linear Regression (Cost function Intuition)



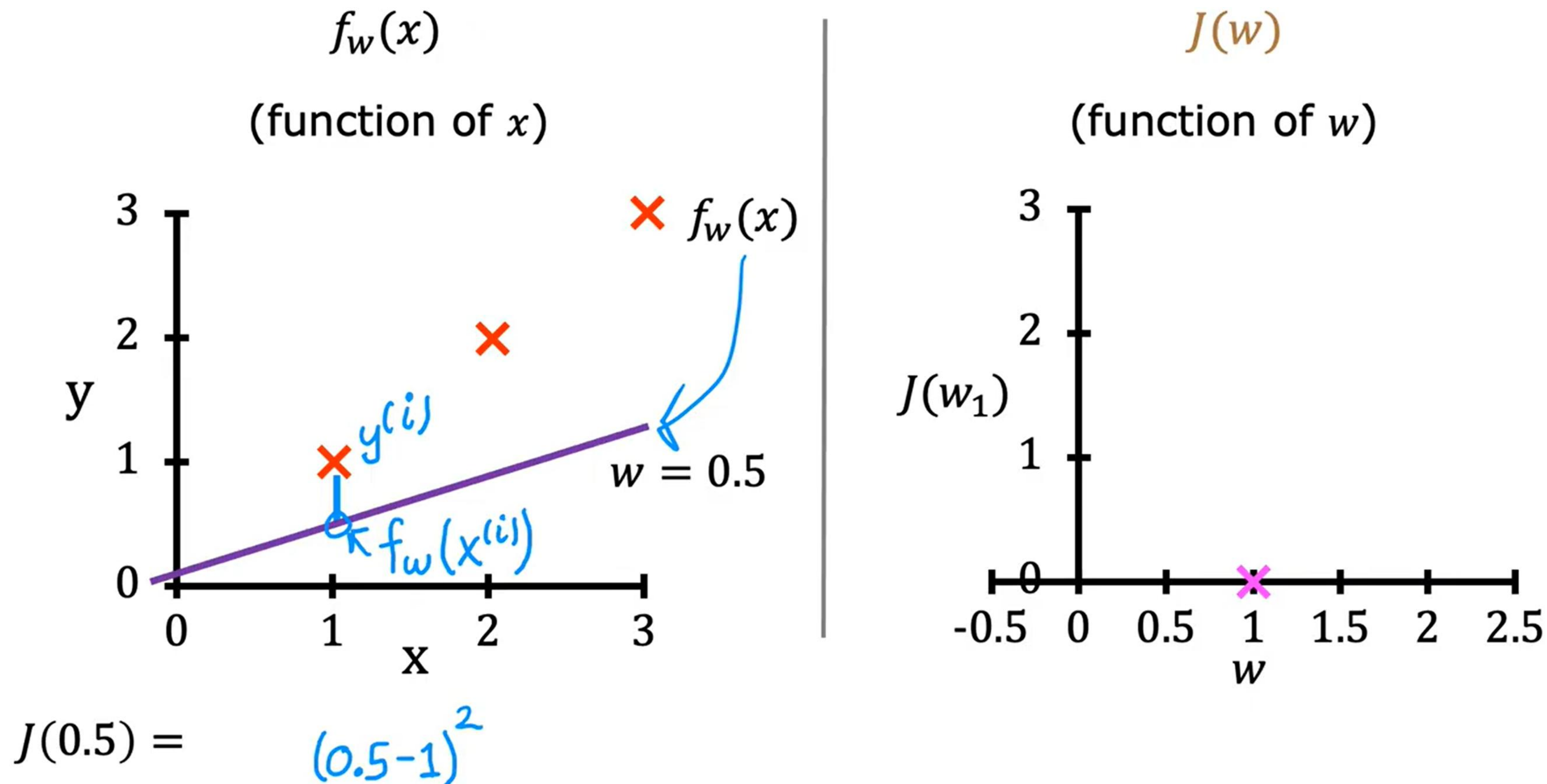
Linear Regression (Cost function Intuition)



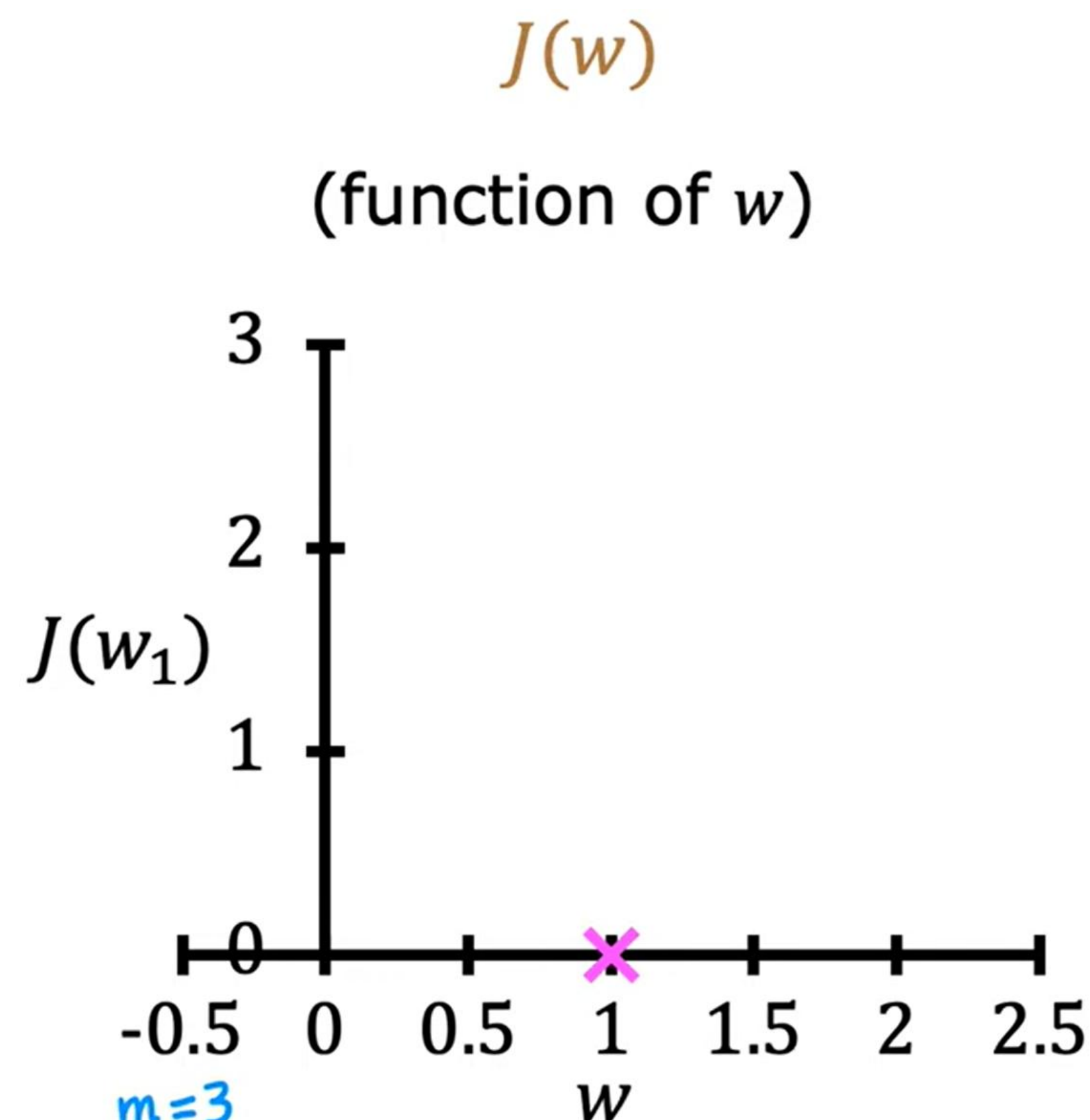
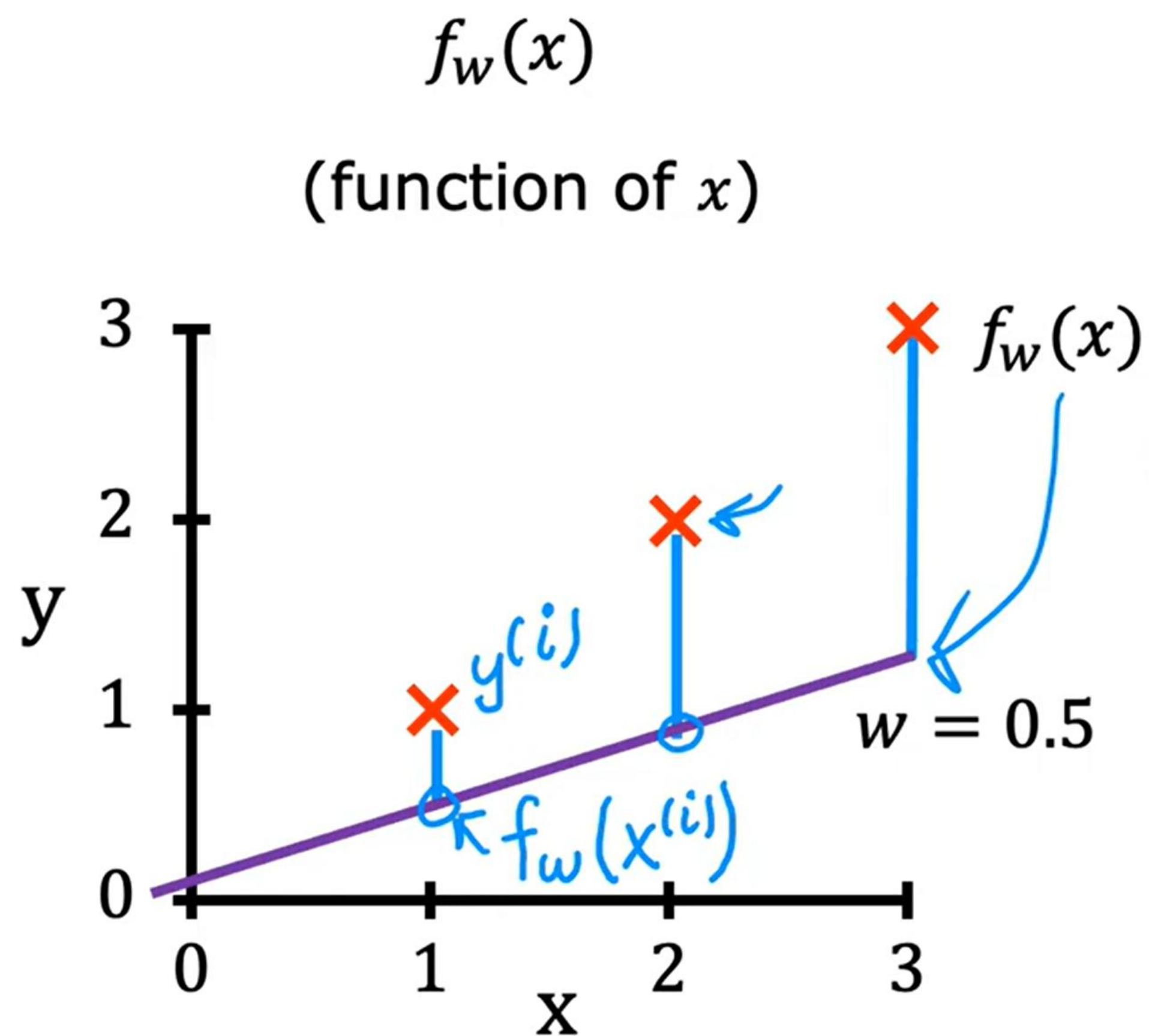
Linear Regression (Cost function Intuition)



Linear Regression (Cost function Intuition)



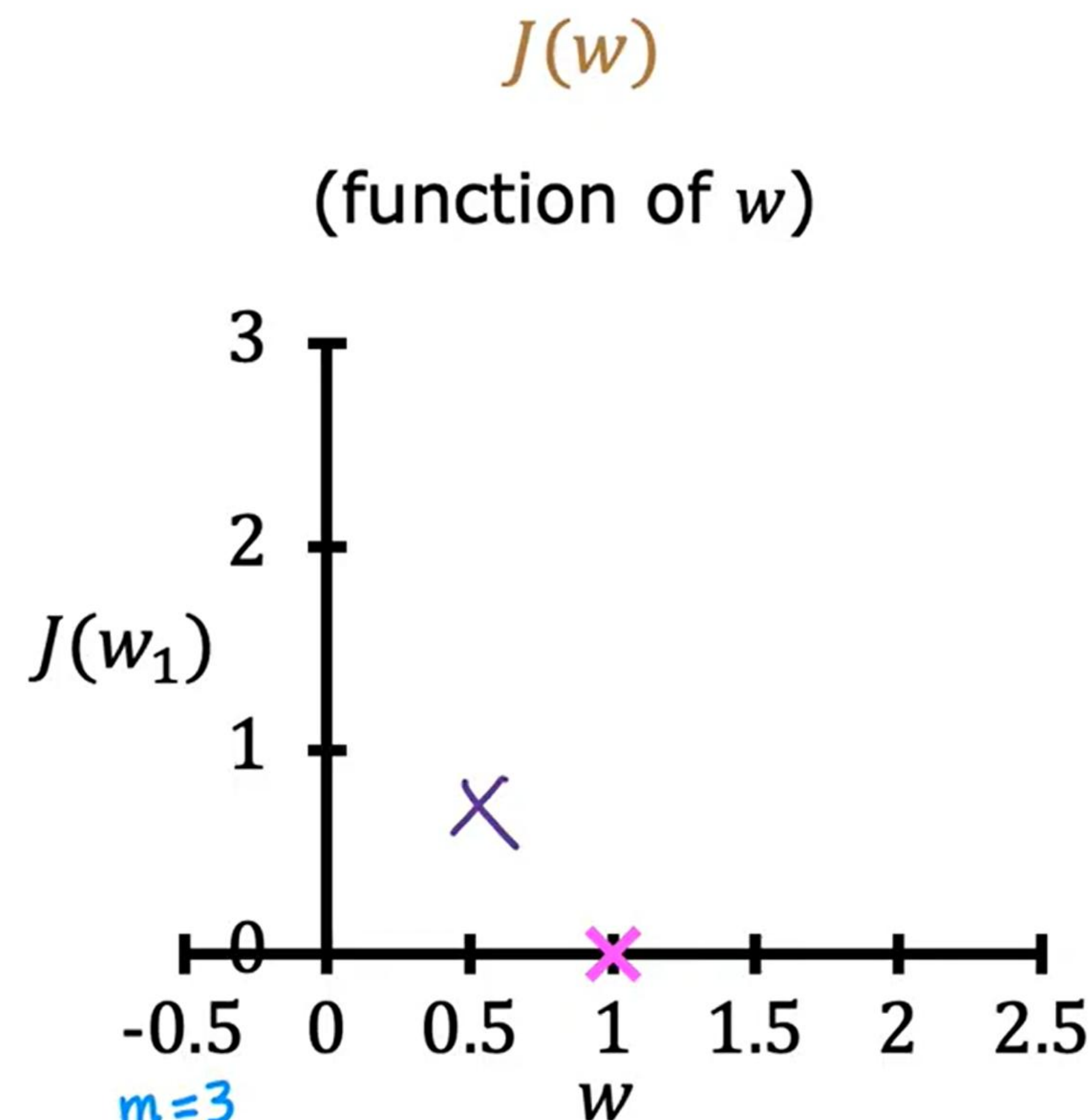
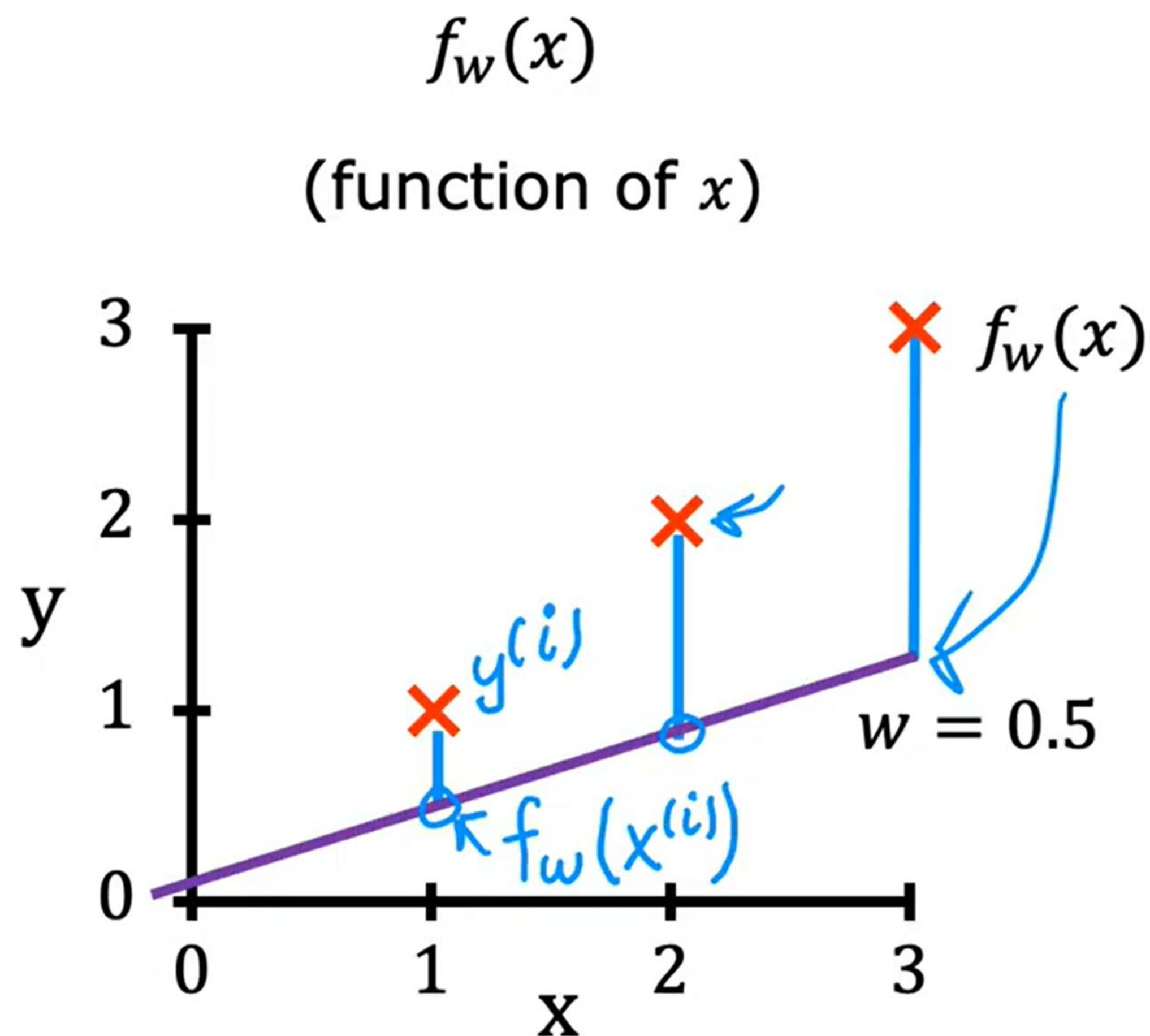
Linear Regression (Cost function Intuition)



$$J(0.5) = \frac{1}{2m} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2] = \frac{1}{2 \times 3} [3.5] = \frac{3.5}{6} \approx 0.58$$

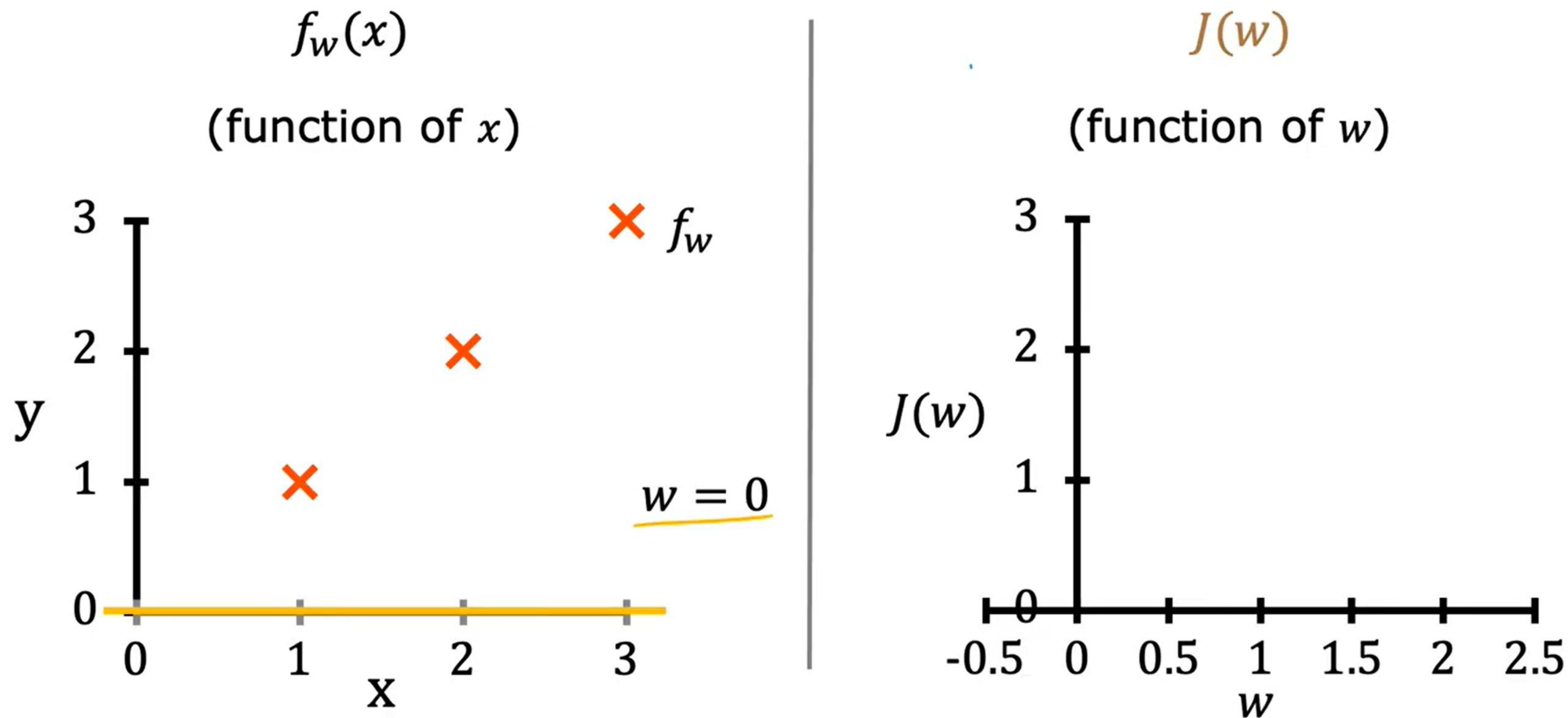
$m=3$

Linear Regression (Cost function Intuition)

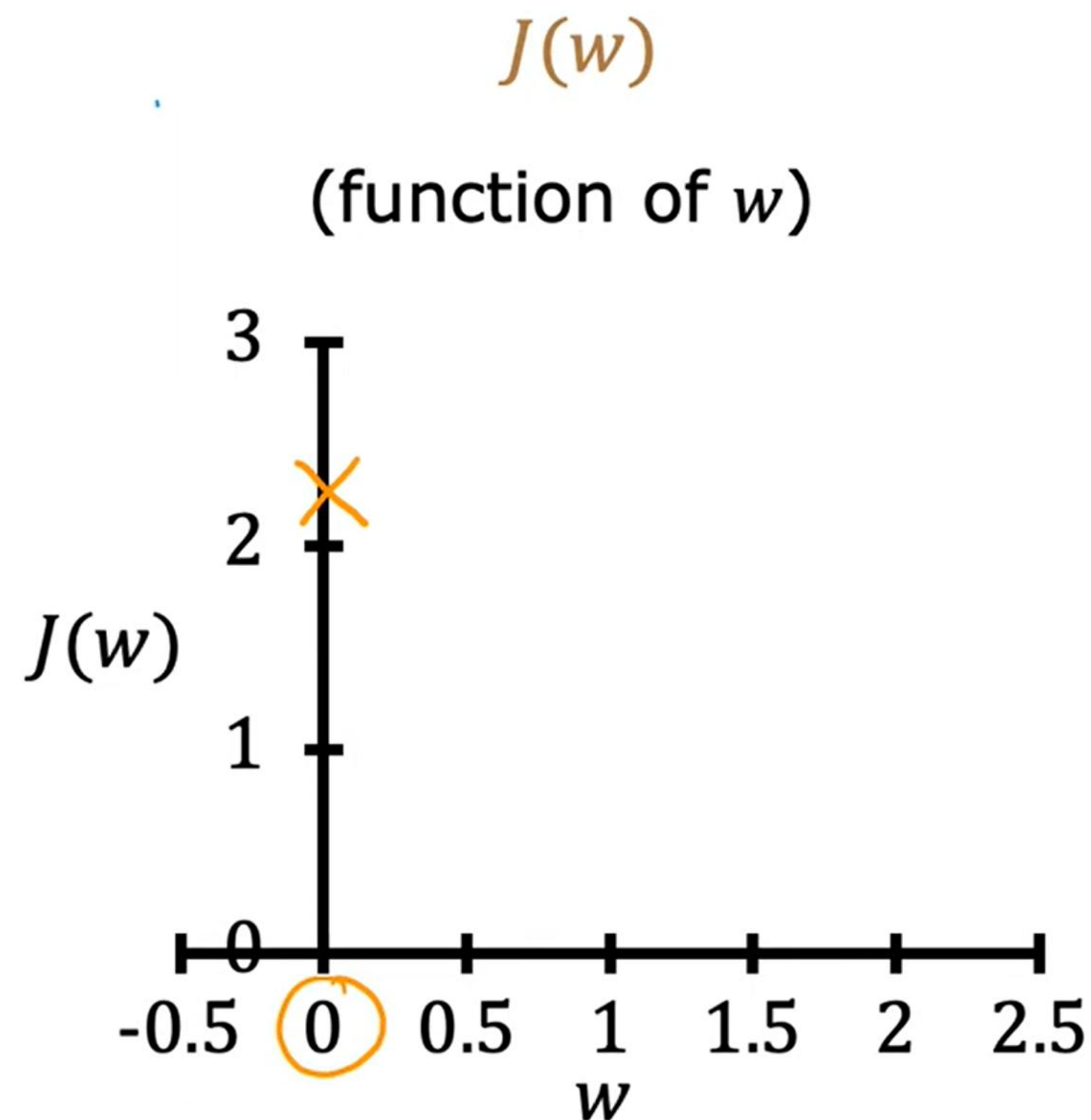
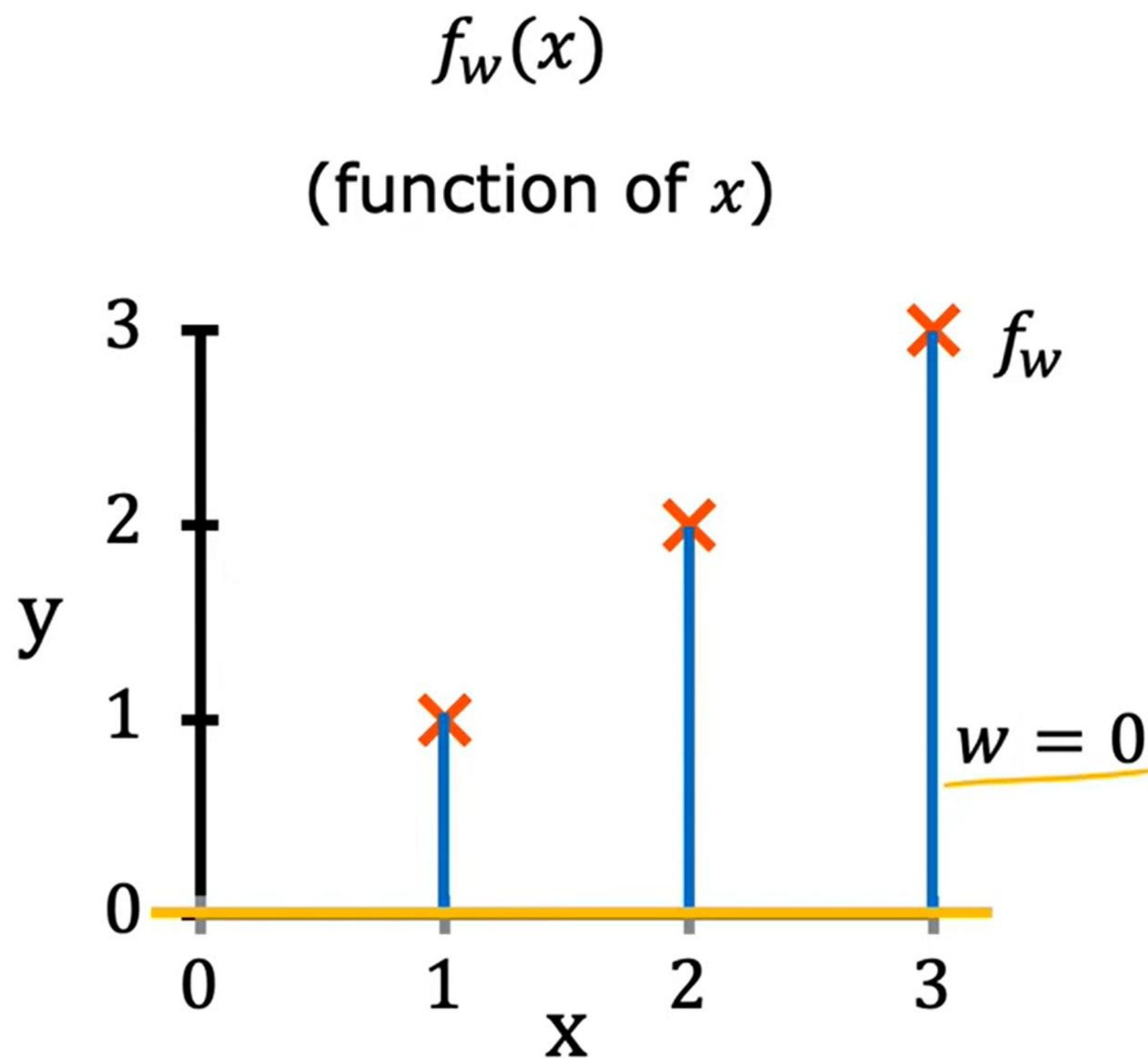


$$J(0.5) = \frac{1}{2m} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2] = \frac{1}{2 \times 3} [3.5] = \frac{3.5}{6} \approx 0.58$$

Linear Regression (Cost function Intuition)

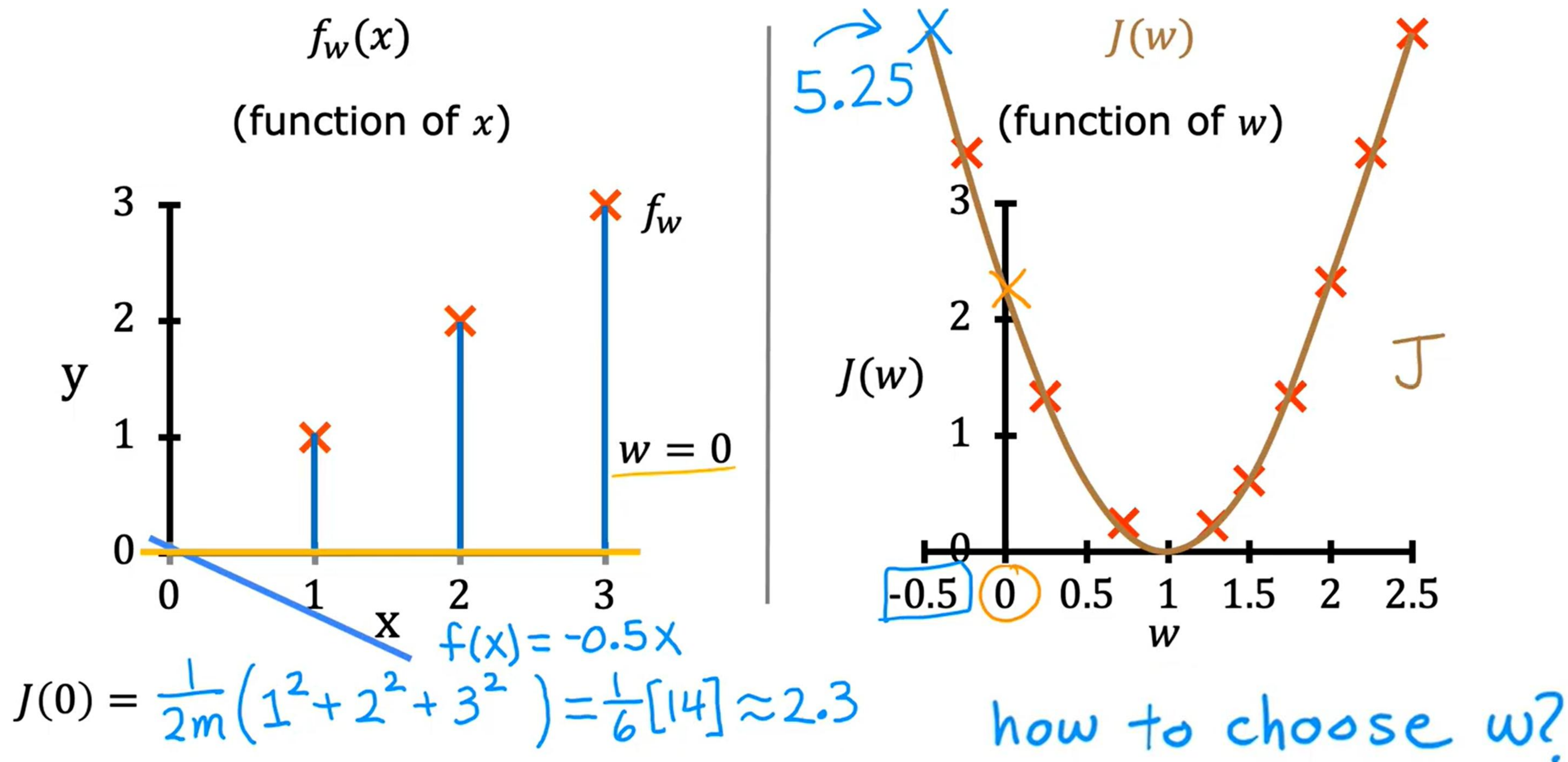


Linear Regression (Cost function Intuition)



$$J(0) = \frac{1}{2m} (1^2 + 2^2 + 3^2) = \frac{1}{6} [14] \approx 2.3$$

Linear Regression (Cost function Intuition)



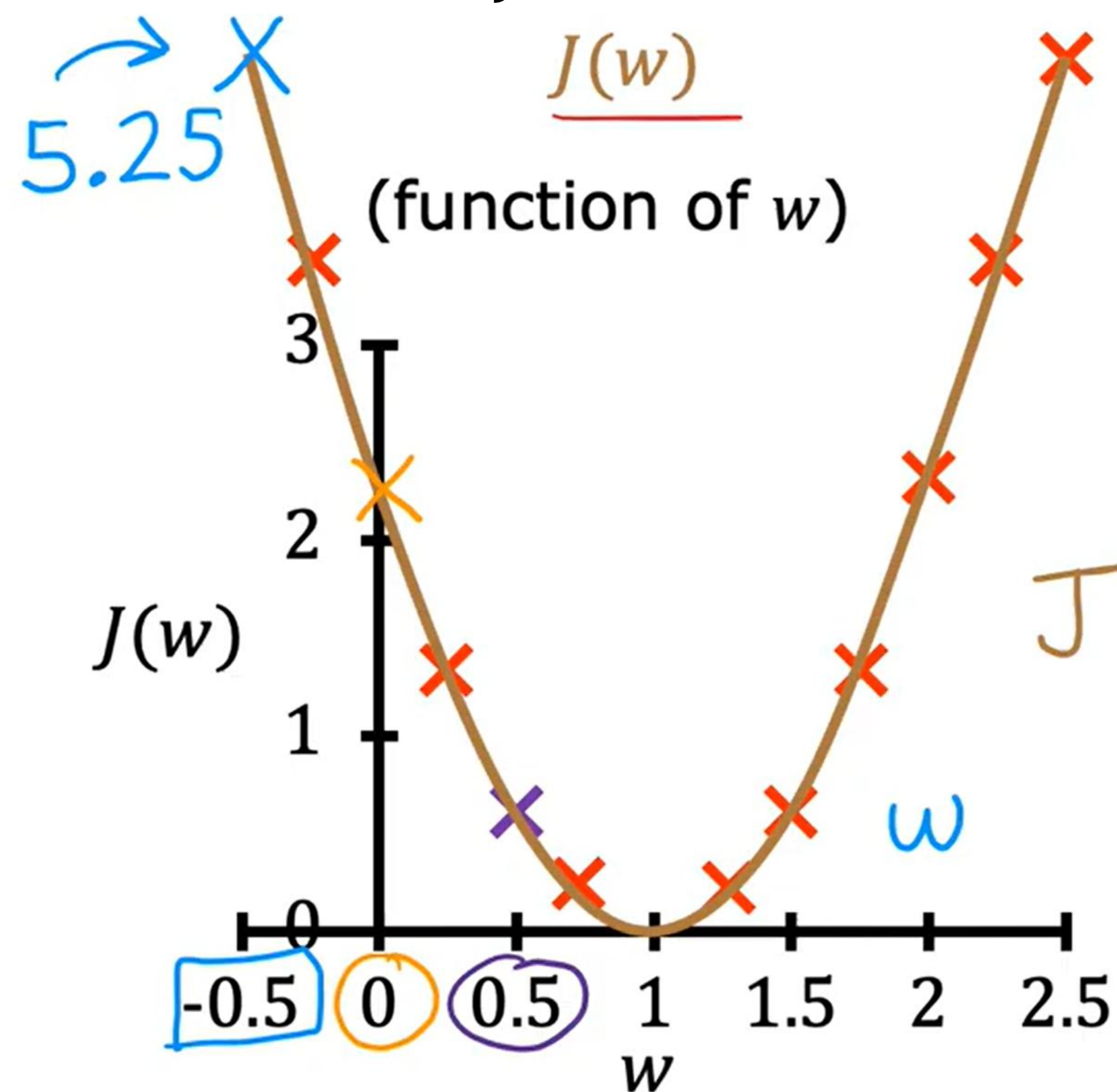
Linear Regression (Cost function Intuition)

goal of linear regression:

$$\underset{w}{\text{minimize}} J(w)$$

general case:

$$\underset{w,b}{\text{minimize}} J(w, b)$$



choose w to minimize $J(w)$

Linear Regression (Visualization)

Model

$$f_{w,b}(x) = wx + b$$

Parameters

w, b ~~before: $b=0$~~

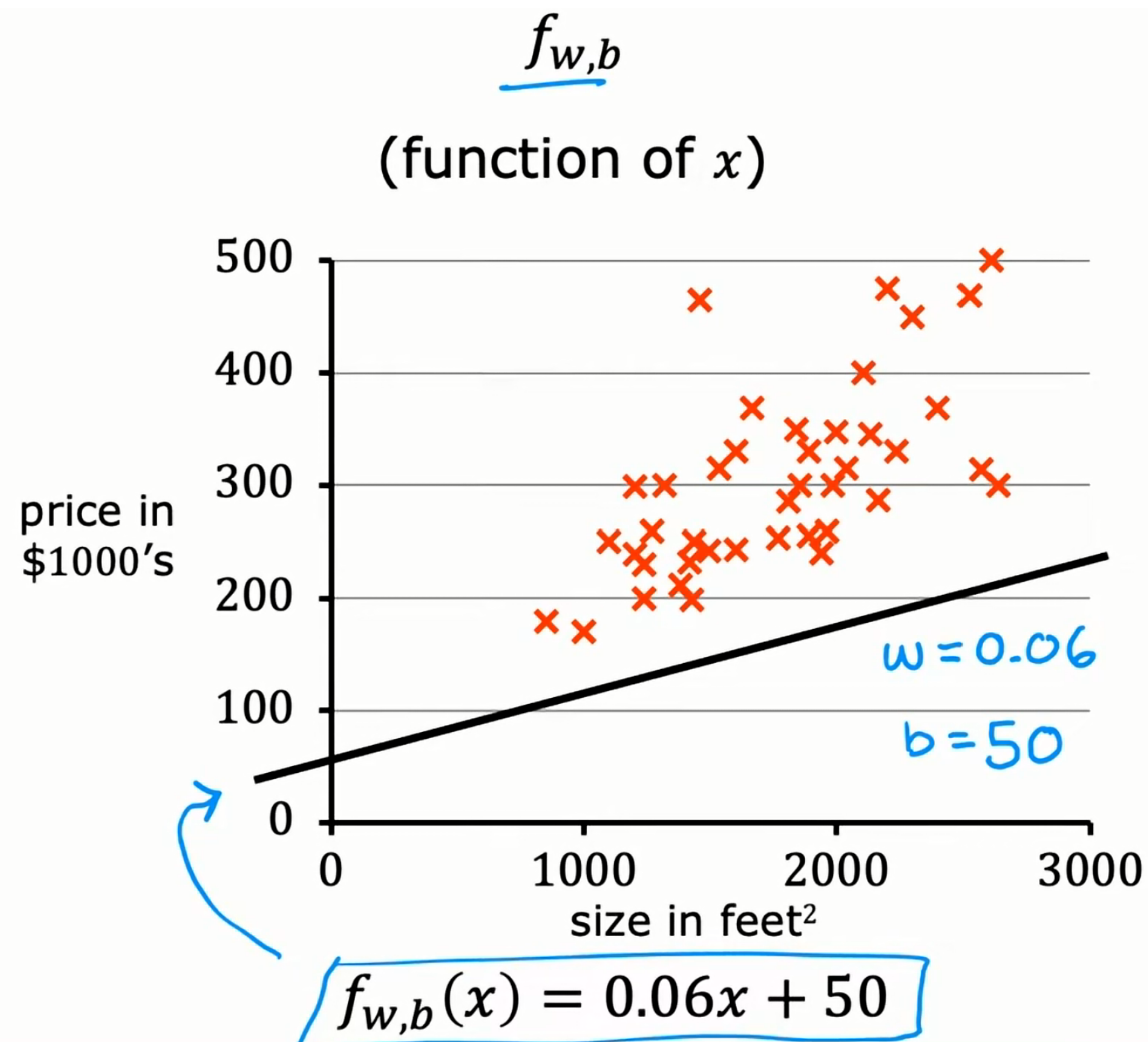
Cost Function

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Objective

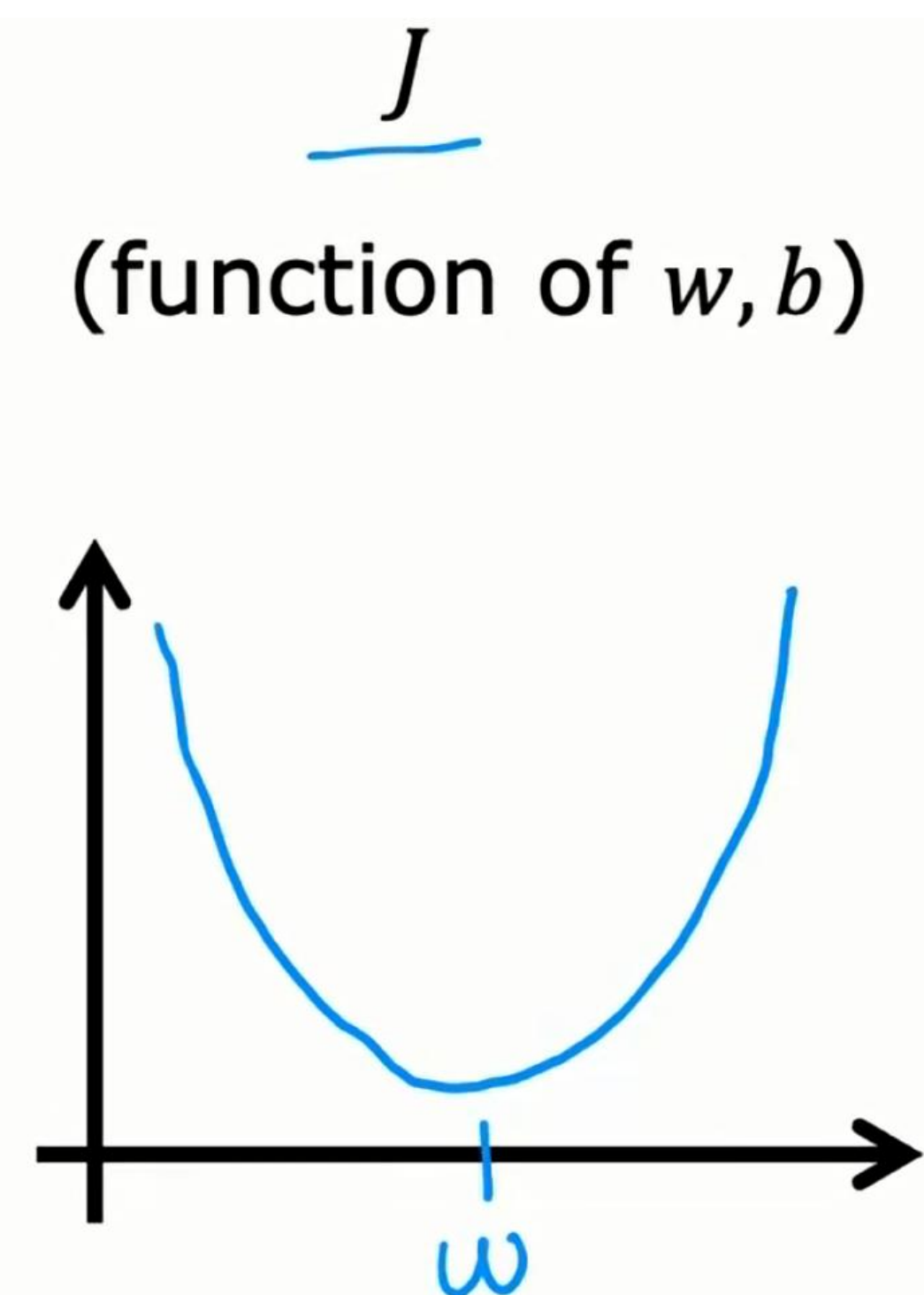
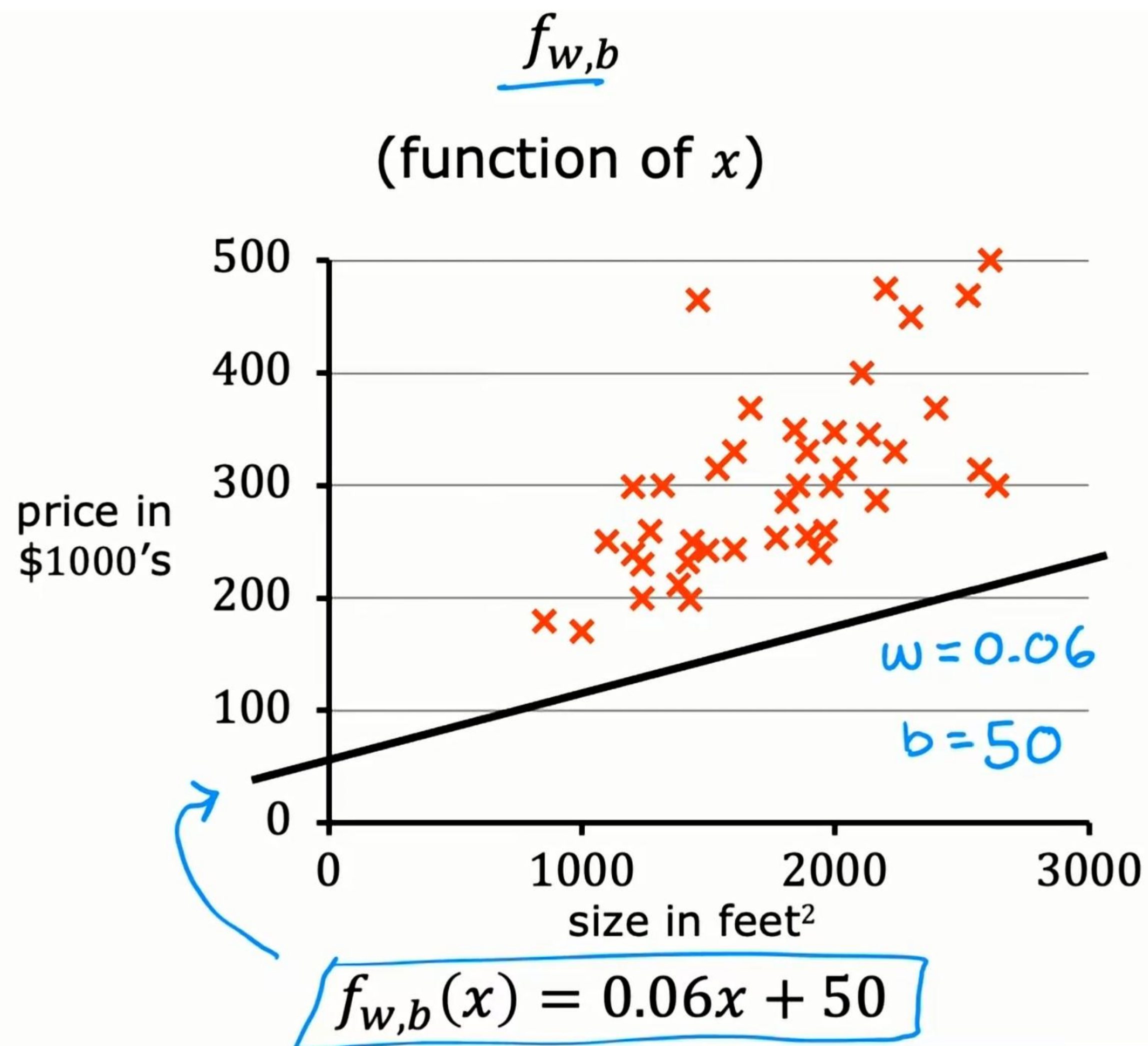
$$\underset{w,b}{\text{minimize}} J(w, b)$$

Linear Regression (Visualization)

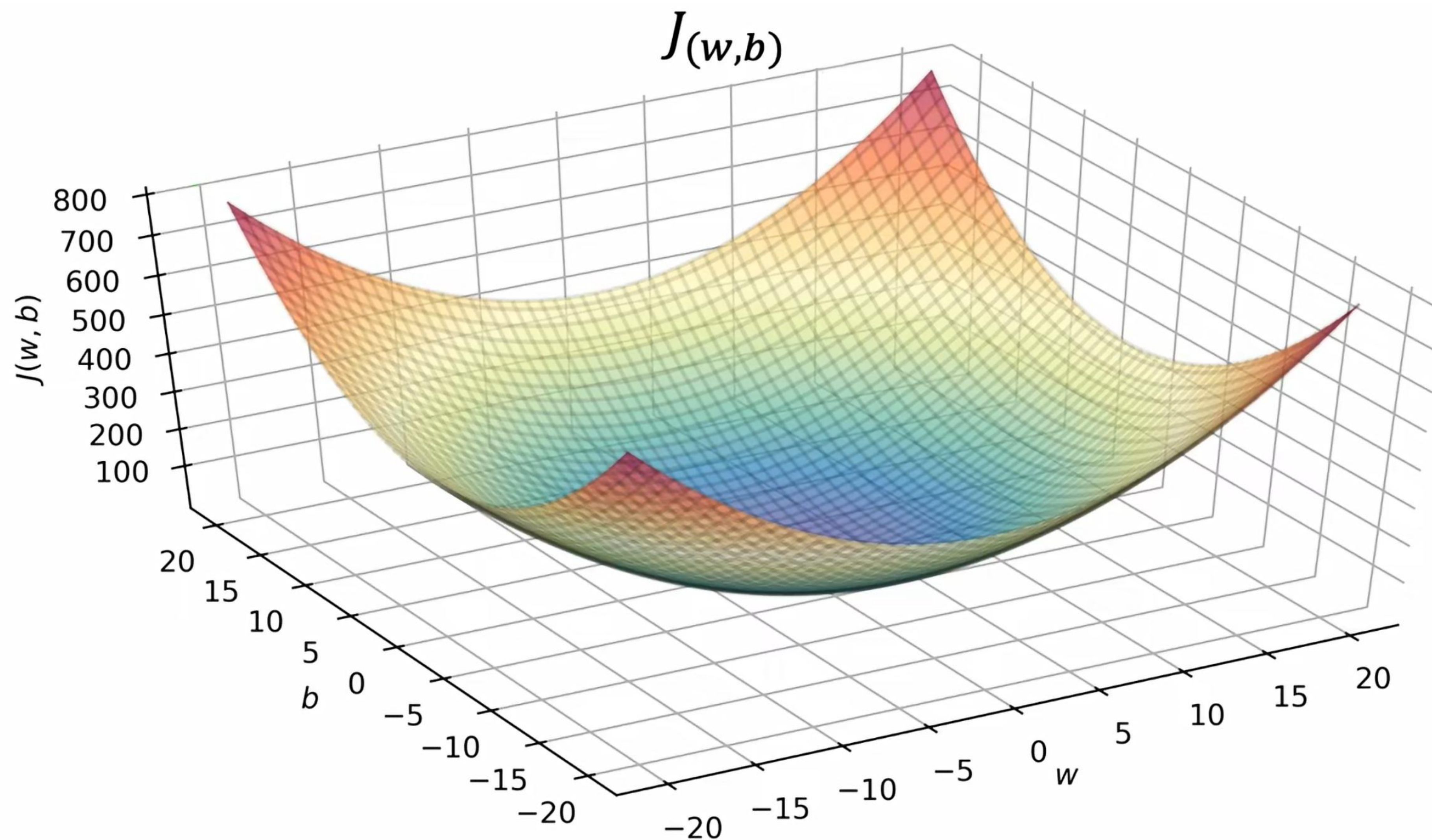


J
(function of w, b)

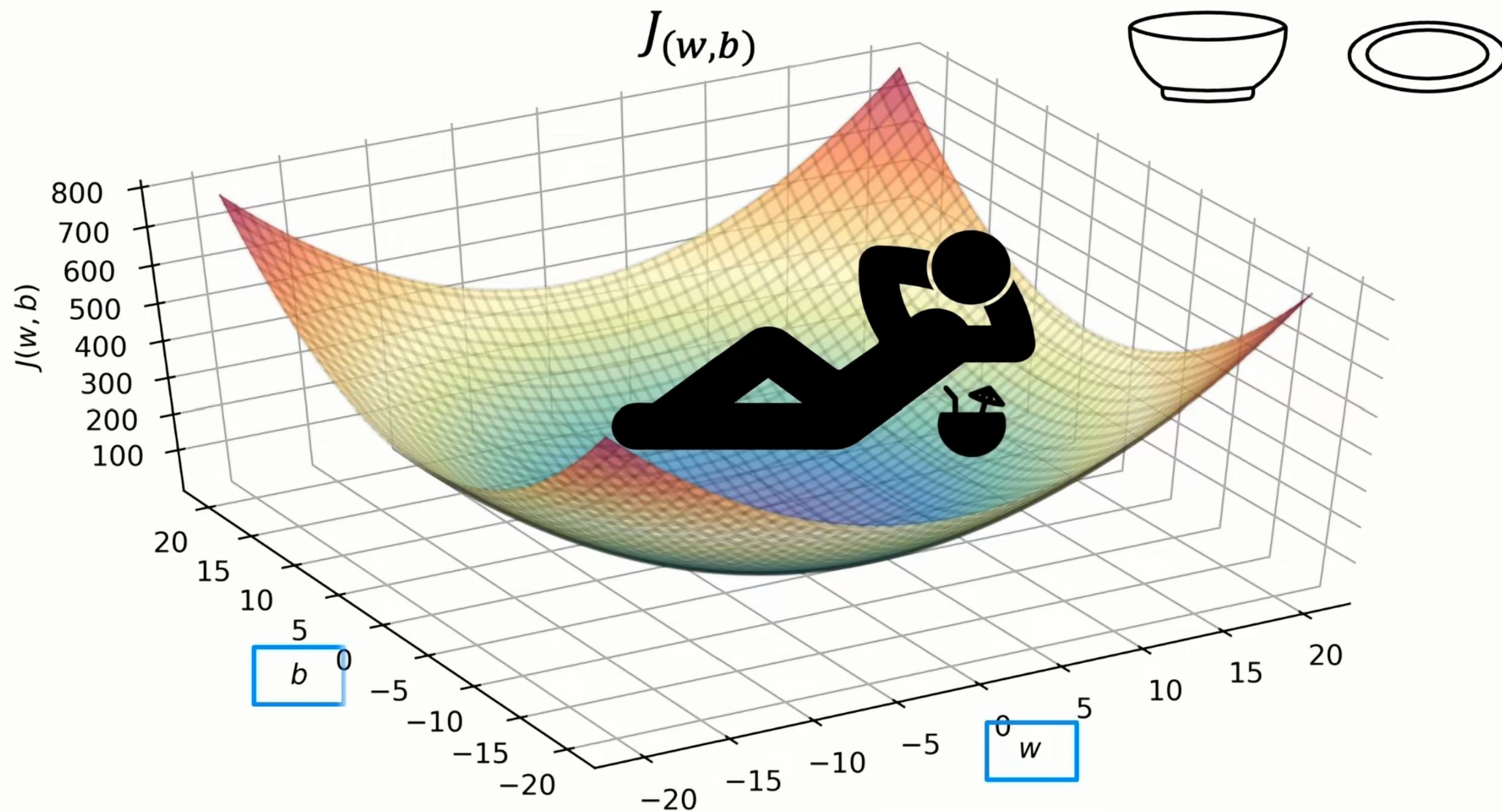
Linear Regression (Visualization)



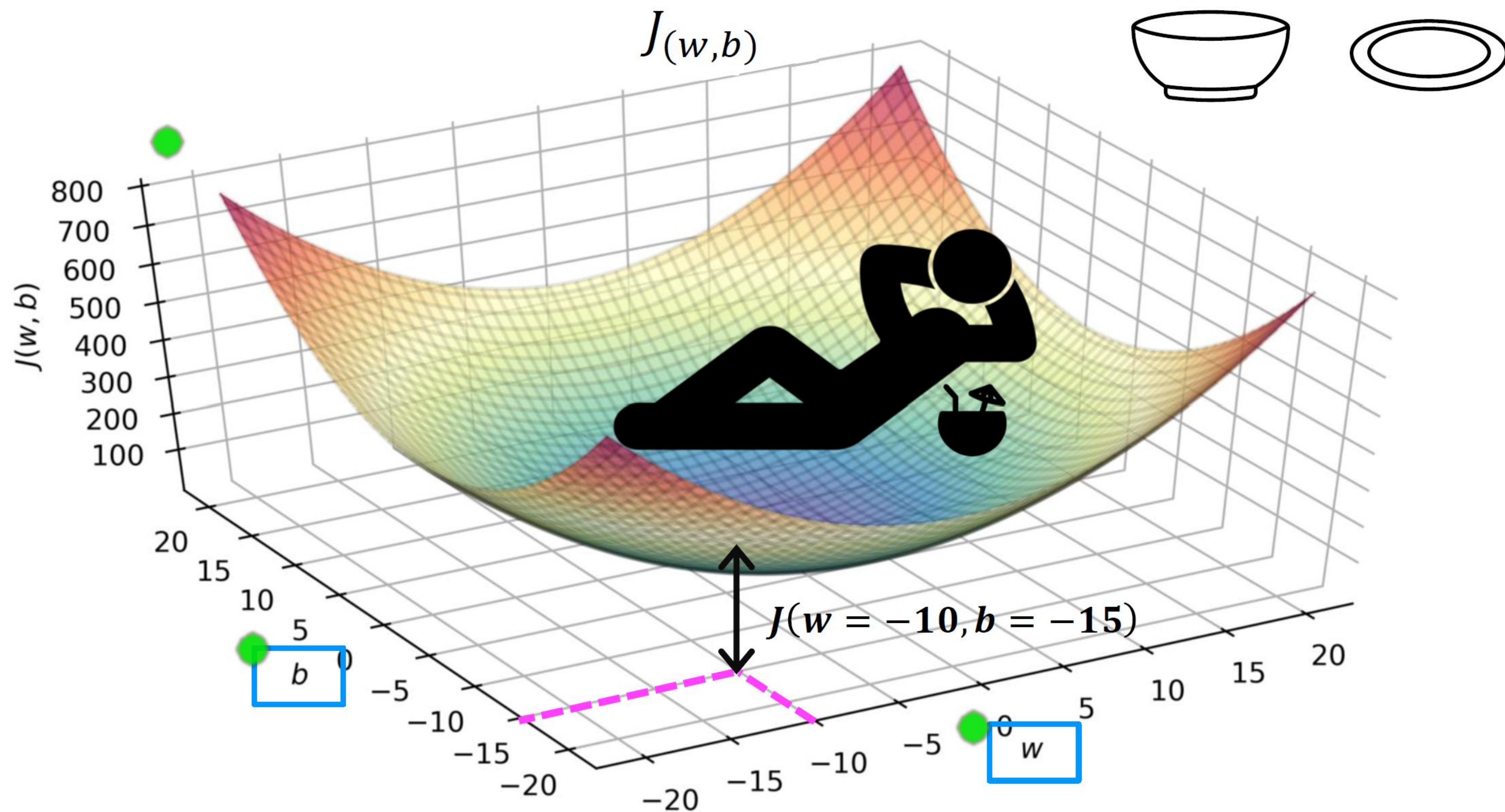
Linear Regression (Visualization)



Linear Regression (Visualization)

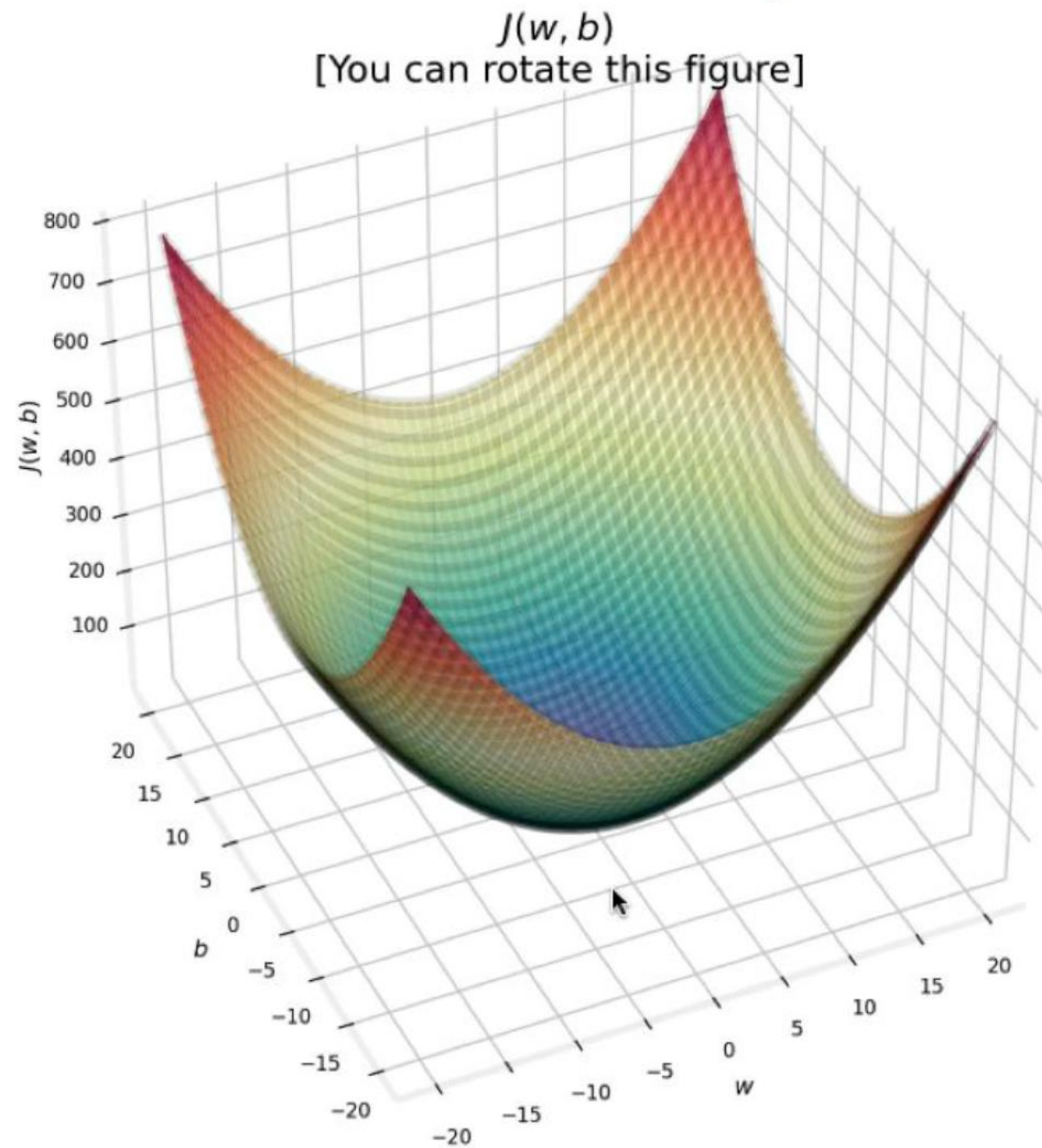


Linear Regression (Visualization)



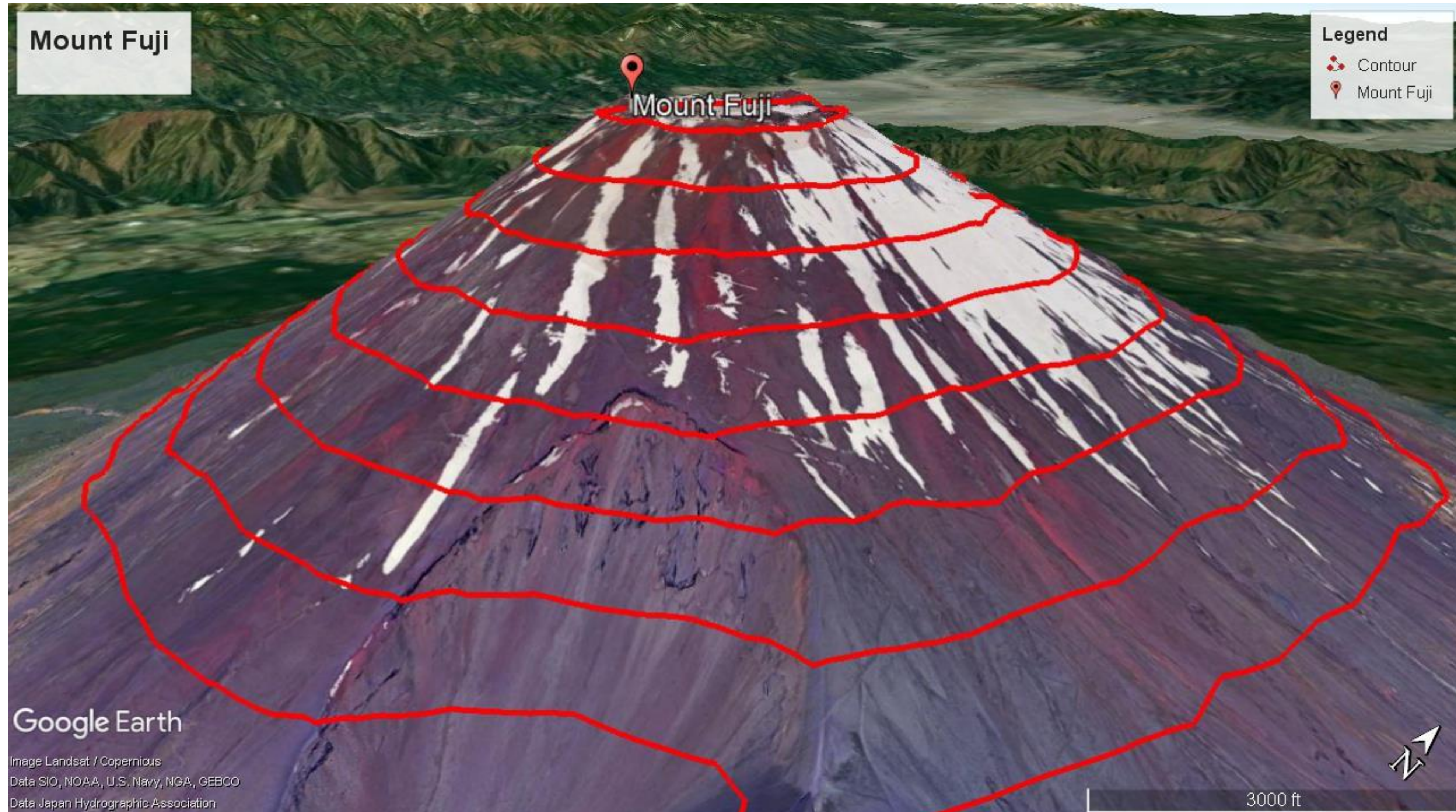
Linear Regression (Visualization)

3D surface plot

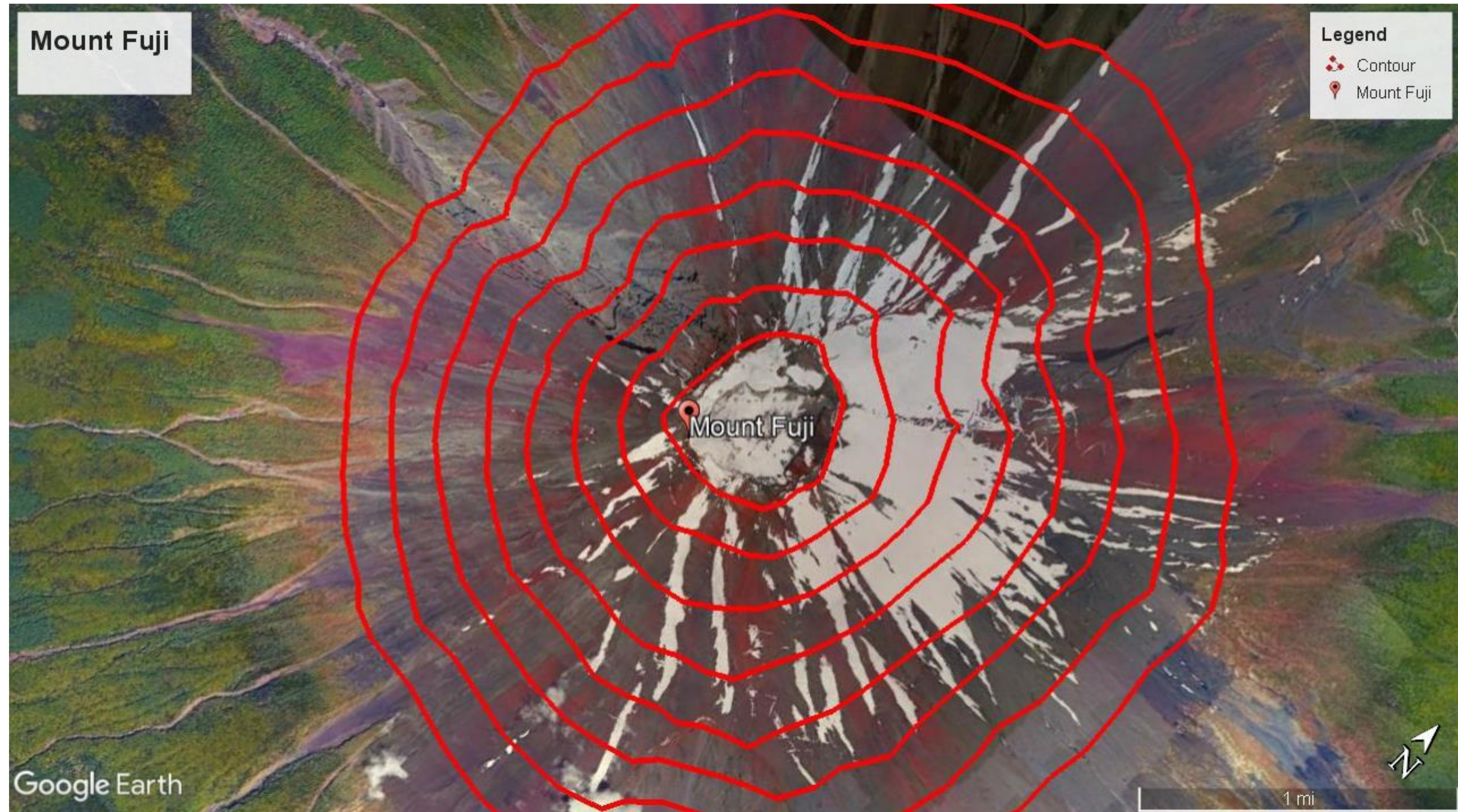


Alternative
contour plot

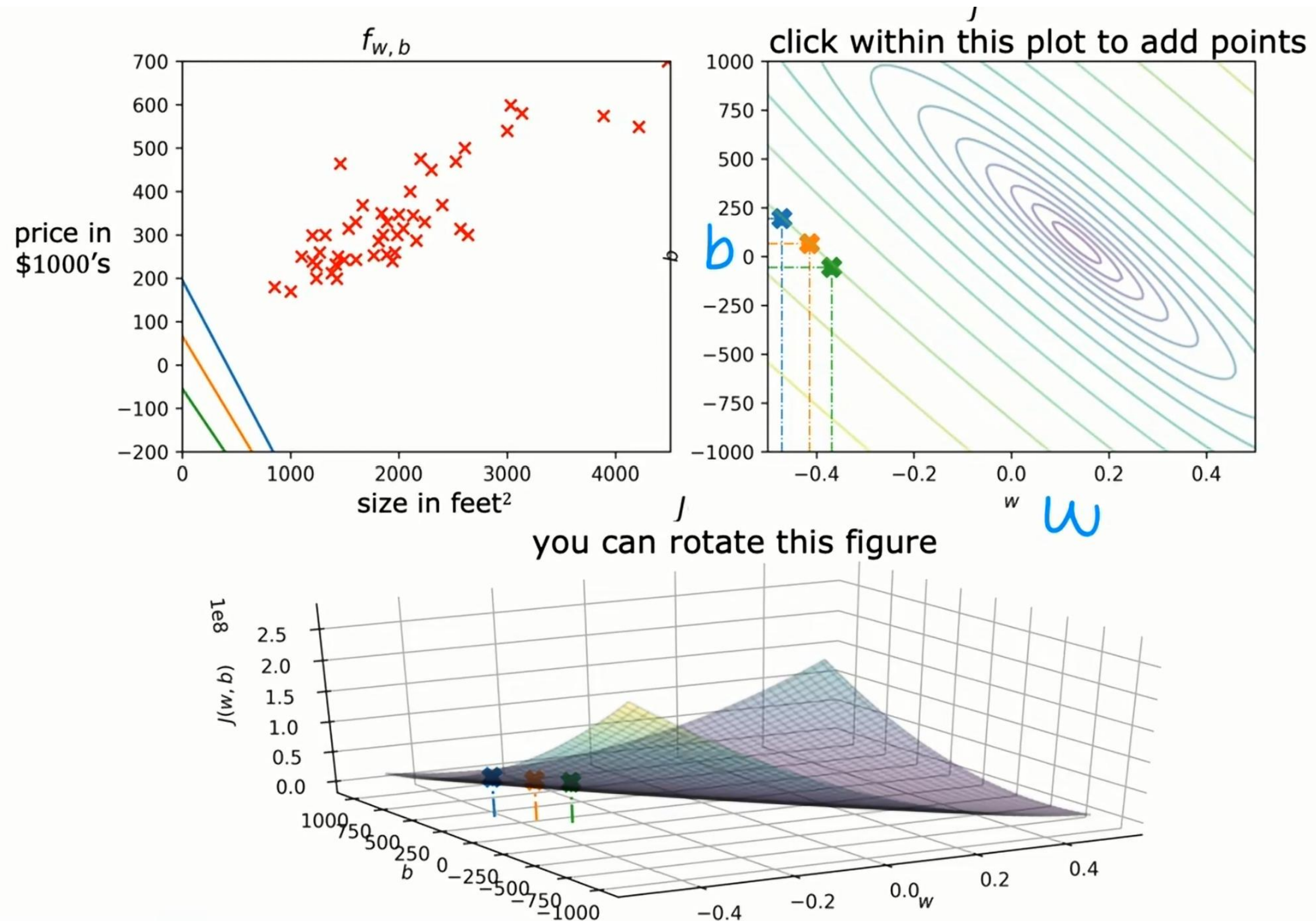
Linear Regression (Visualization)



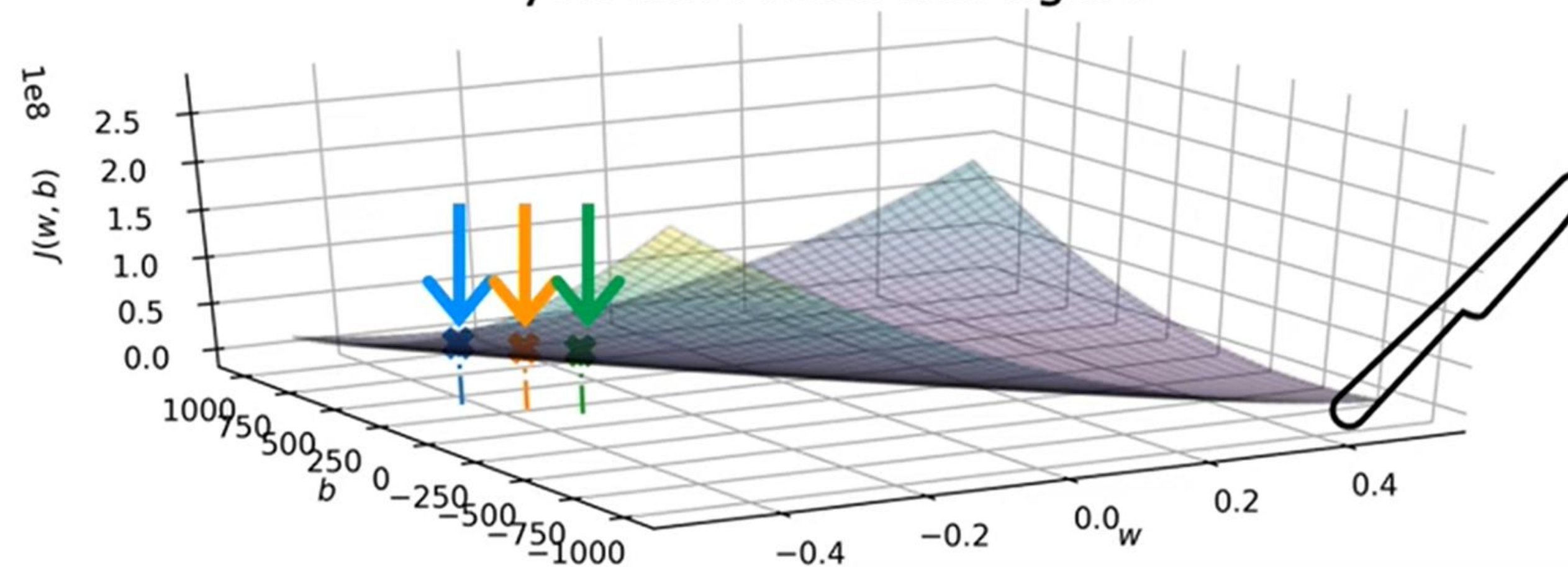
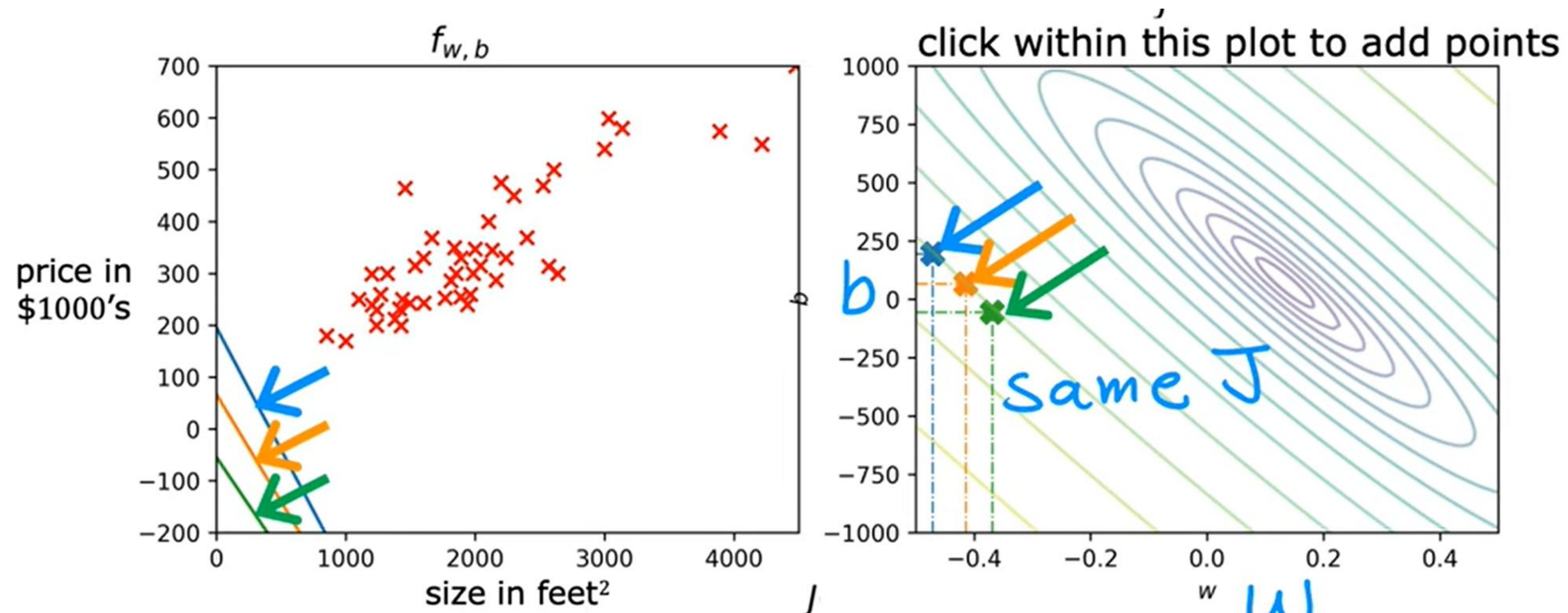
Linear Regression (Visualization)



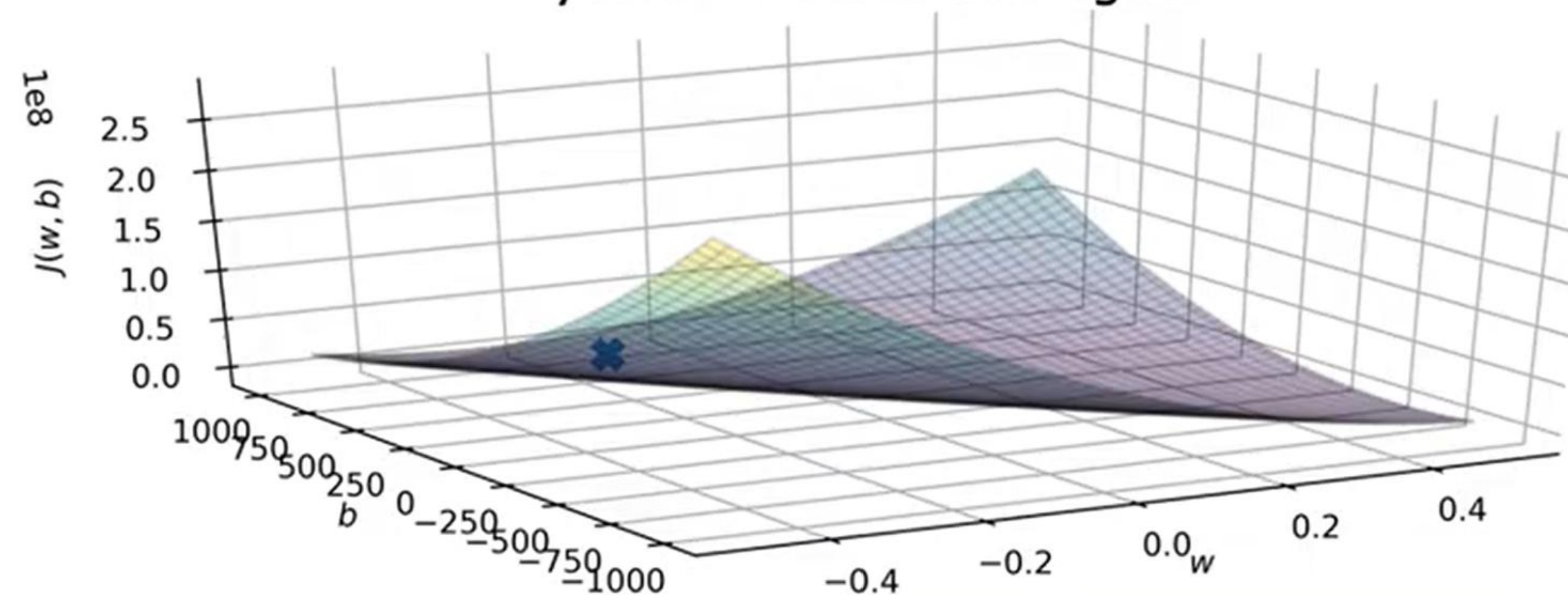
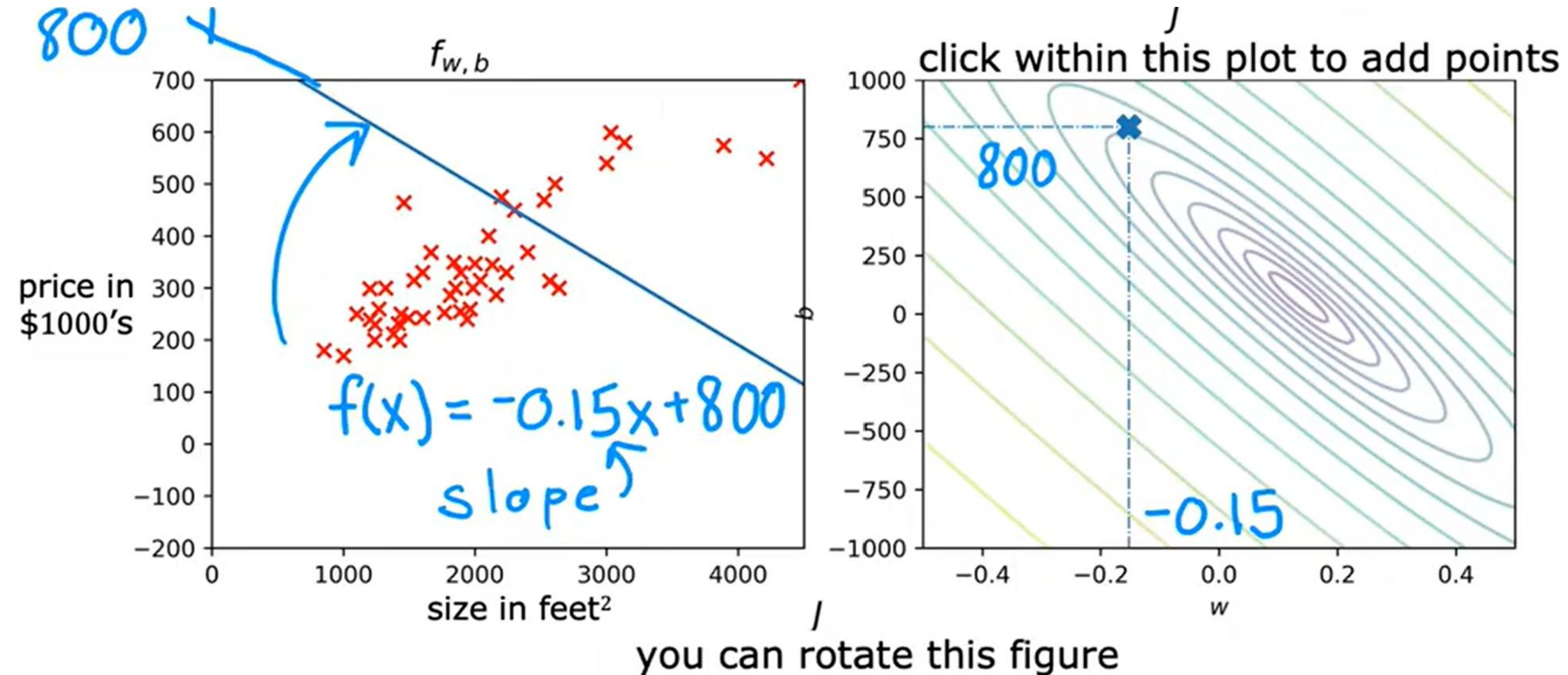
Linear Regression (Visualization)



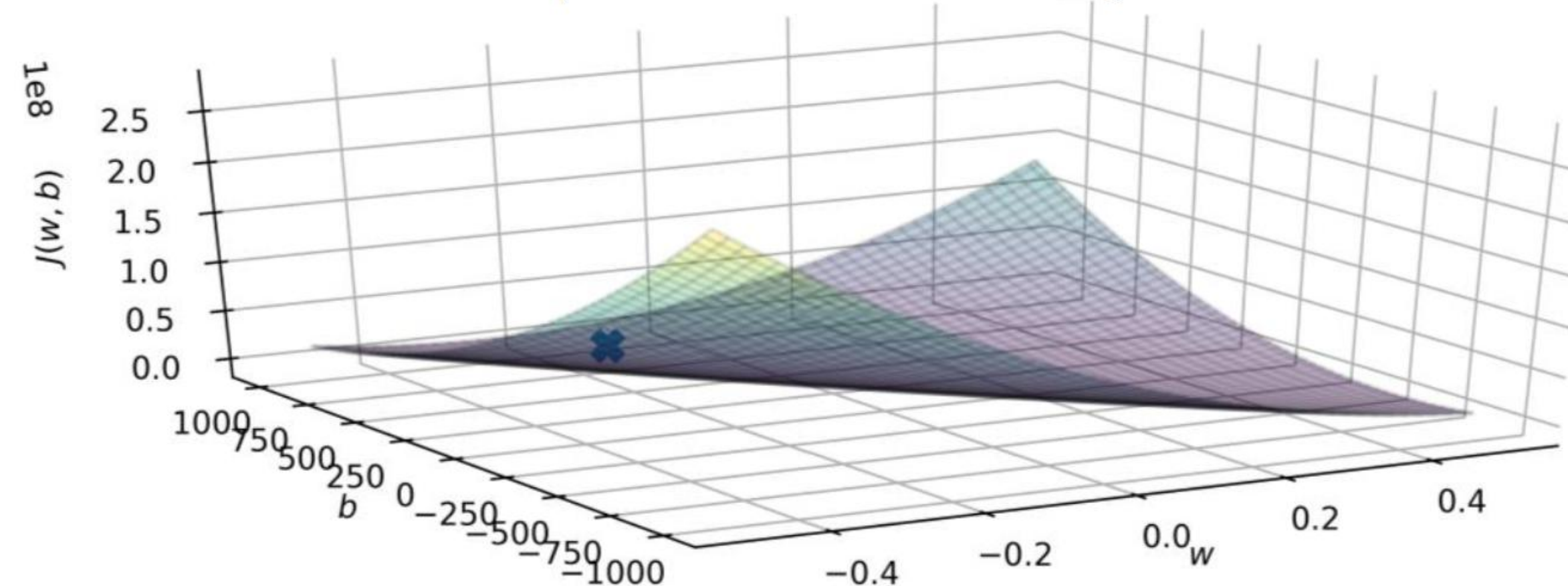
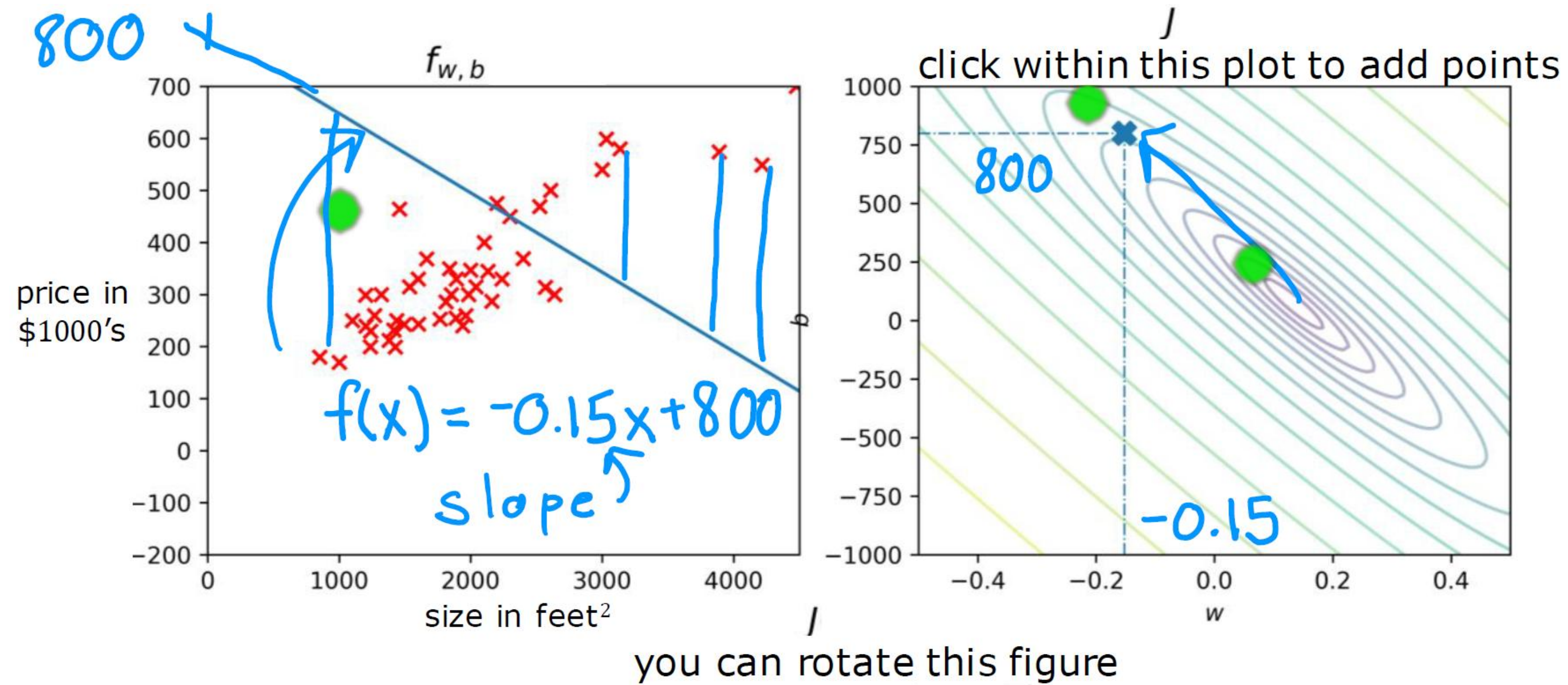
Linear Regression (Visualization)



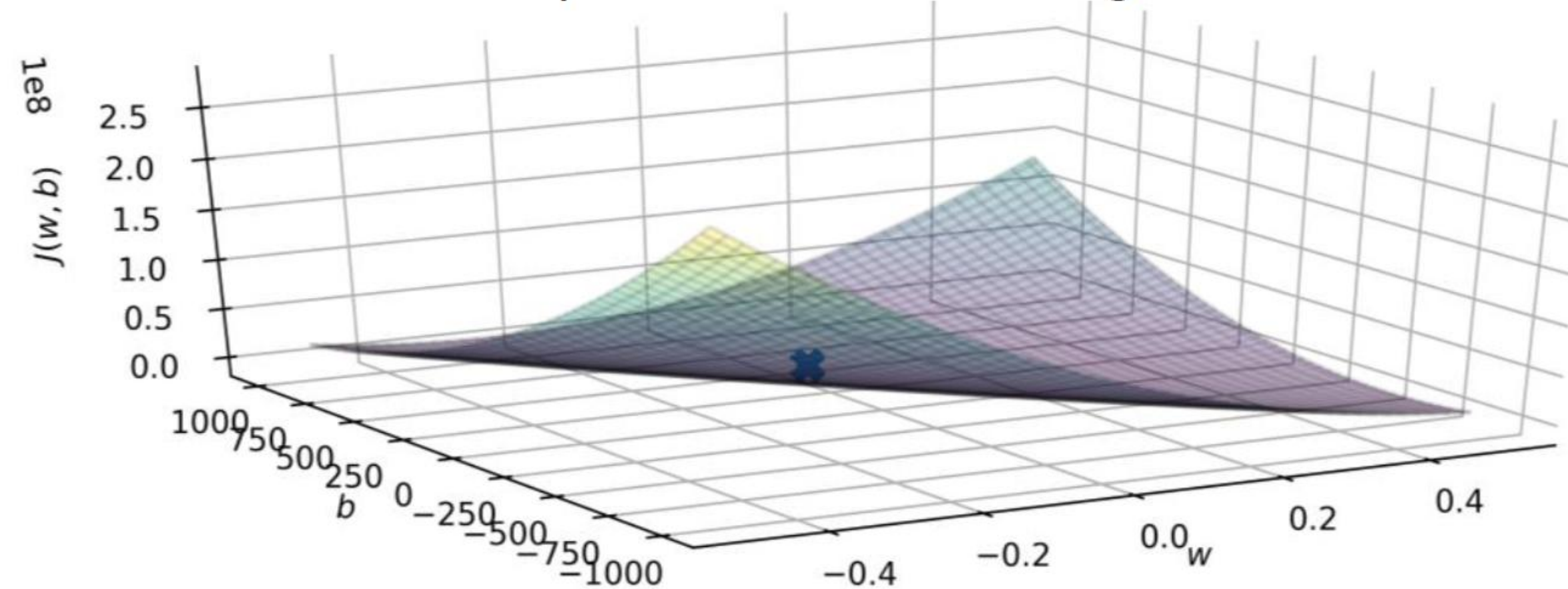
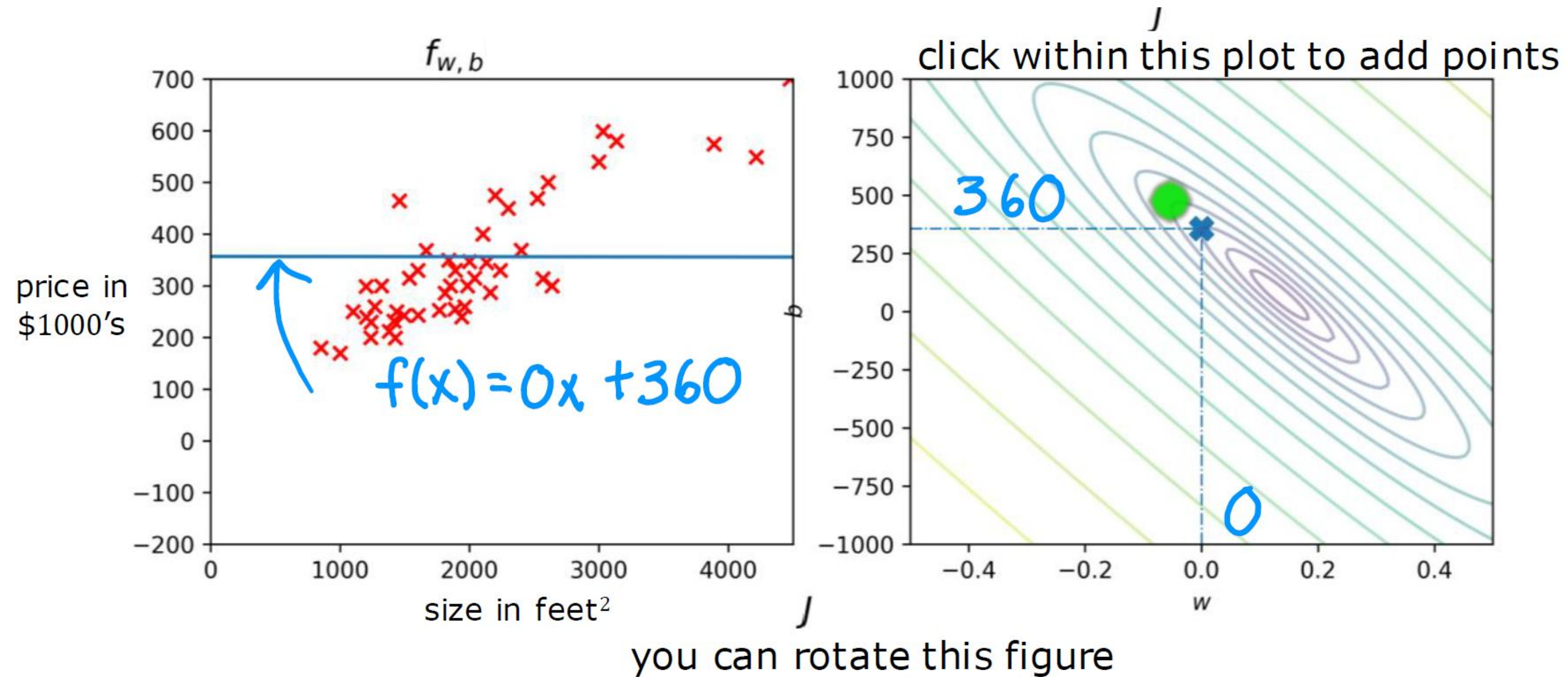
Linear Regression (Visualization)



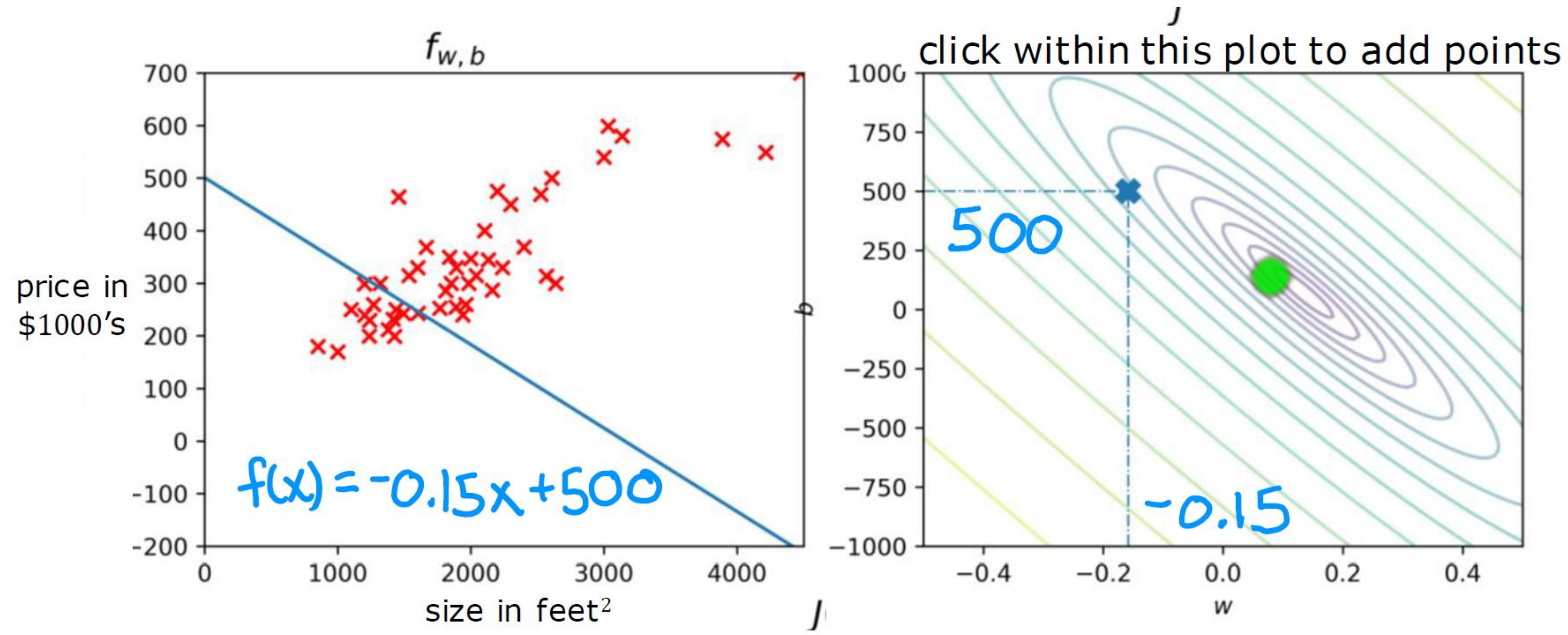
Linear Regression (Visualization)



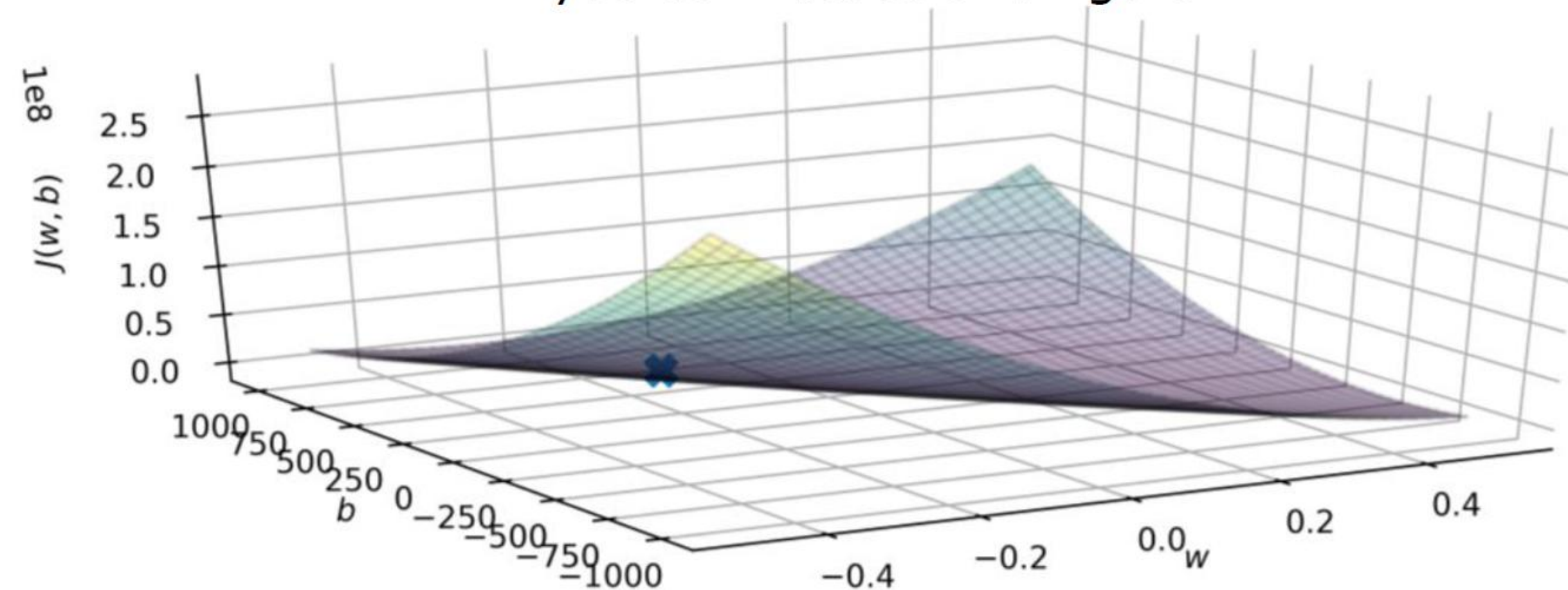
Linear Regression (Visualization)



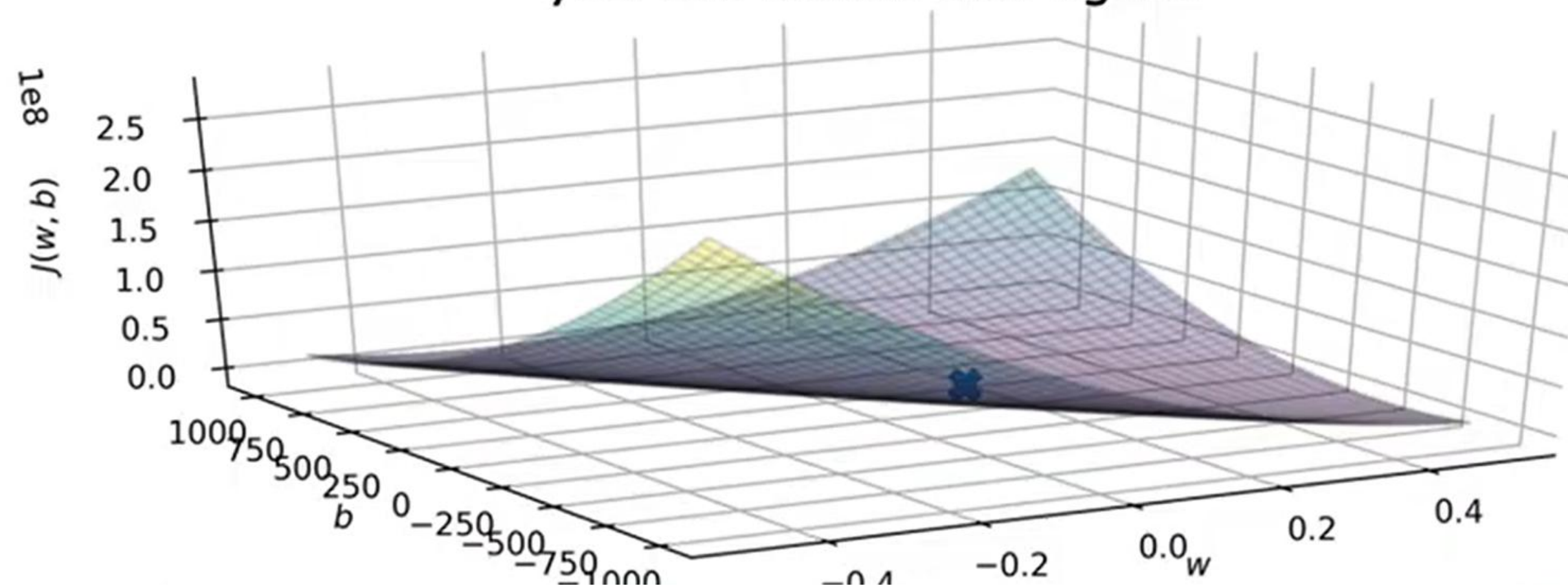
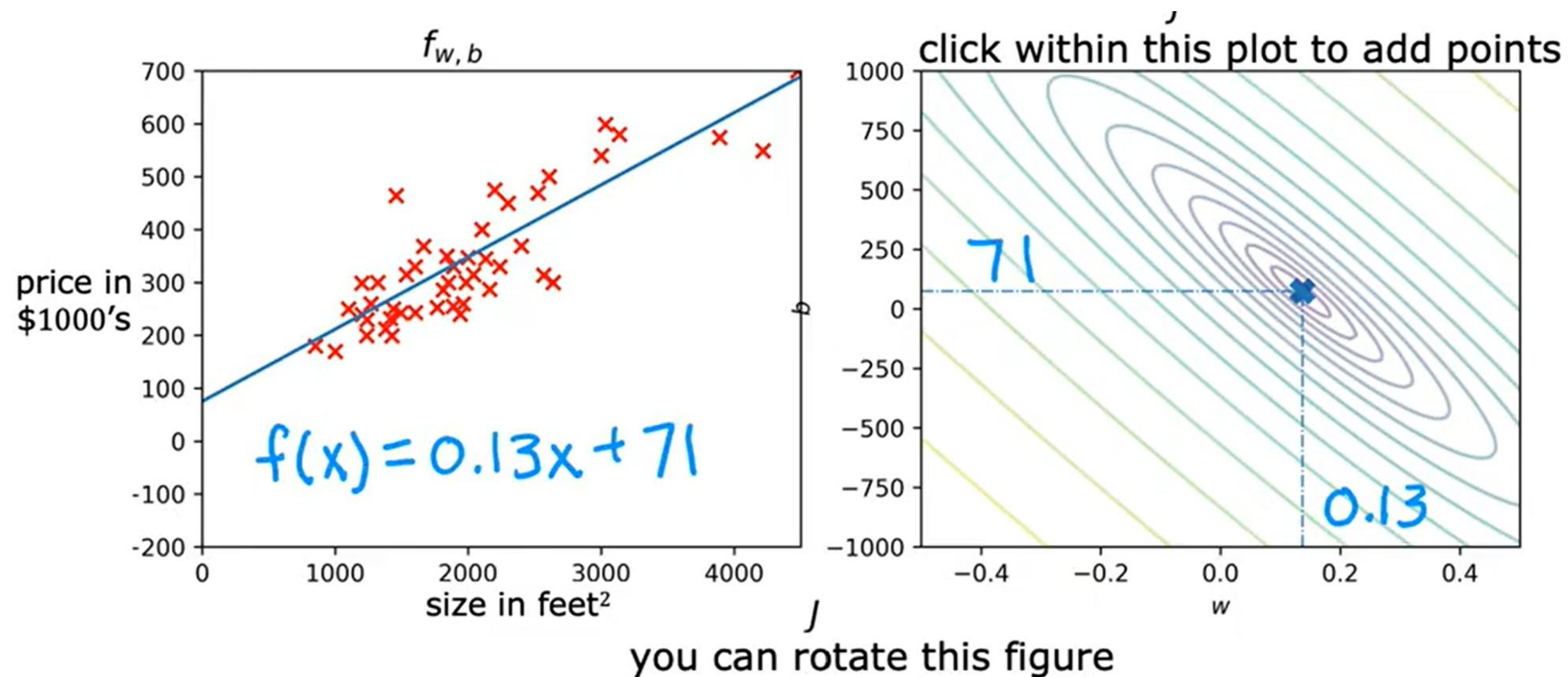
Linear Regression (Visualization)



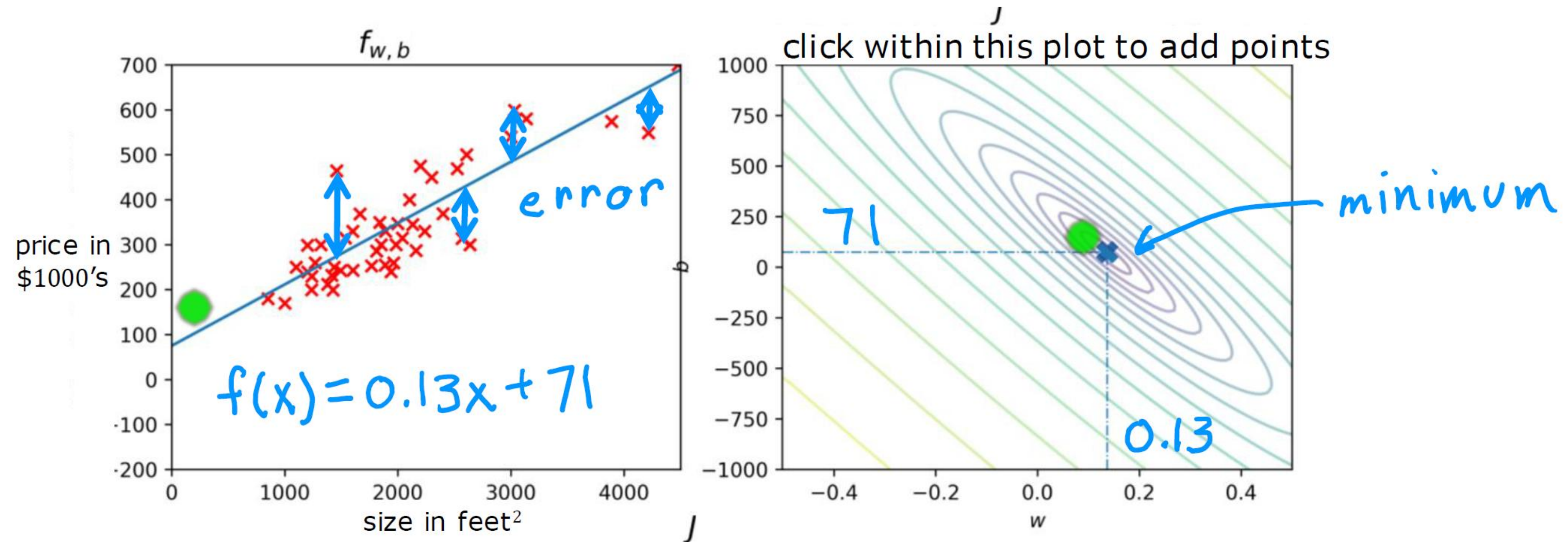
you can rotate this figure



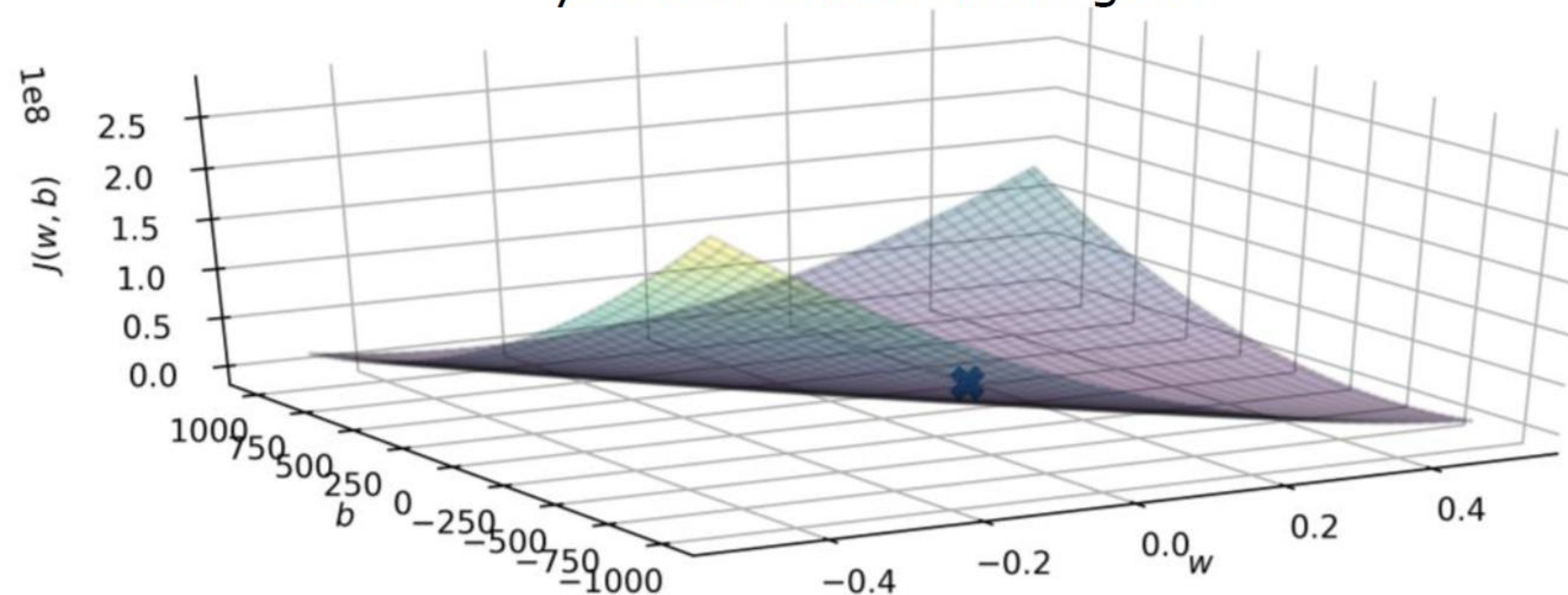
Linear Regression (Visualization)



Linear Regression (Visualization)



you can rotate this figure



Gradient Descent

Have some function $J(w, b)$ *for linear regression
or any function*

Want $\min_{w, b} J(w, b)$

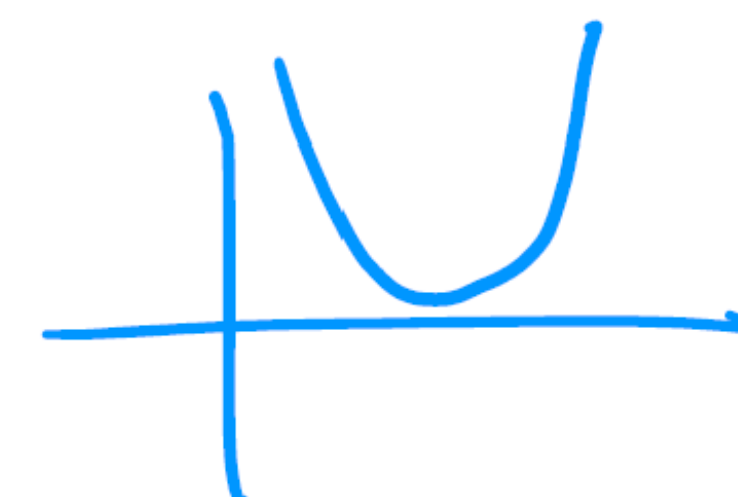
Outline:

Start with some w, b *(set $w=0, b=0$)*

Keep changing w, b to reduce $J(w, b)$

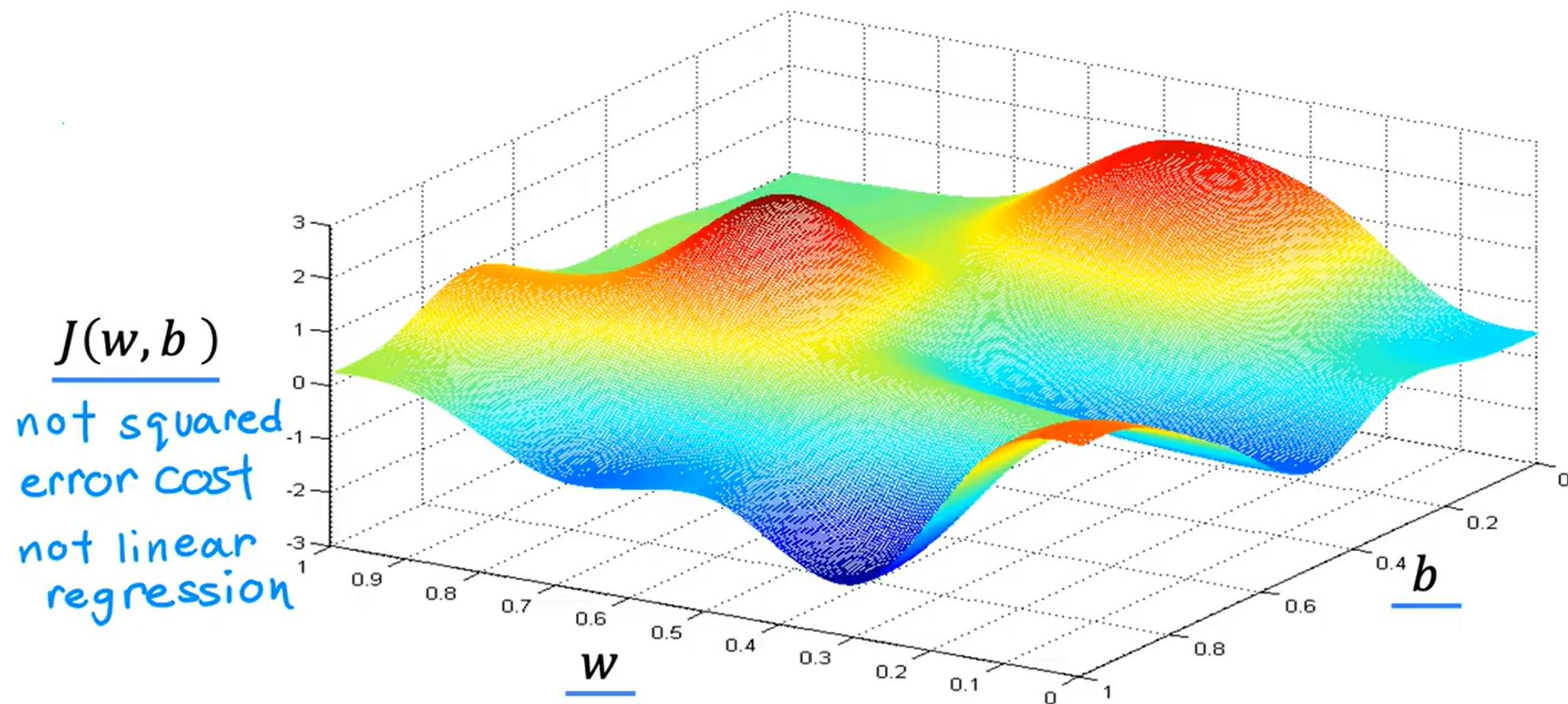
Until we settle at or near a minimum
may have >1 minimum

J not always



Gradient Descent

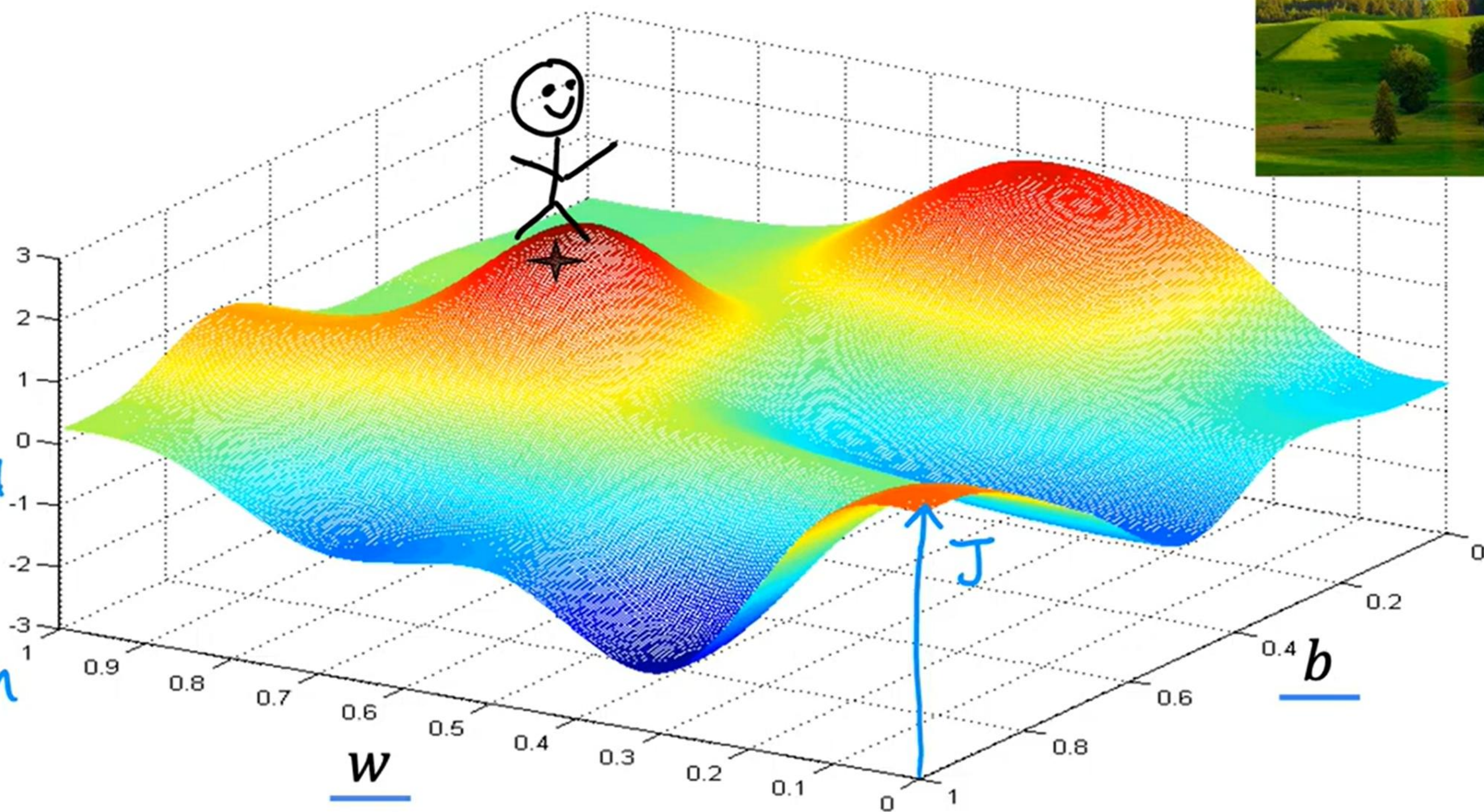
gradient descent



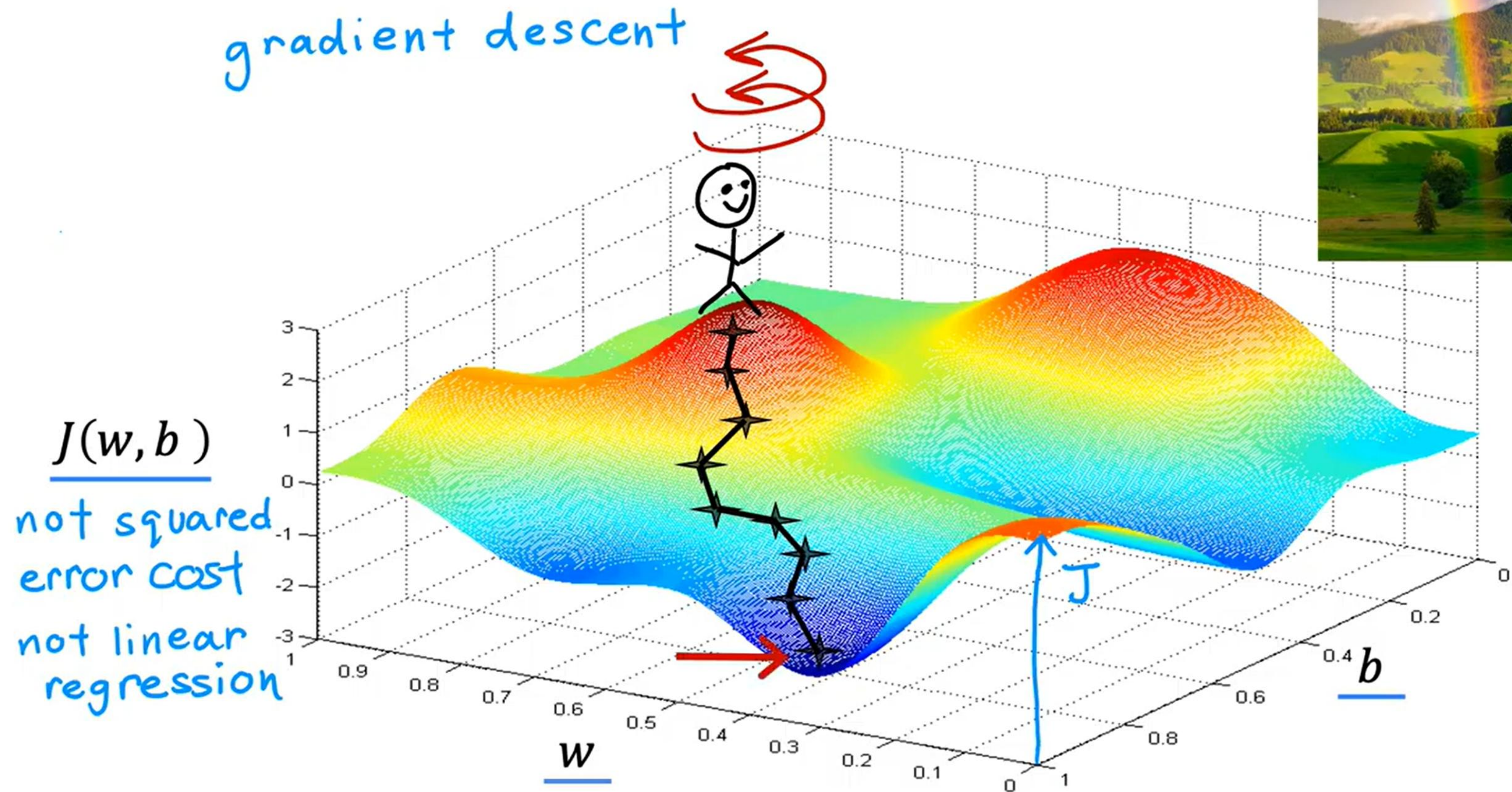
Gradient Descent

gradient descent

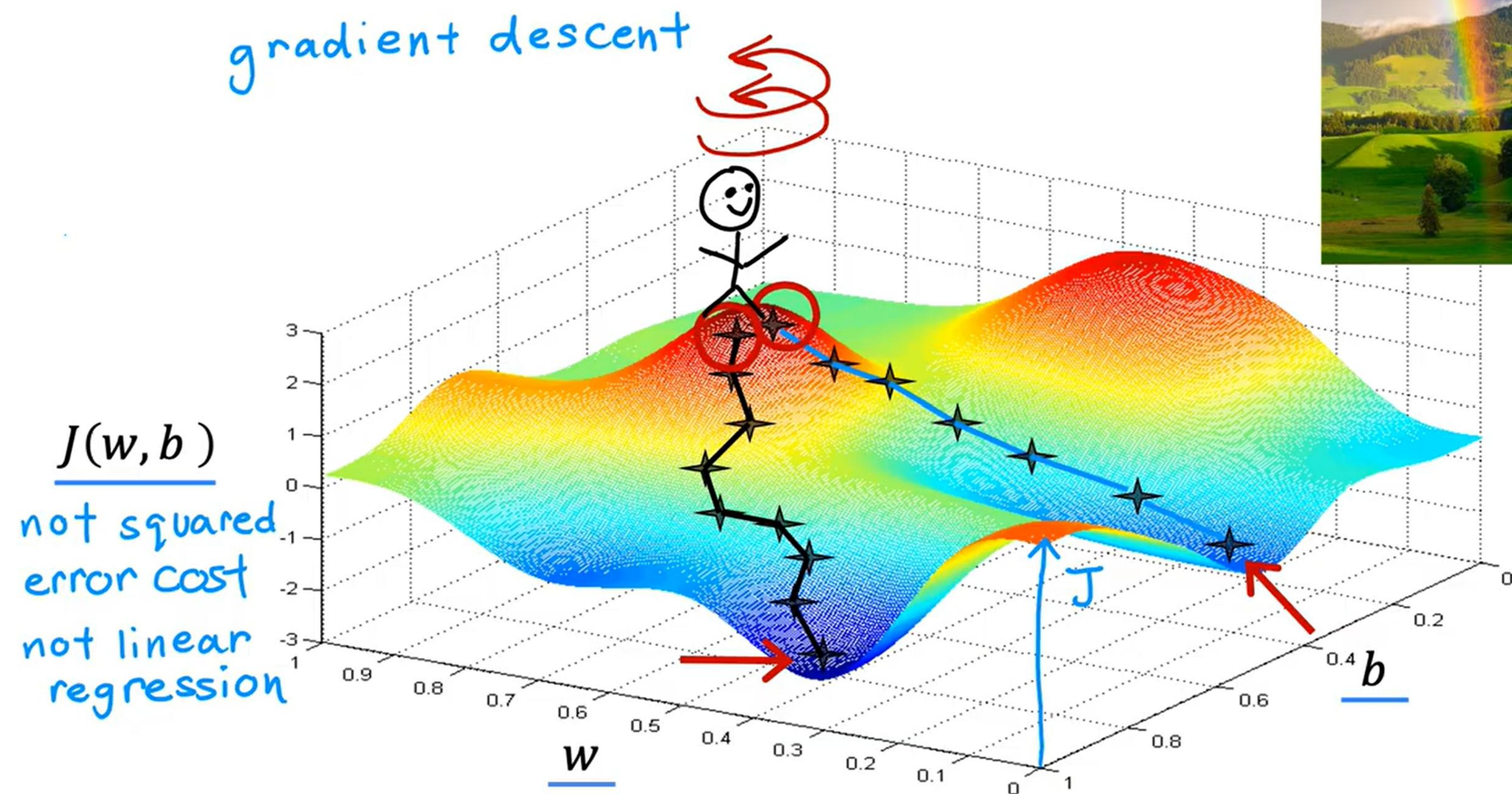
$J(w, b)$
not squared
error cost
not linear
regression



Gradient Descent



Gradient Descent



Gradient Descent

Repeat until convergence

$$w \leftarrow w - \alpha \frac{\partial}{\partial w} J(w, b)$$

Learning rate
Derivative

$$b \leftarrow b - \alpha \frac{\partial}{\partial b} J(w, b)$$

Simultaneously
update w and b

Correct: Simultaneous update

$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$w = tmp_w$$

$$b = tmp_b$$

Incorrect

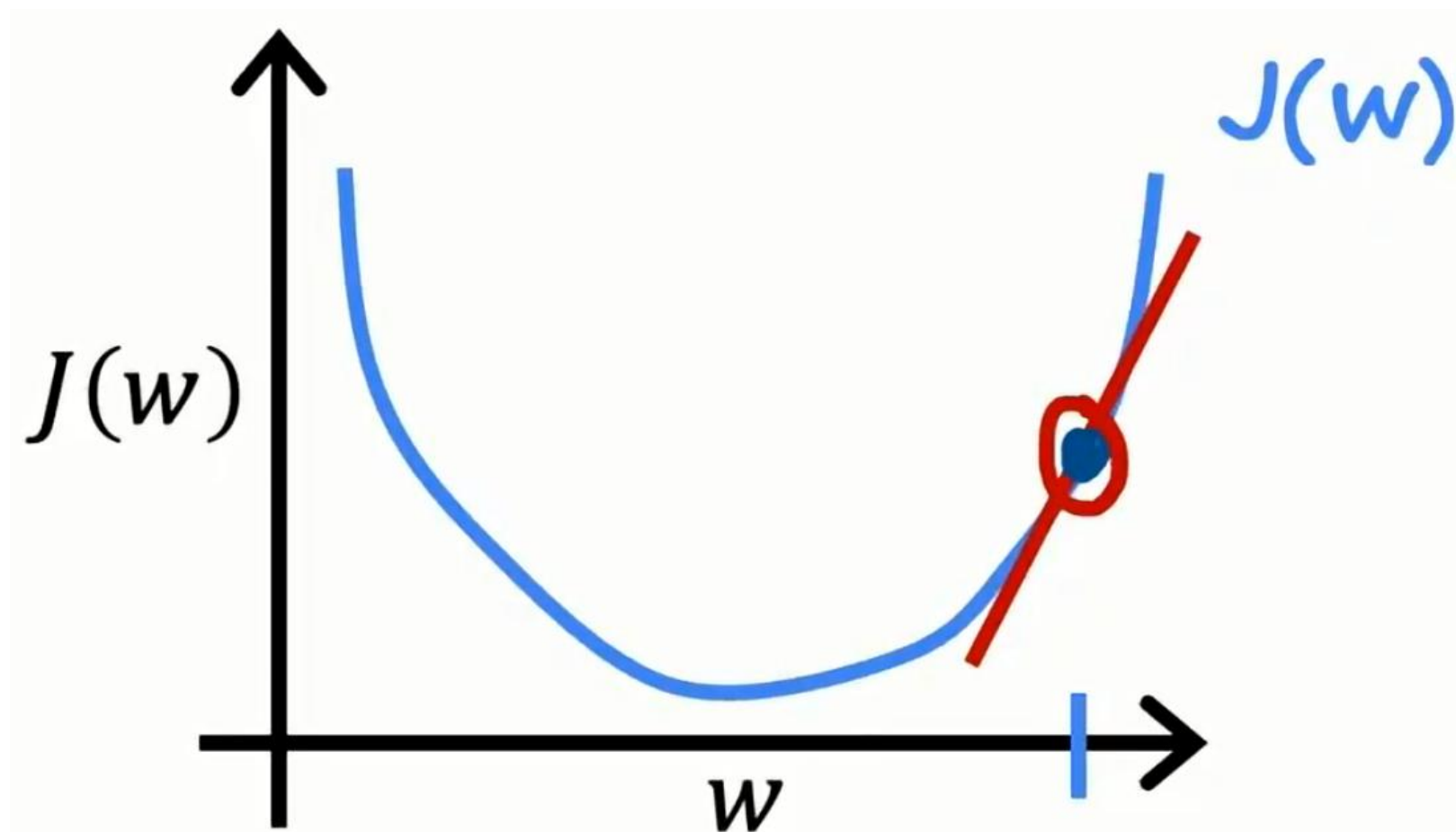
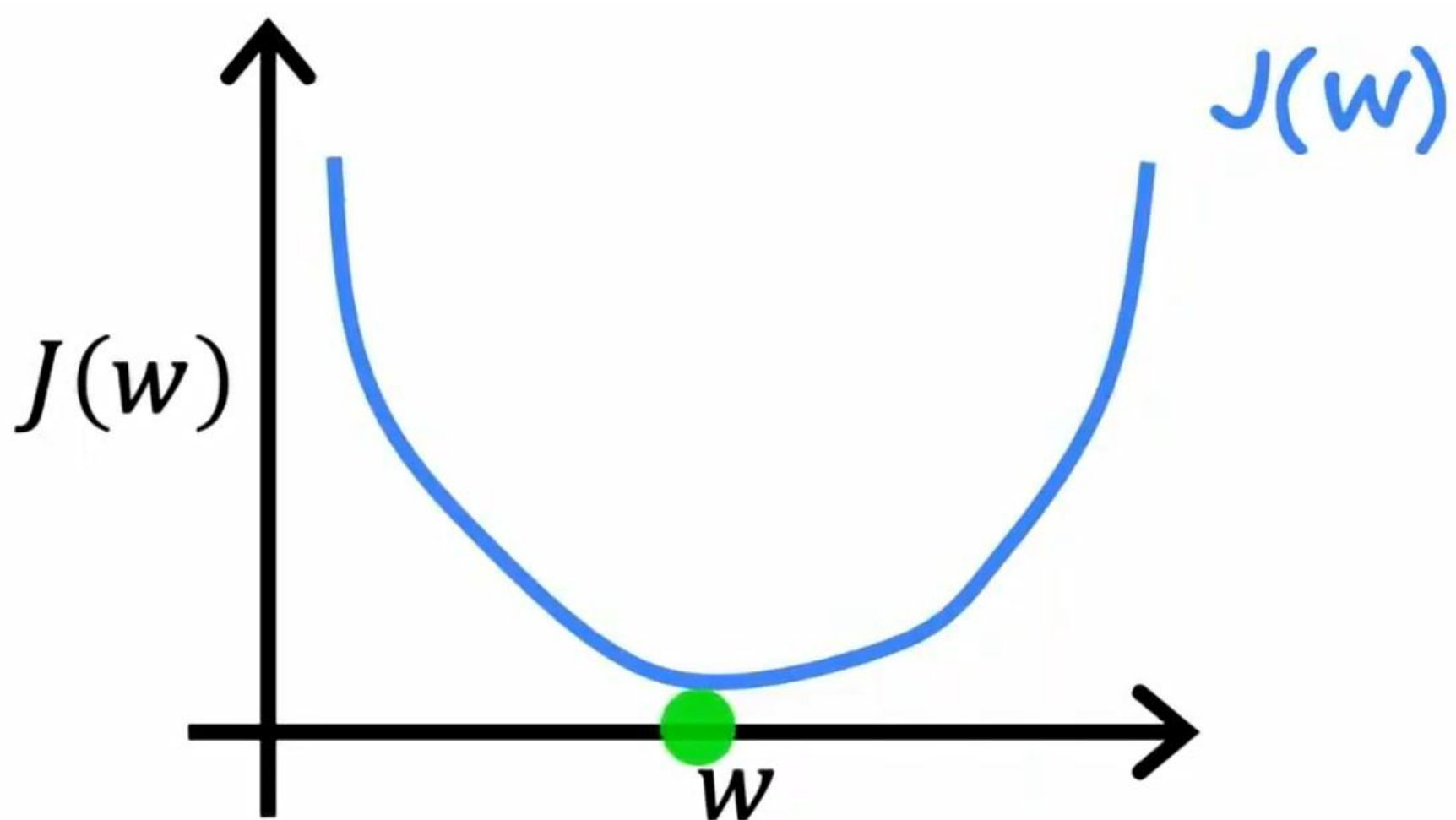
$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

$$w = tmp_w$$

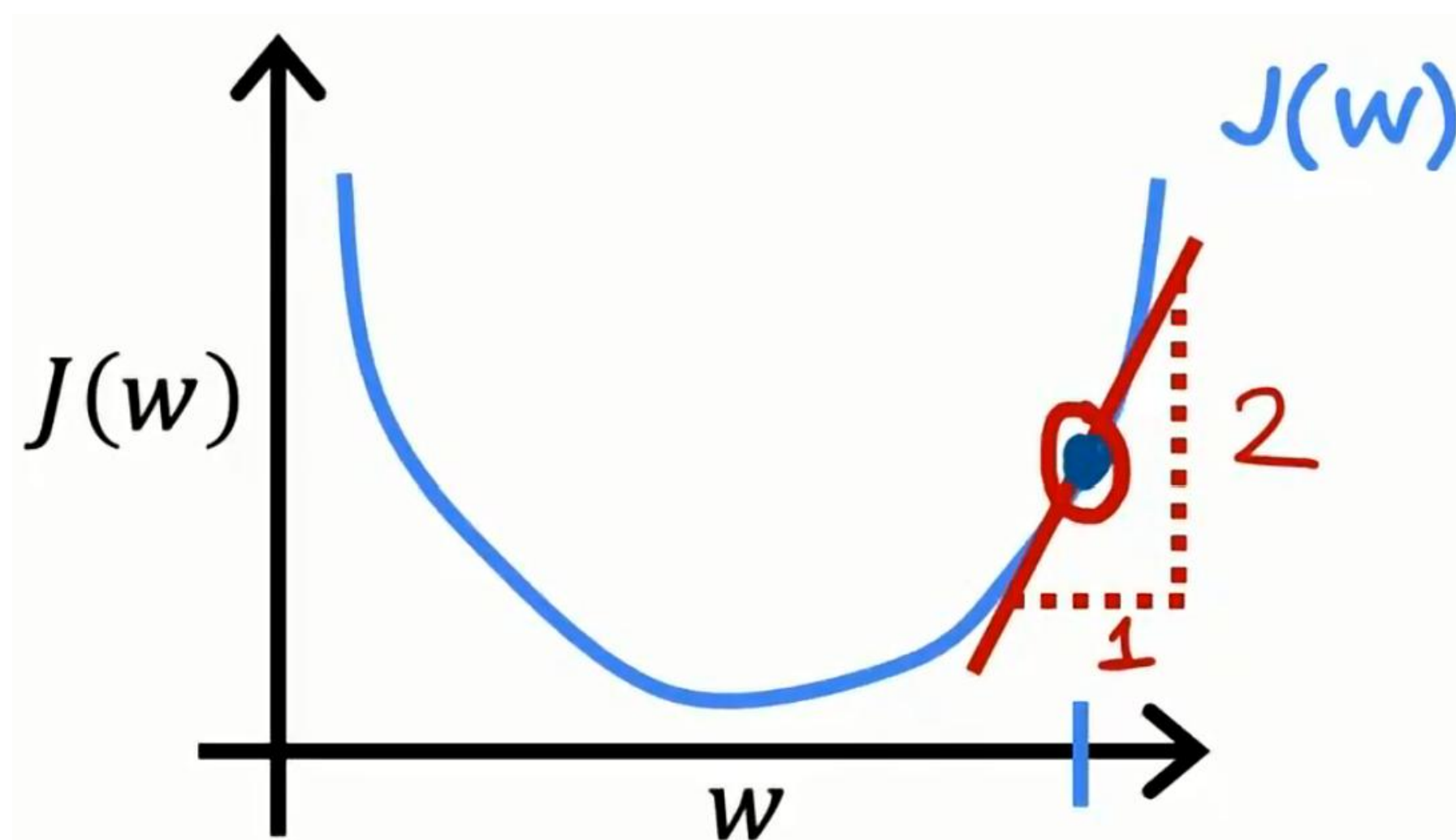
$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$b = tmp_b$$

Gradient Descent Intuition

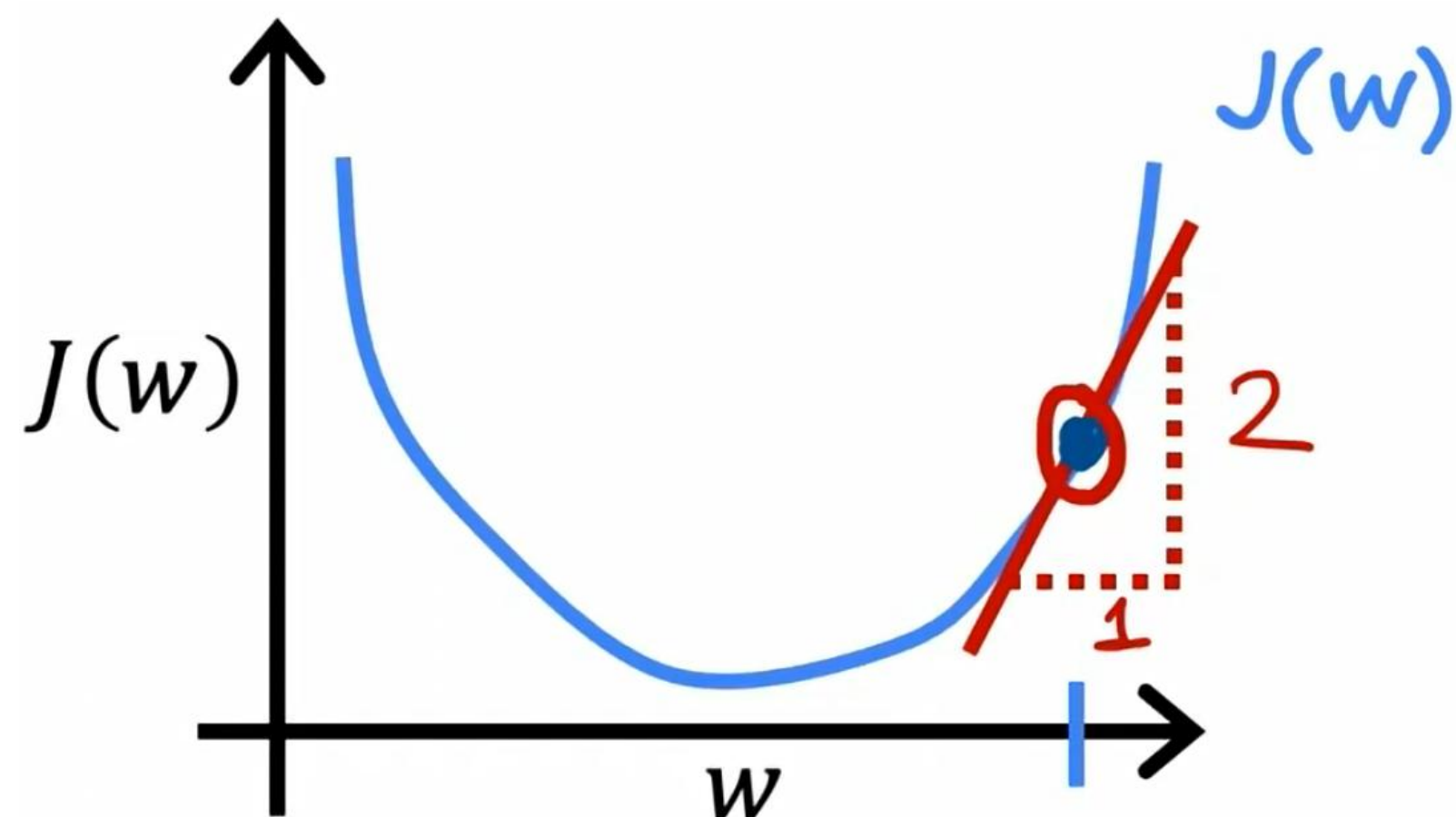


$$w = w - \alpha \frac{d}{dw} J(w)$$



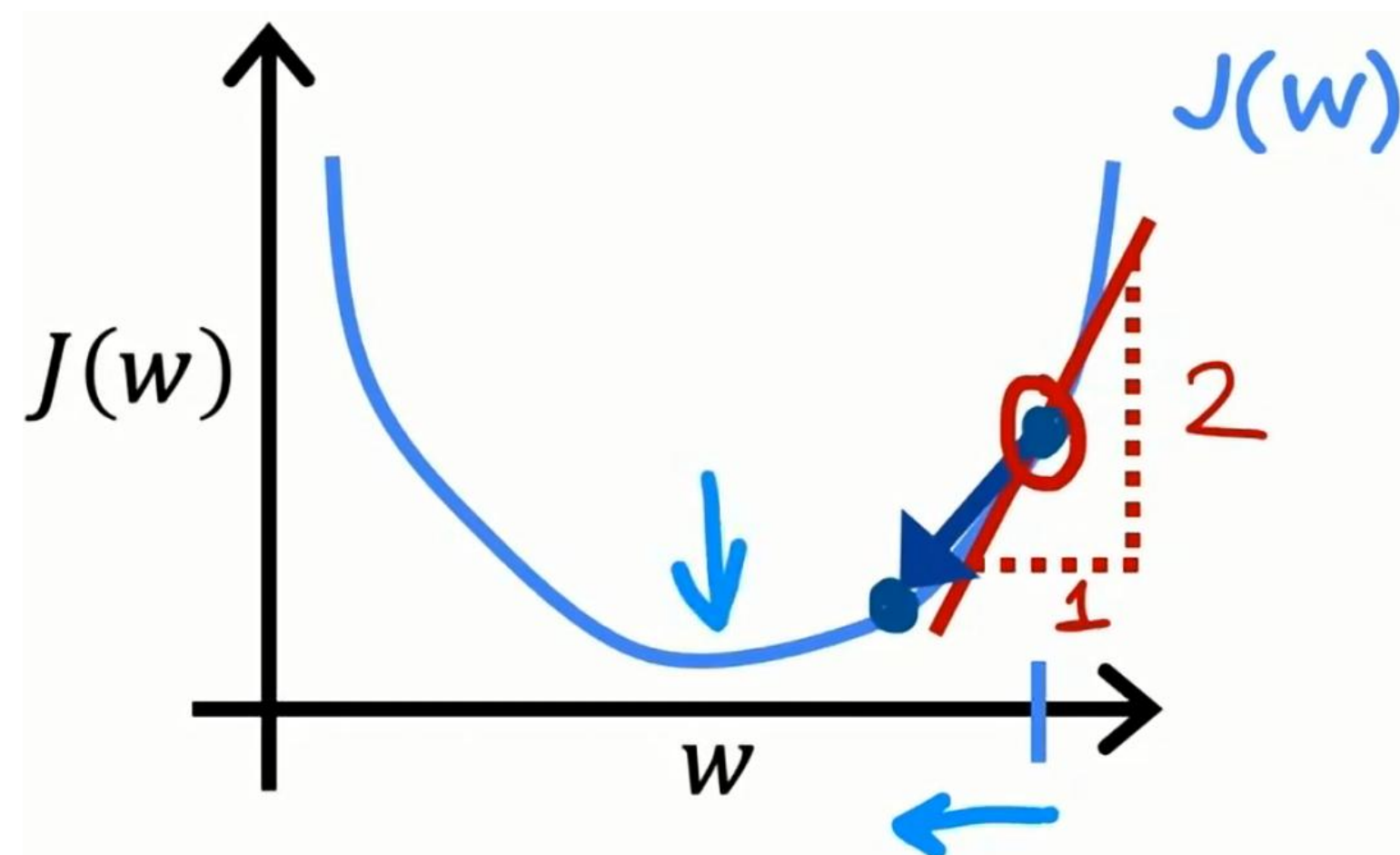
$$w = w - \alpha \frac{d}{dw} J(w)$$

Gradient Descent Intuition



$$w = w - \alpha \underbrace{\frac{d}{dw} J(w)}_{>0}$$

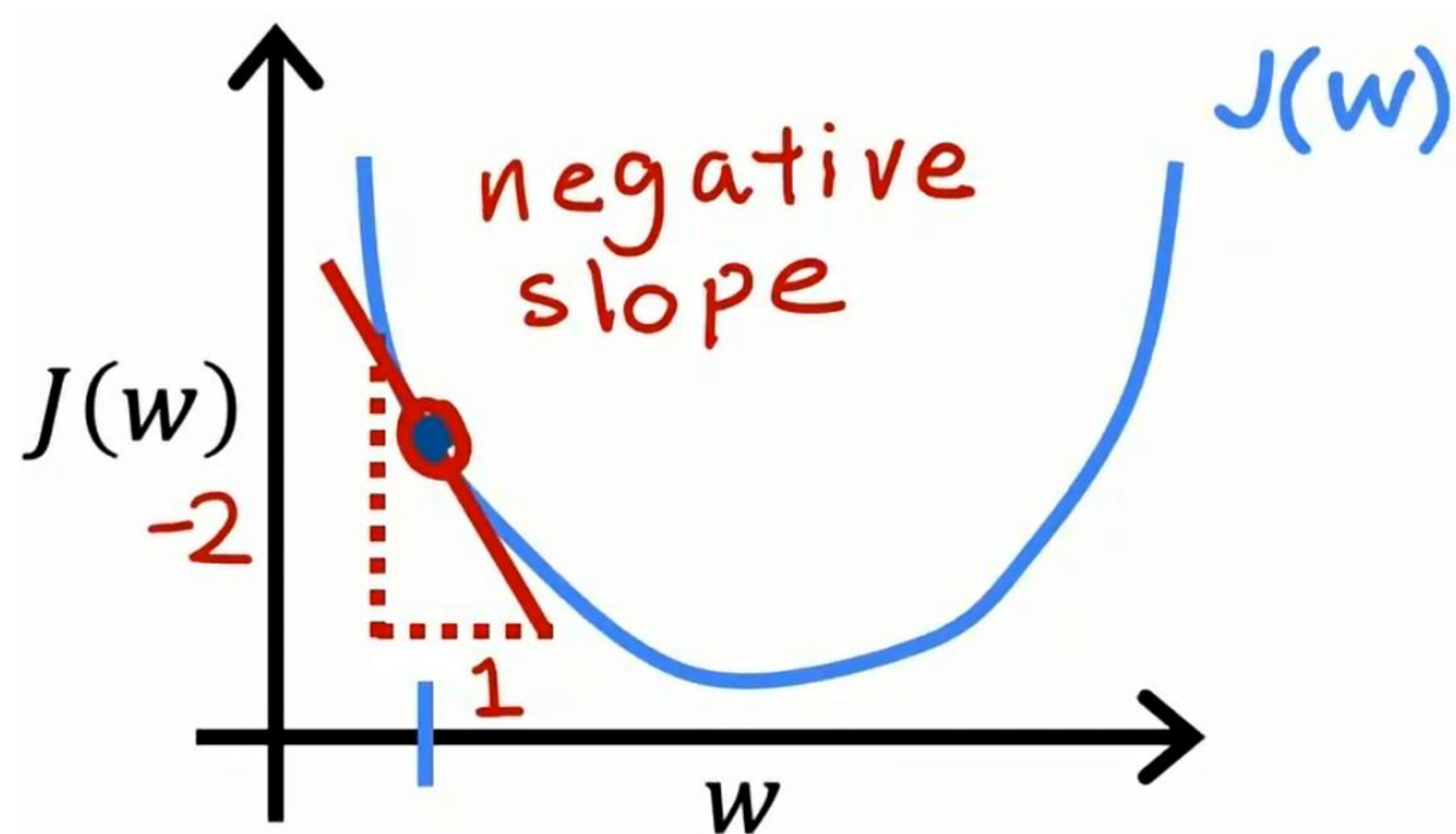
$$w = w - \alpha \cdot (\text{positive number})$$



$$w = w - \alpha \underbrace{\frac{d}{dw} J(w)}_{>0}$$

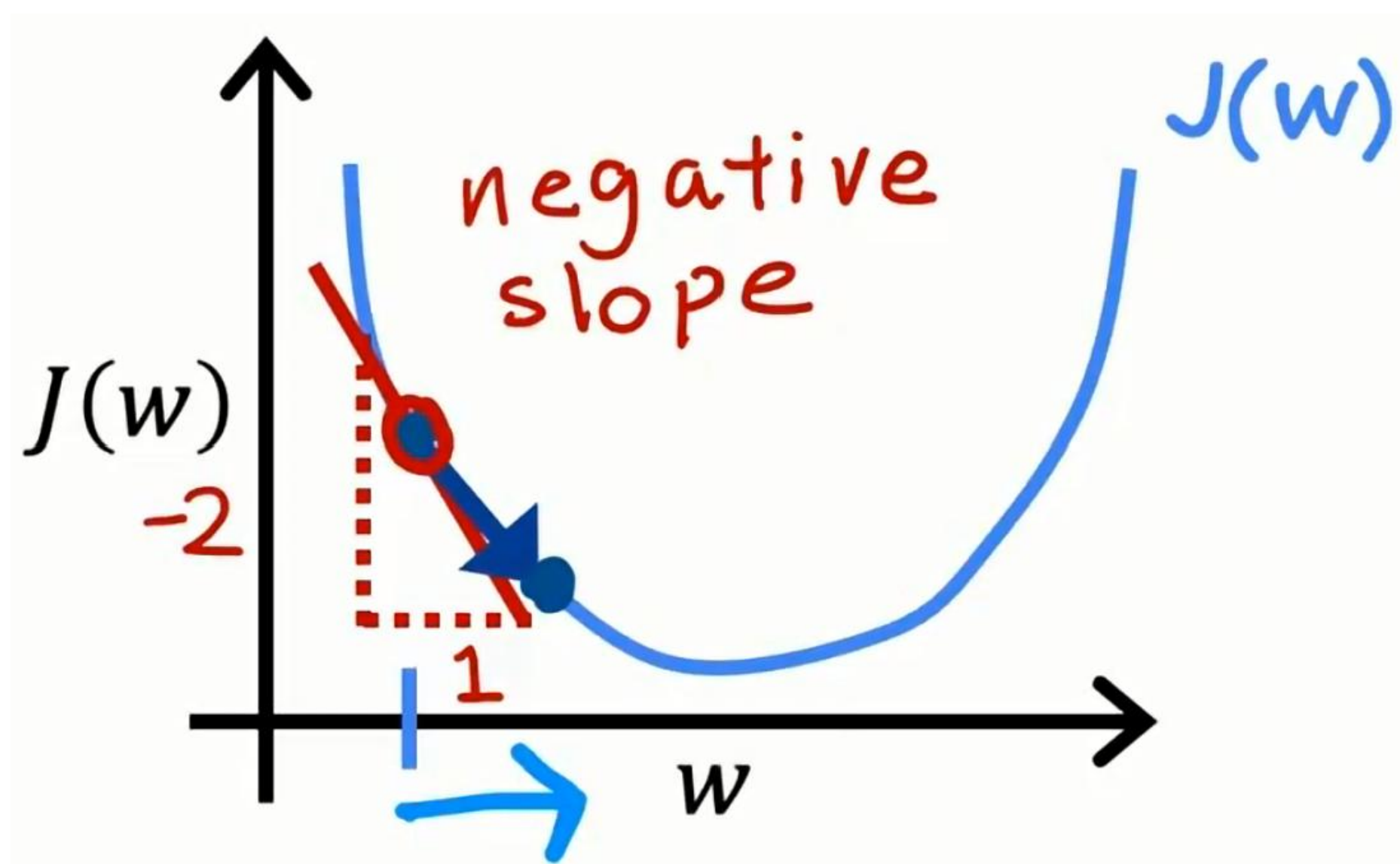
$$w = w - \underline{\alpha} \cdot (\text{positive number})$$

Gradient Descent Intuition



$$\frac{d}{dw} J(w) < 0$$

$$w = w - \alpha \cdot (\text{negative number})$$

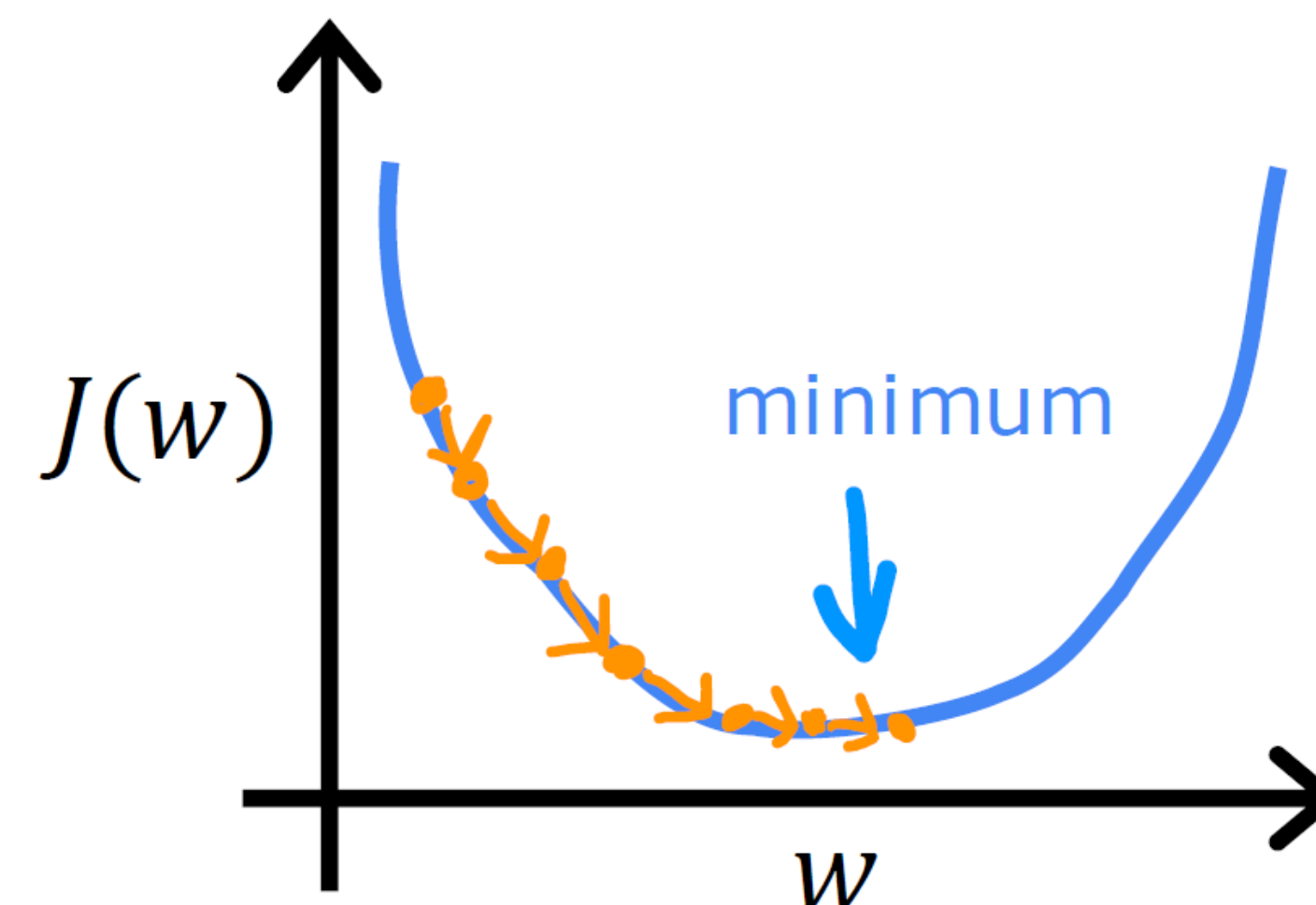


Gradient Descent (Learning rate)

$$w = w - \alpha \frac{d}{dw} J(w)$$

If α is too small...

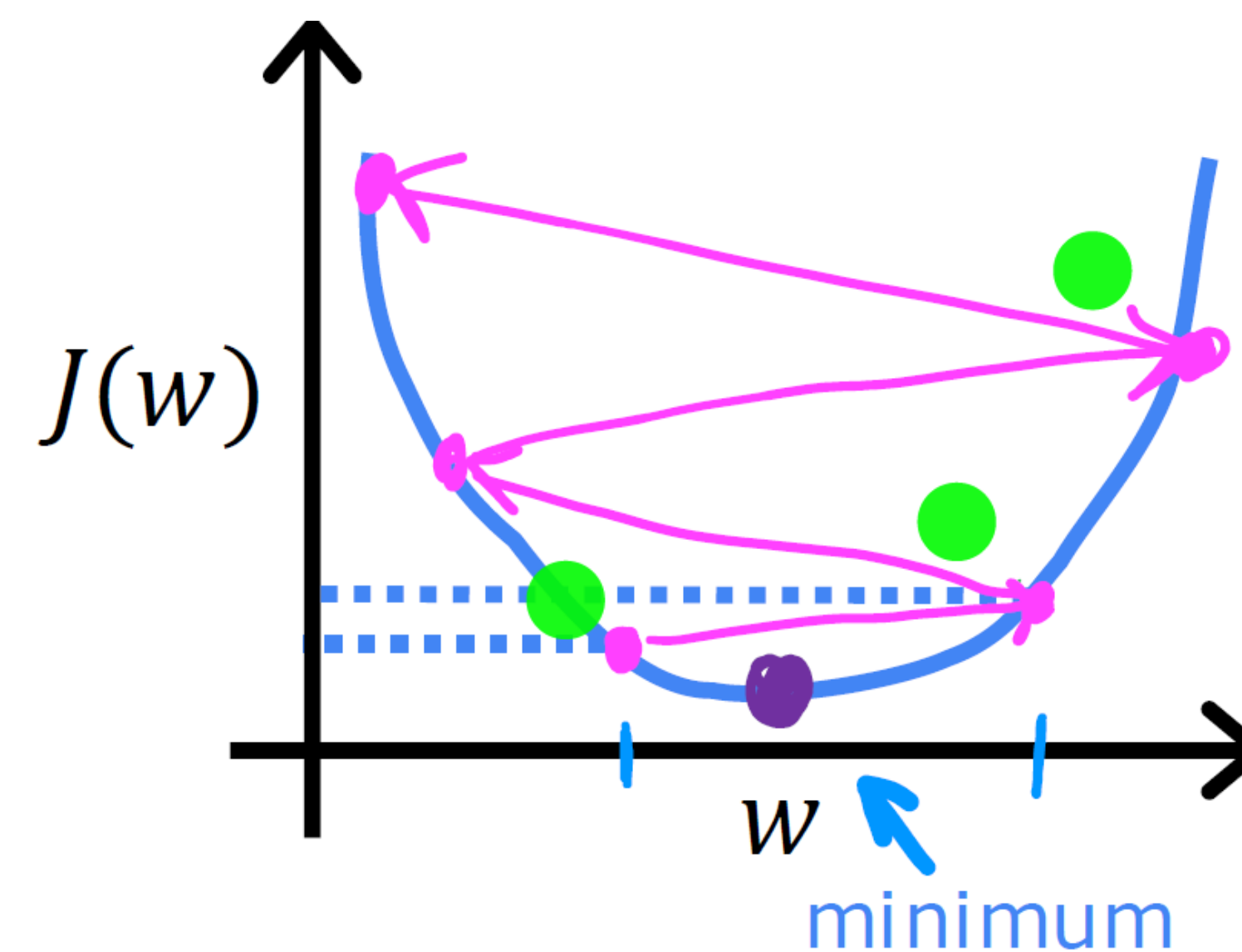
Gradient descent may be slow.



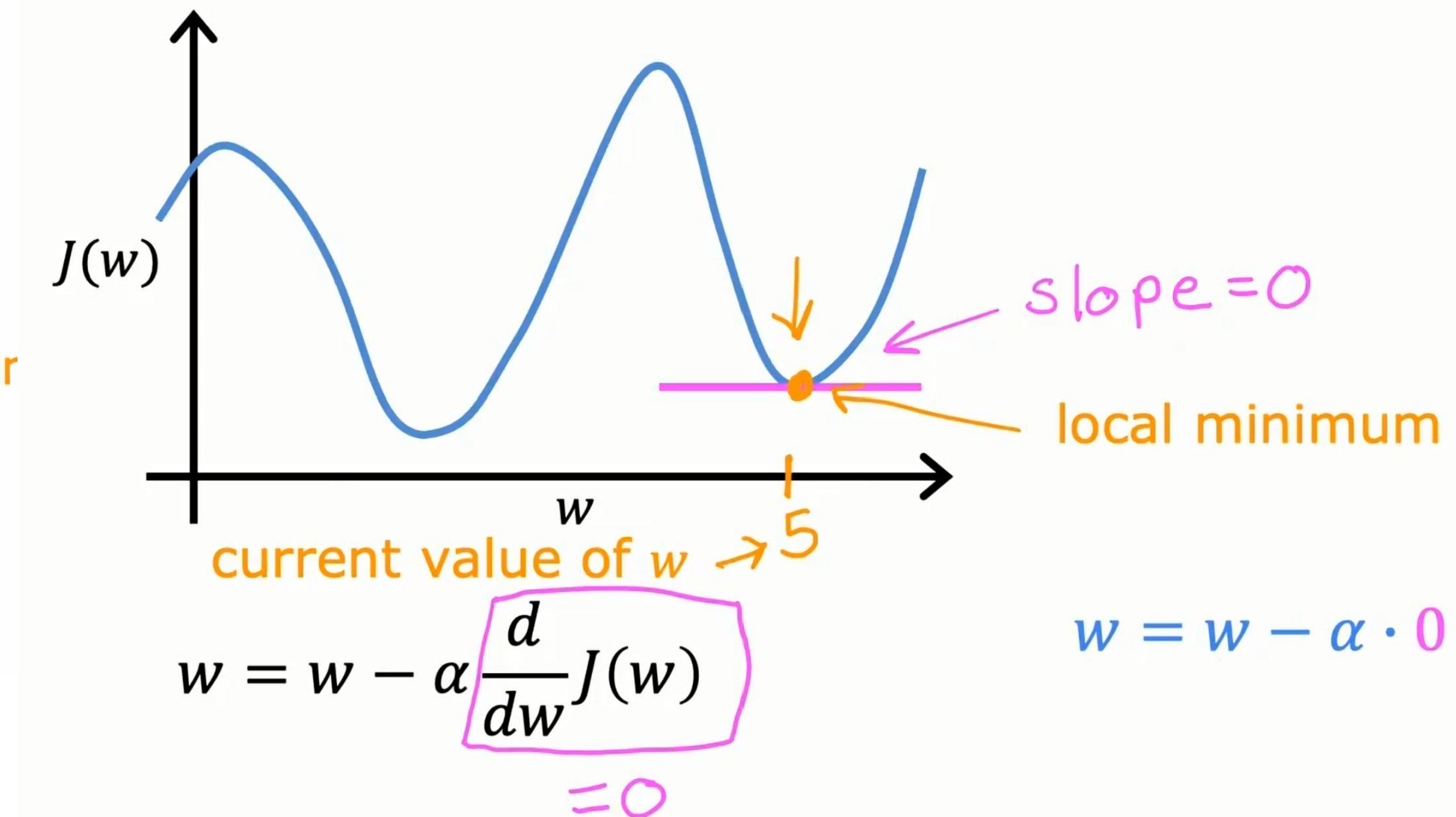
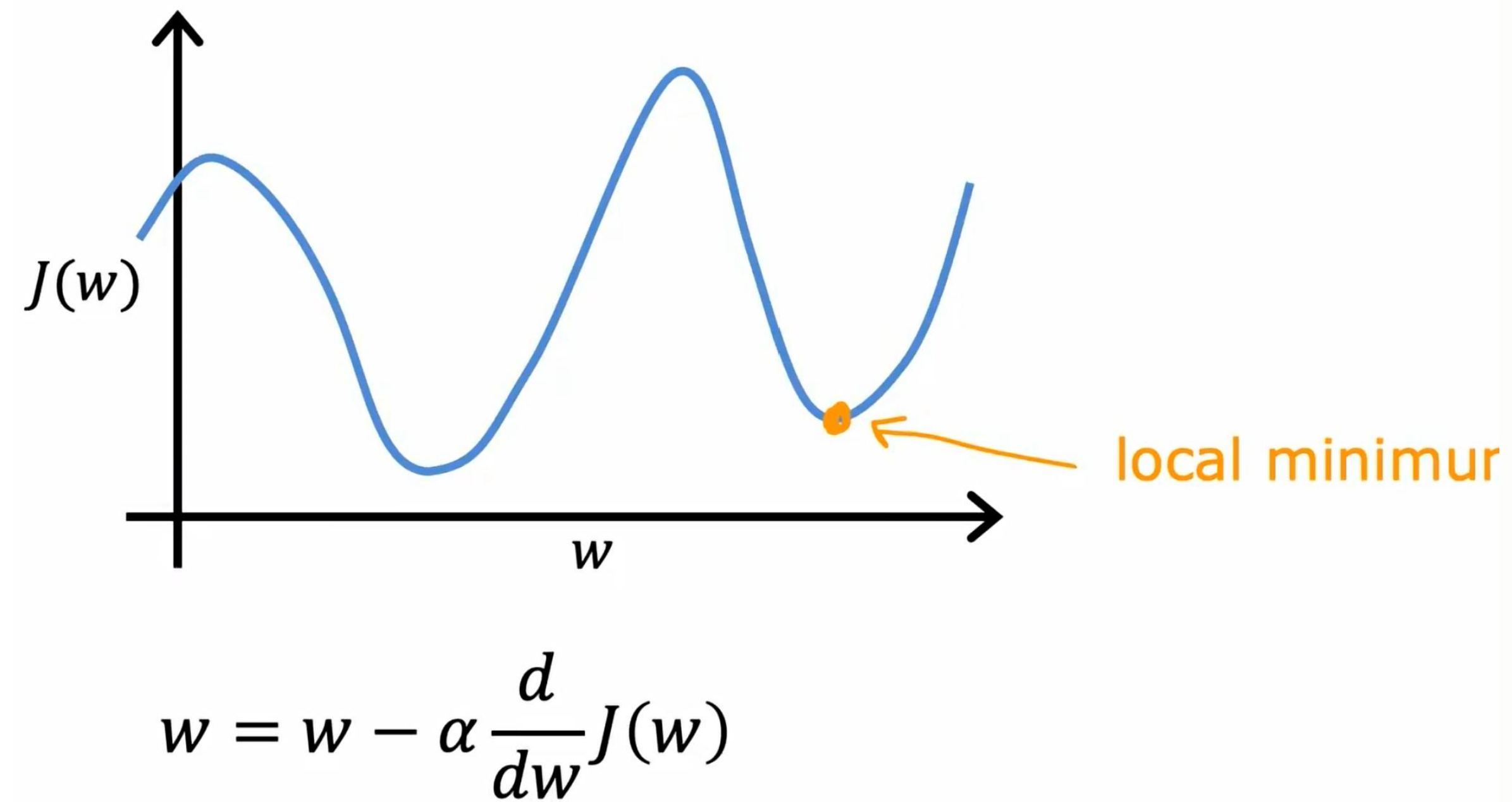
If α is too large...

Gradient descent may:

- Overshoot, never reach minimum
- Fail to converge, diverge



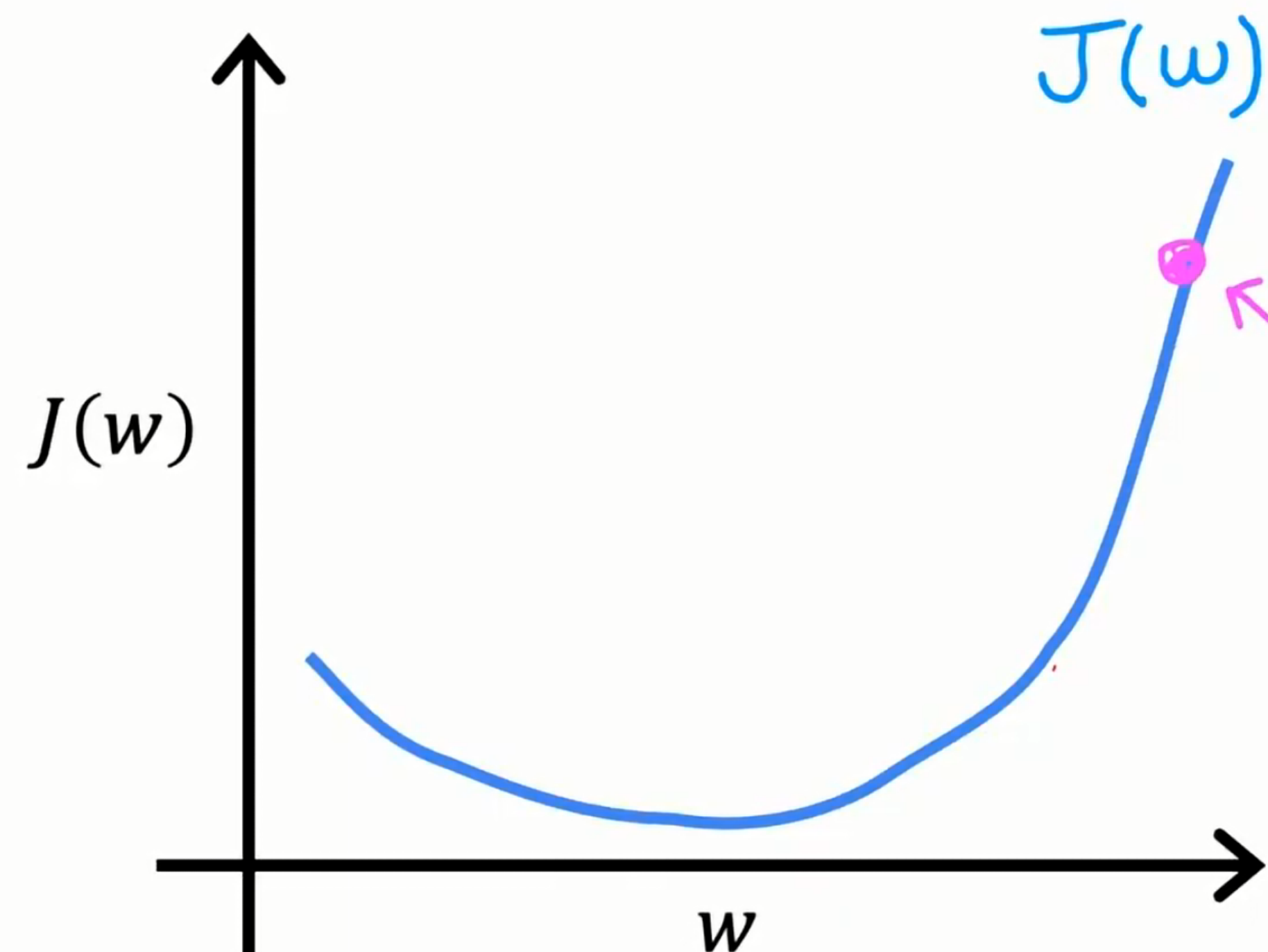
Gradient Descent (Learning rate)



Gradient Descent (Learning rate)

Can reach local minimum with fixed learning rate
 α

$$w = w - \alpha \frac{d}{dw} J(w)$$



Gradient Descent (Learning rate)

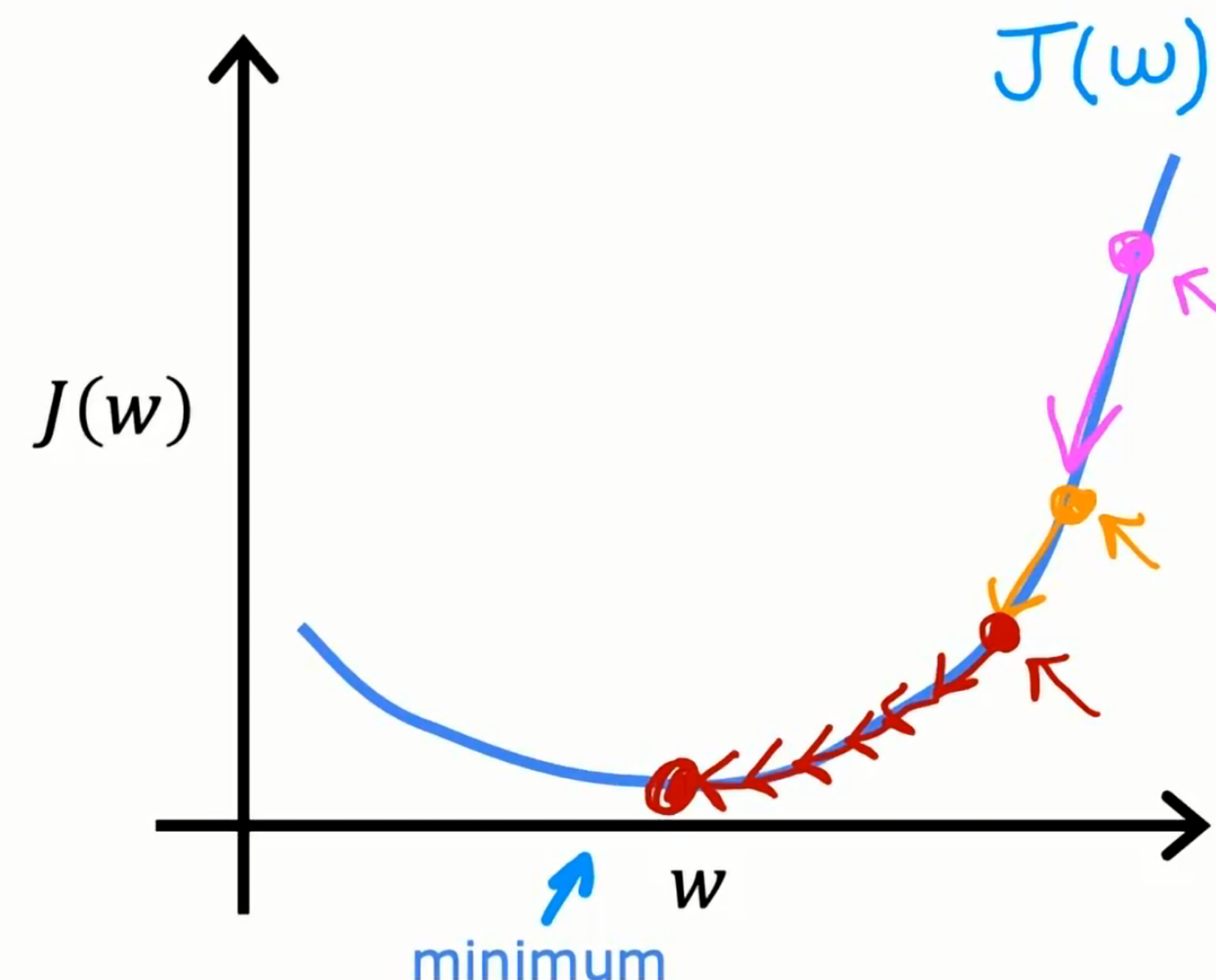
Can reach local minimum with fixed learning rate

$$w = w - \underbrace{\alpha}_{\text{smaller}} \underbrace{\frac{d}{dw} J(w)}_{\text{not as large}} \underbrace{J(w)}_{\text{large}}$$

Near a local minimum,

- Derivative becomes smaller
- Update steps become smaller

Can reach minimum without decreasing learning rate α



Gradient Descent for Linear Regression

Linear regression model

$$f_{w,b}(x) = wx + b$$

Cost function

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Gradient descent algorithm

repeat until convergence {

$$w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

}

$$\rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$\rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

Gradient Descent for Linear Regression

(Optional)

$$\frac{\partial}{\partial w} J(w, b) = \frac{d}{dw} \frac{1}{2m} \sum_{i=1}^m \left(\underline{f_{w,b}(x^{(i)})} - y^{(i)} \right)^2 = \frac{d}{dw} \frac{1}{2m} \sum_{i=1}^m \left(\underline{wx^{(i)} + b} - y^{(i)} \right)^2$$

$$= \cancel{\frac{1}{2m}} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)}) \cancel{2} x^{(i)} = \boxed{\frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}}$$

$$\frac{\partial}{\partial b} J(w, b) = \frac{d}{db} \frac{1}{2m} \sum_{i=1}^m \left(f_{w,b}(x^{(i)}) - y^{(i)} \right)^2 = \frac{d}{db} \frac{1}{2m} \sum_{i=1}^m \left(\underline{wx^{(i)} + b} - y^{(i)} \right)^2$$

$$= \cancel{\frac{1}{2m}} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)}) \cancel{2} = \boxed{\frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})}$$

no $x^{(i)}$

Gradient Descent for Linear Regression

Gradient descent algorithm

repeat until convergence {

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

}

$$\frac{\partial}{\partial w} J(w, b)$$

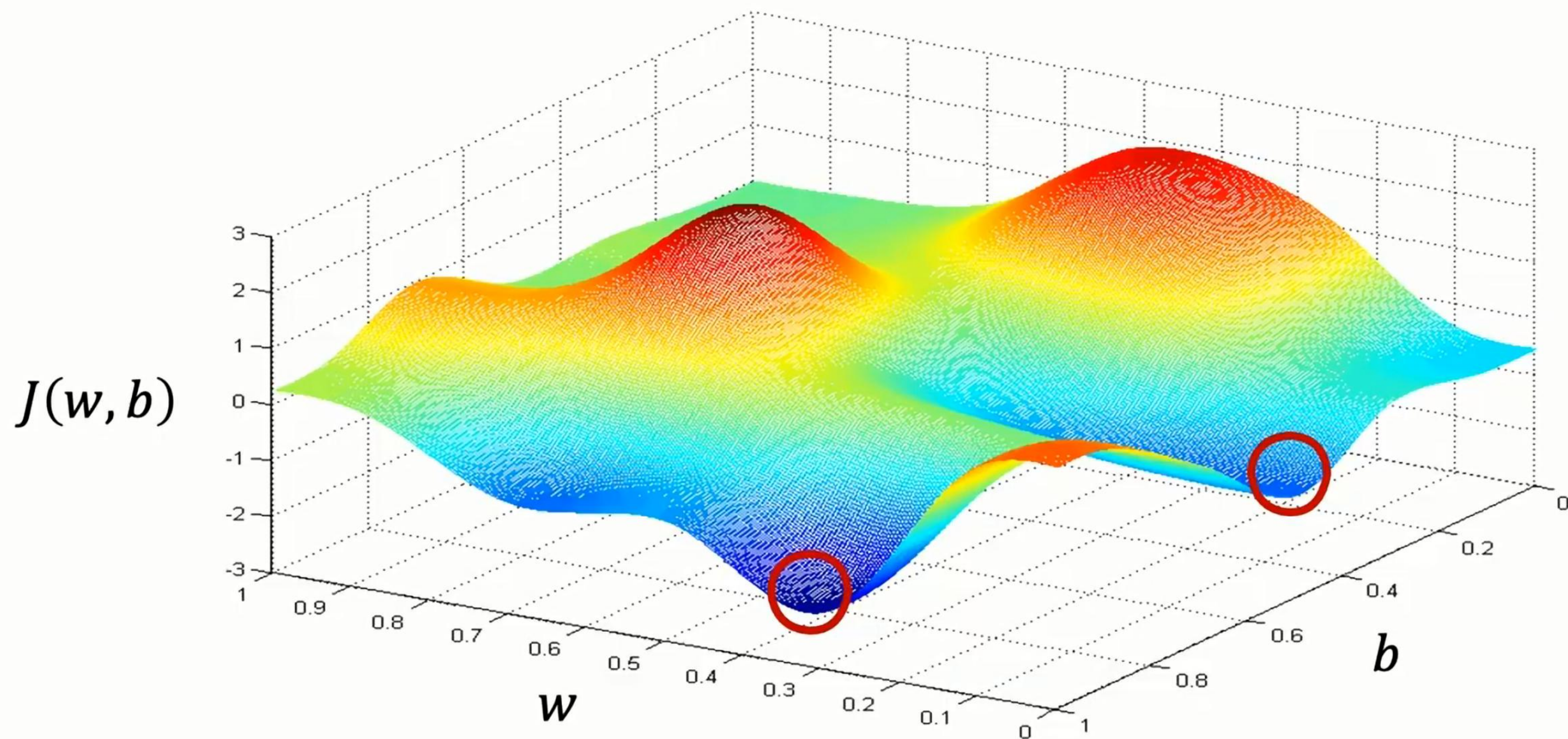
$$\frac{\partial}{\partial b} J(w, b)$$

Update
w and b
simultaneously

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

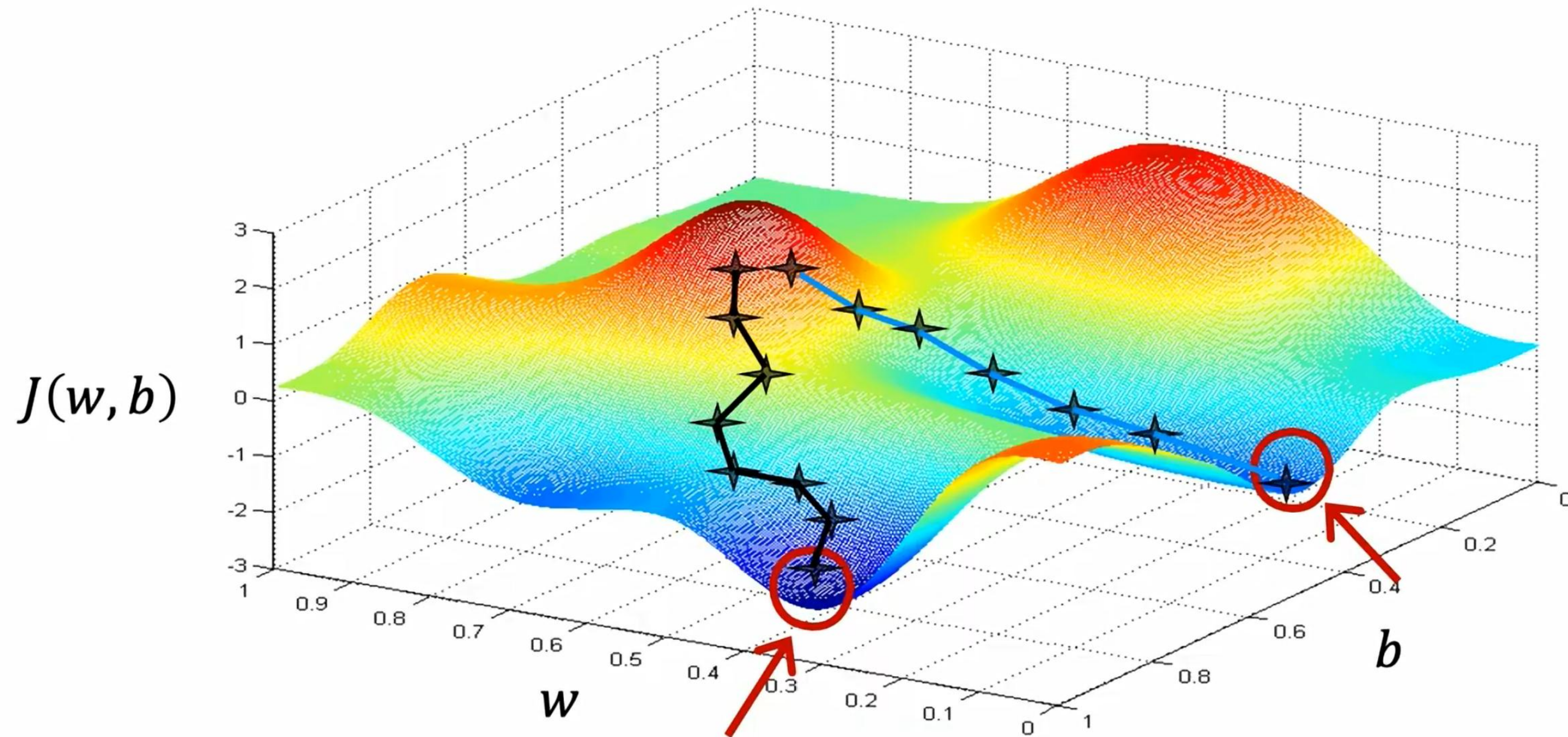
Gradient Descent for Linear Regression

More than one local minimum

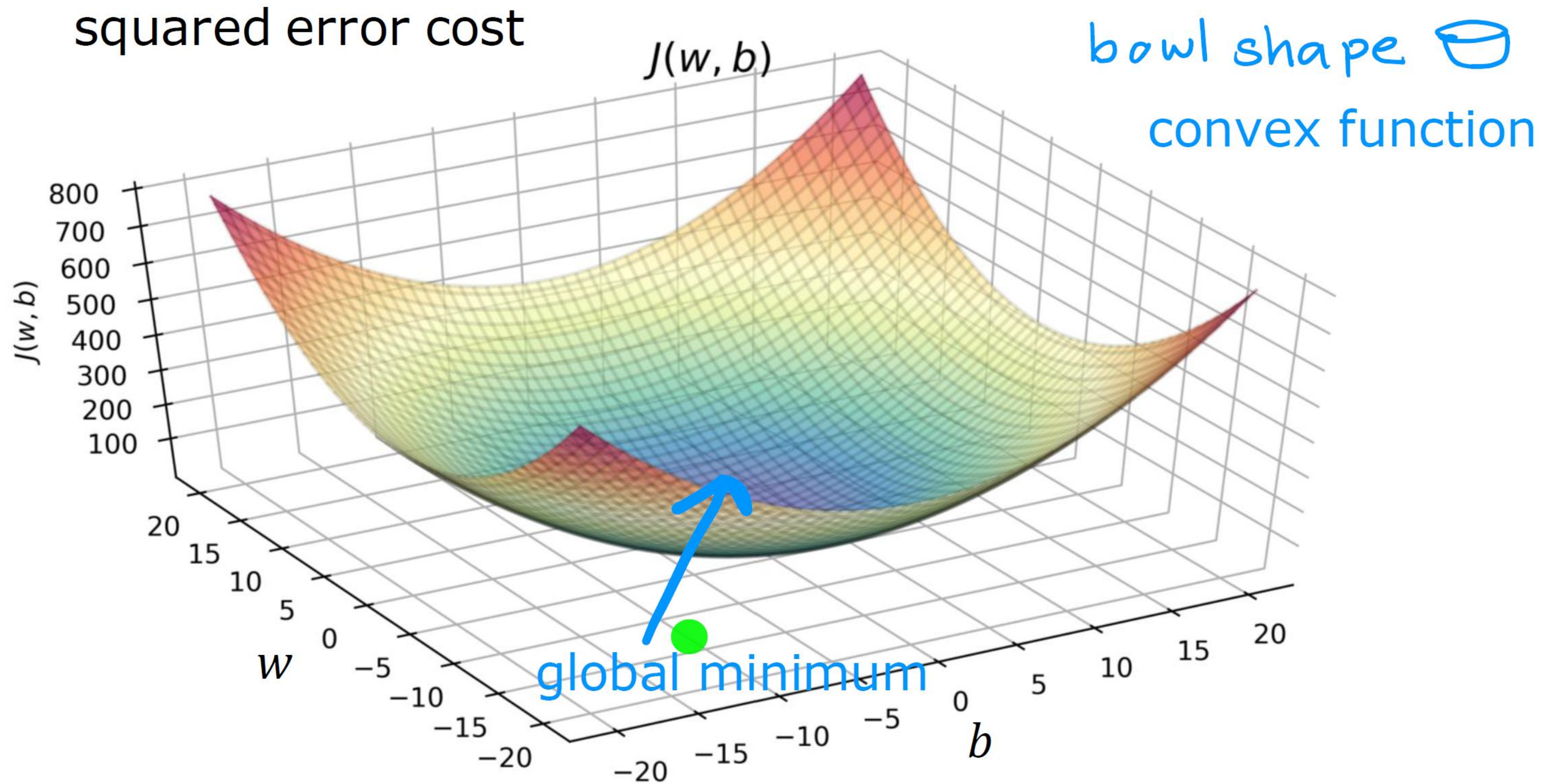


Gradient Descent for Linear Regression

More than one local minimum



Gradient Descent for Linear Regression

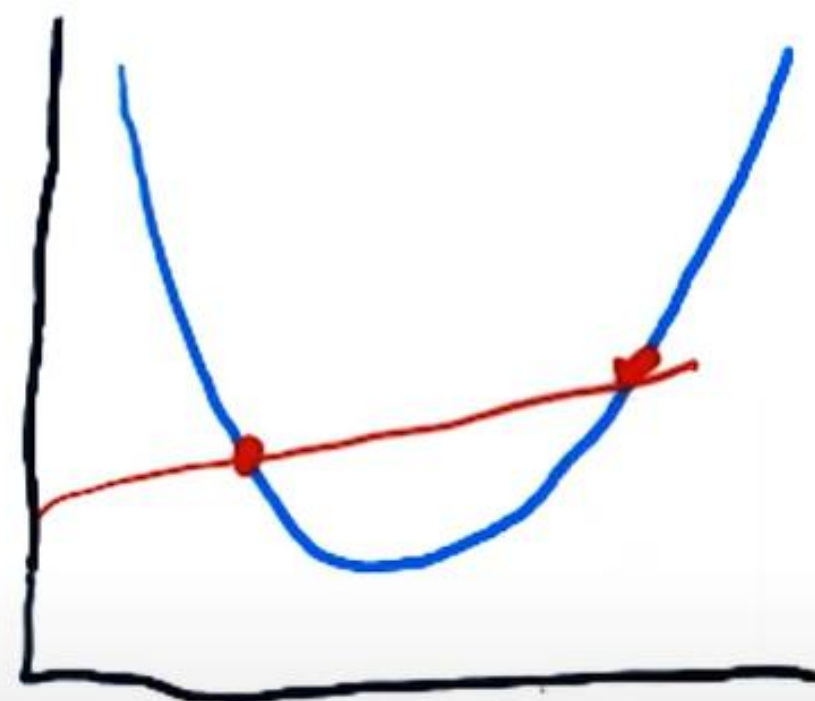


Convexity

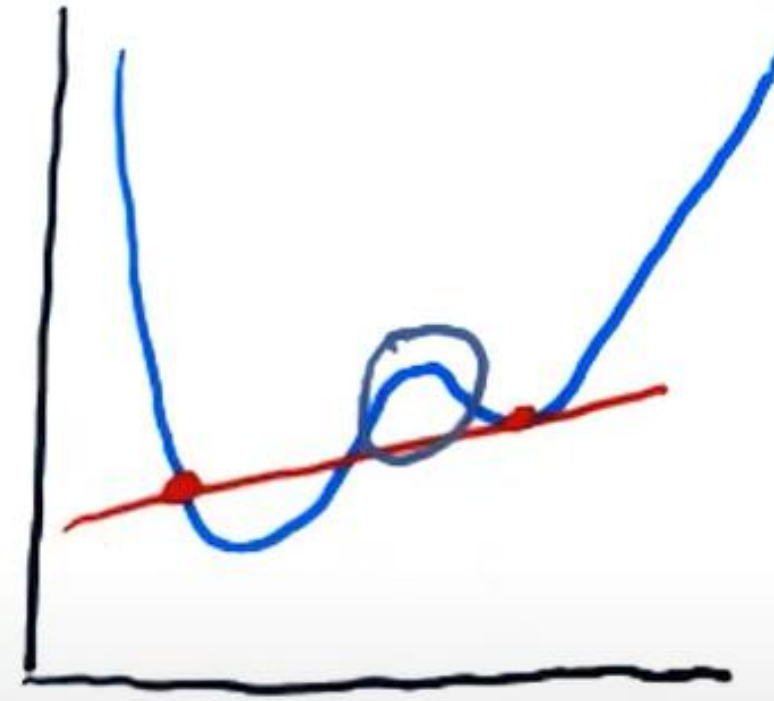
In mathematics, a real-valued function is called convex if the line segment between any two distinct points on the graph of the function lies above or on the graph between the two points.

Convex problems

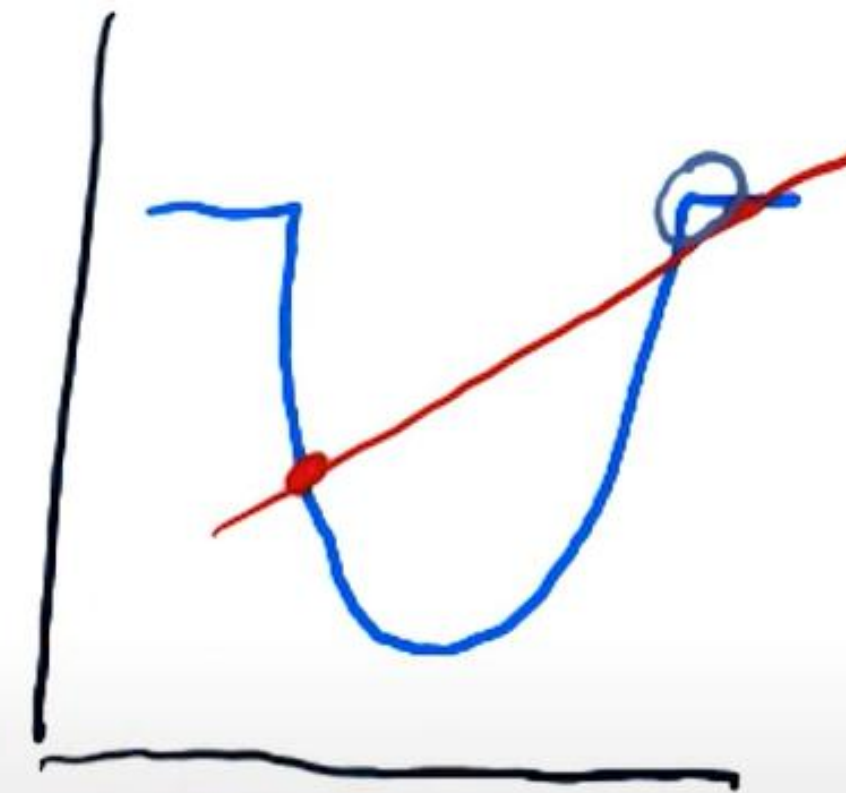
- Choose two points, draw line
- Convex if line is above graph



convex

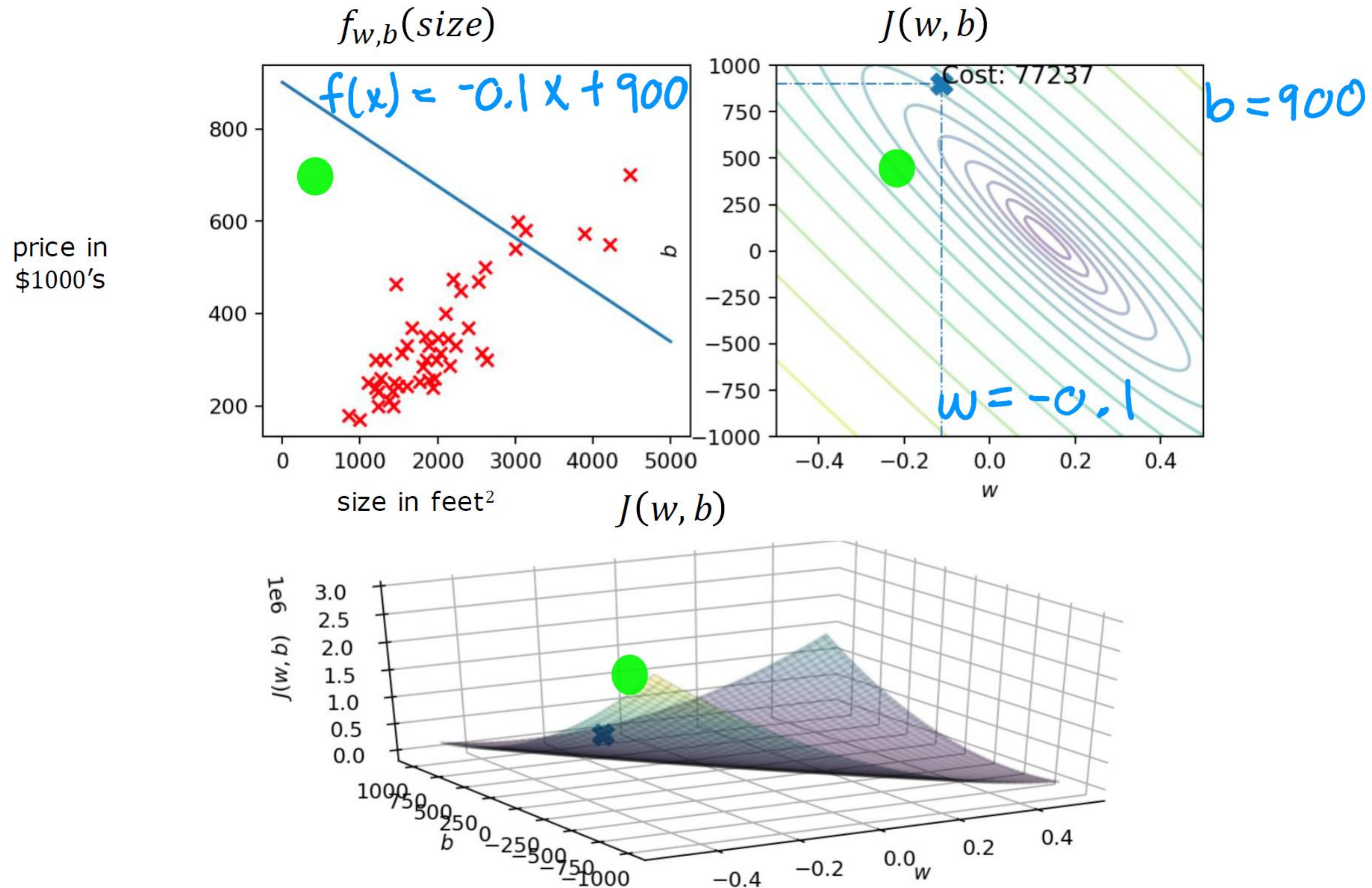


non convex

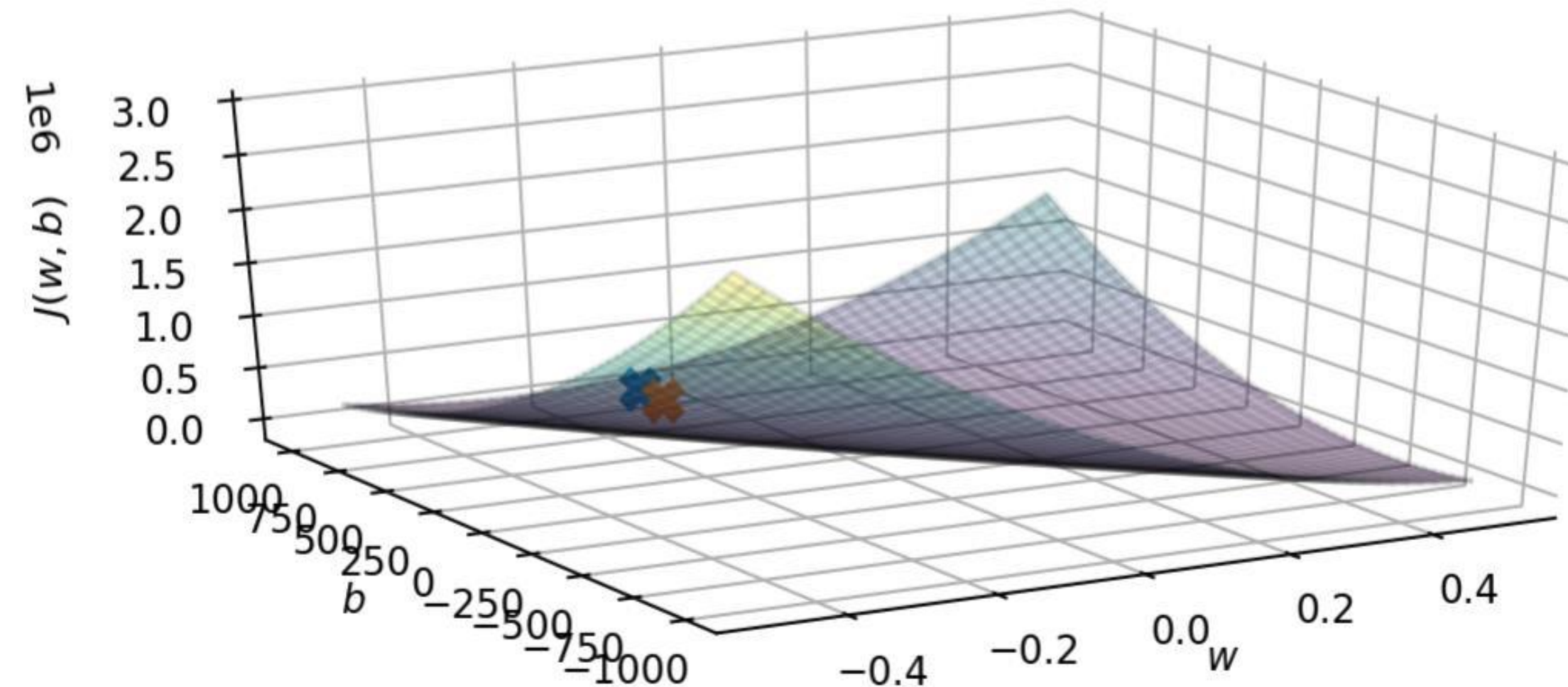
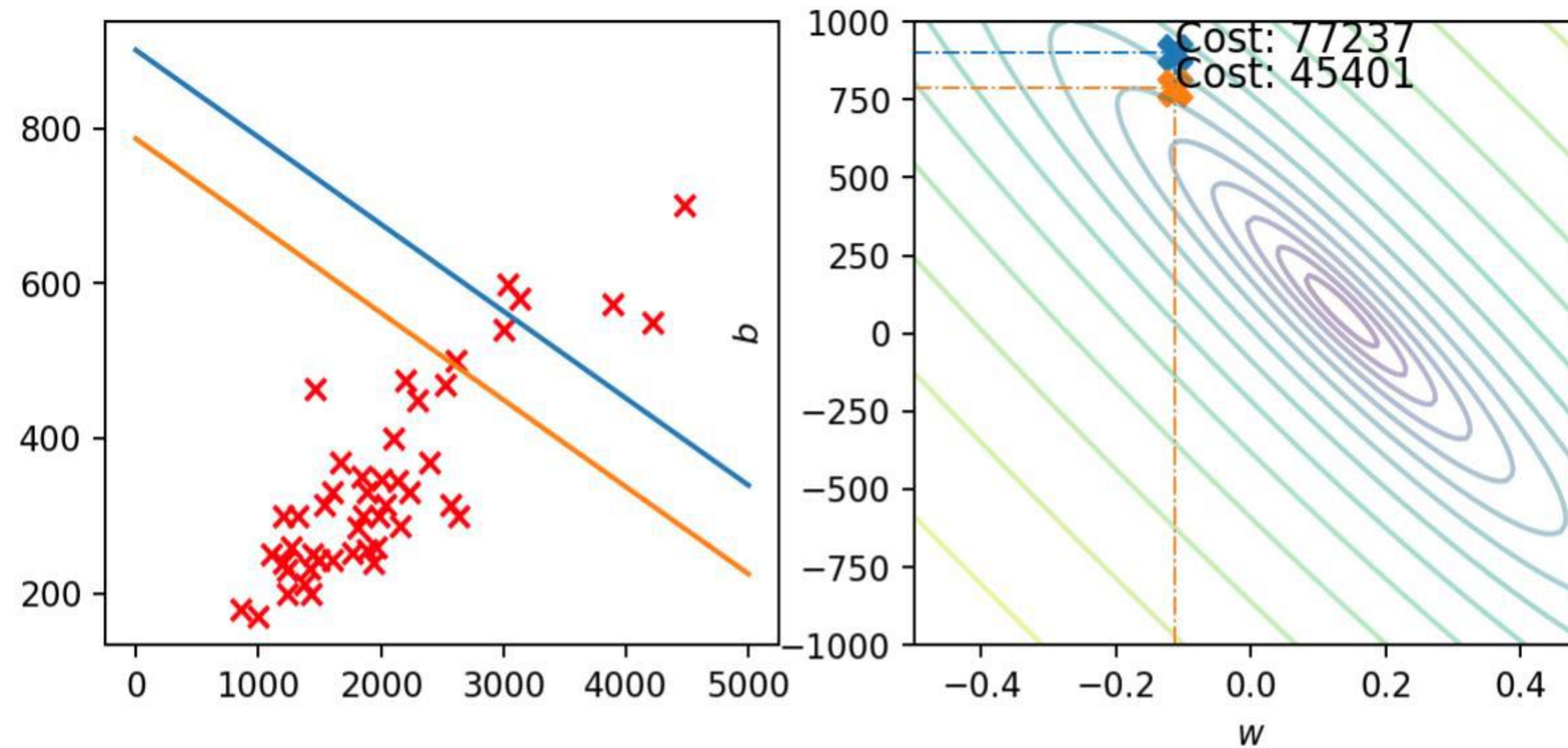


non convex

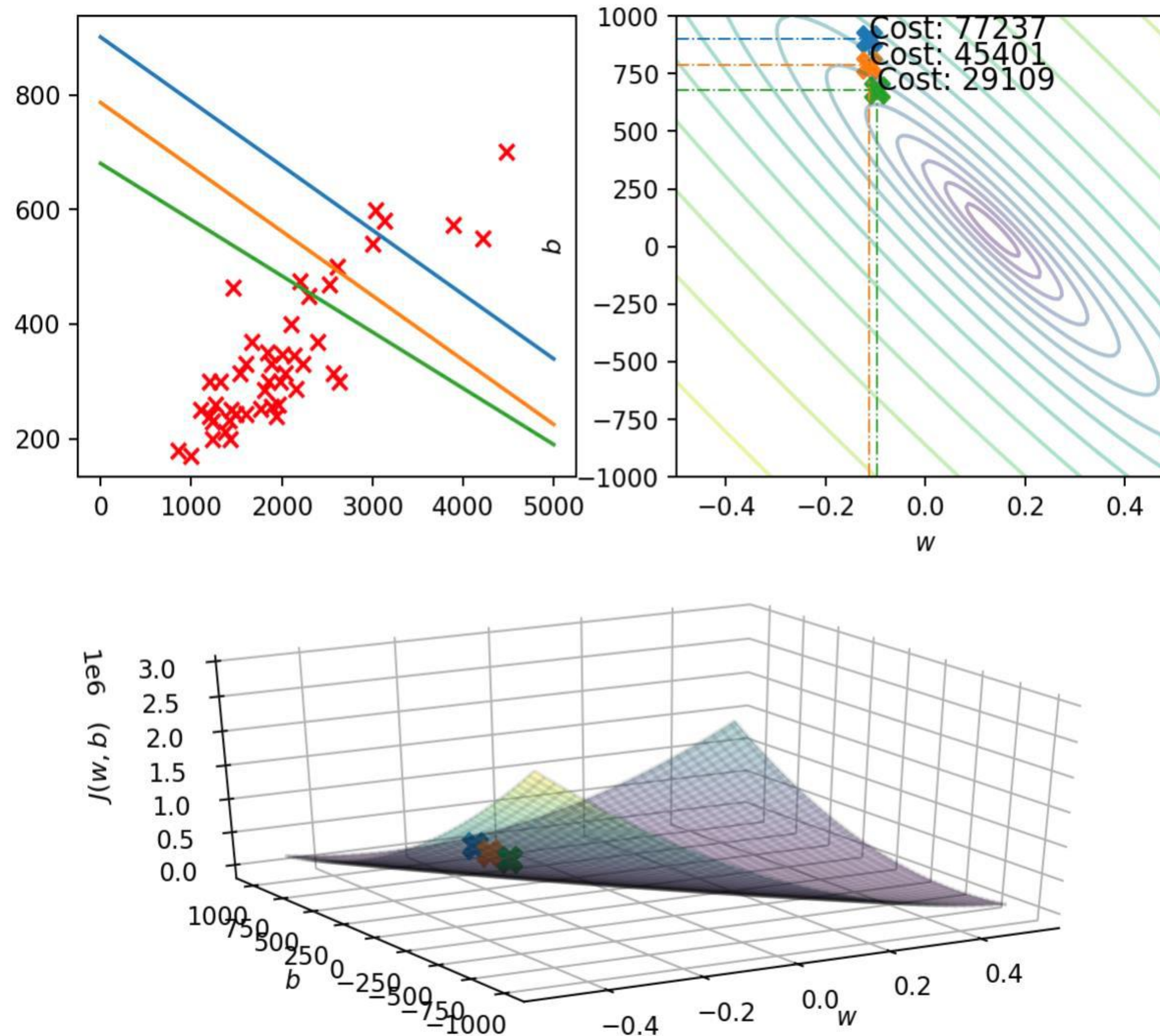
Gradient Descent for Linear Regression



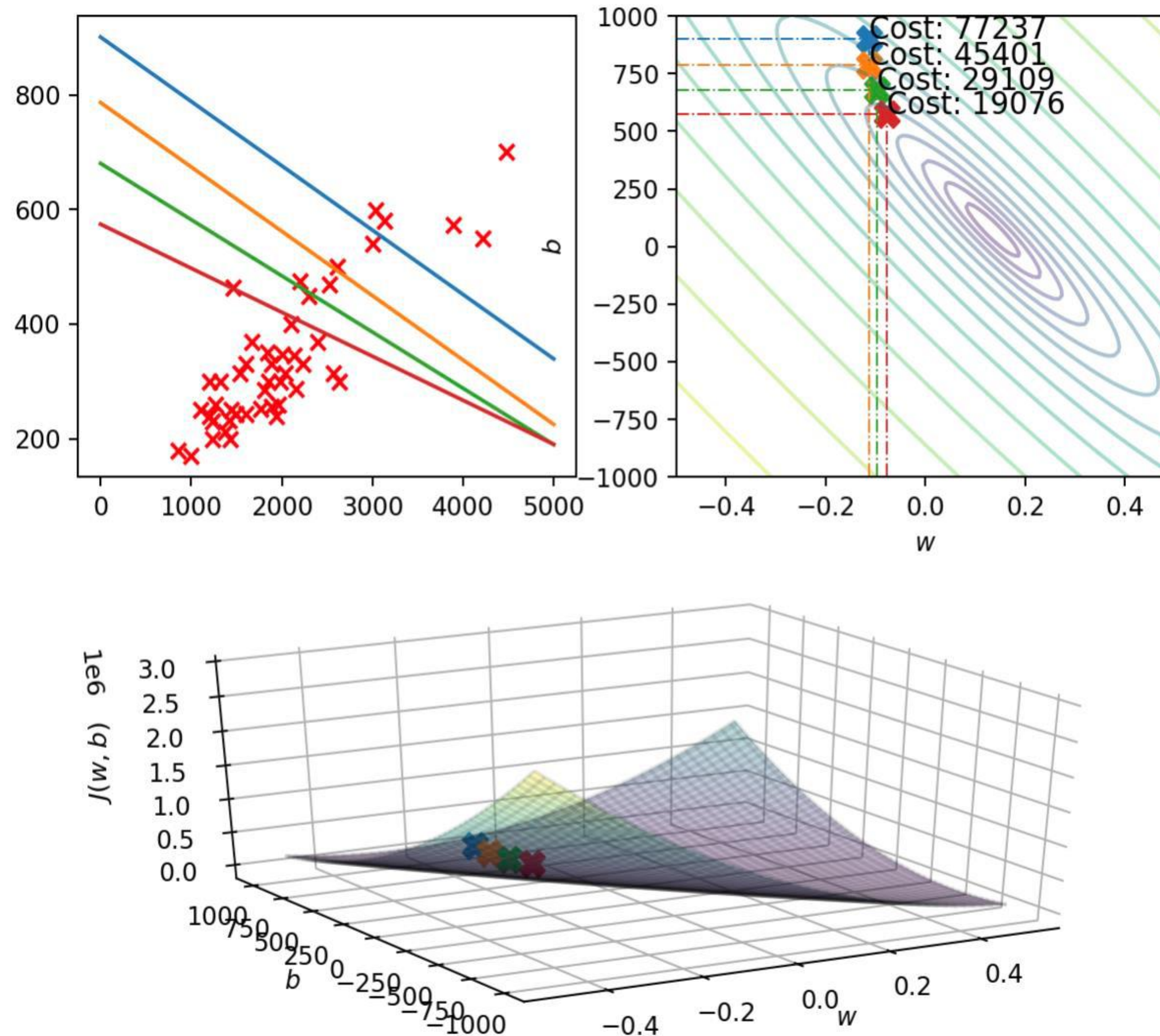
Gradient Descent for Linear Regression



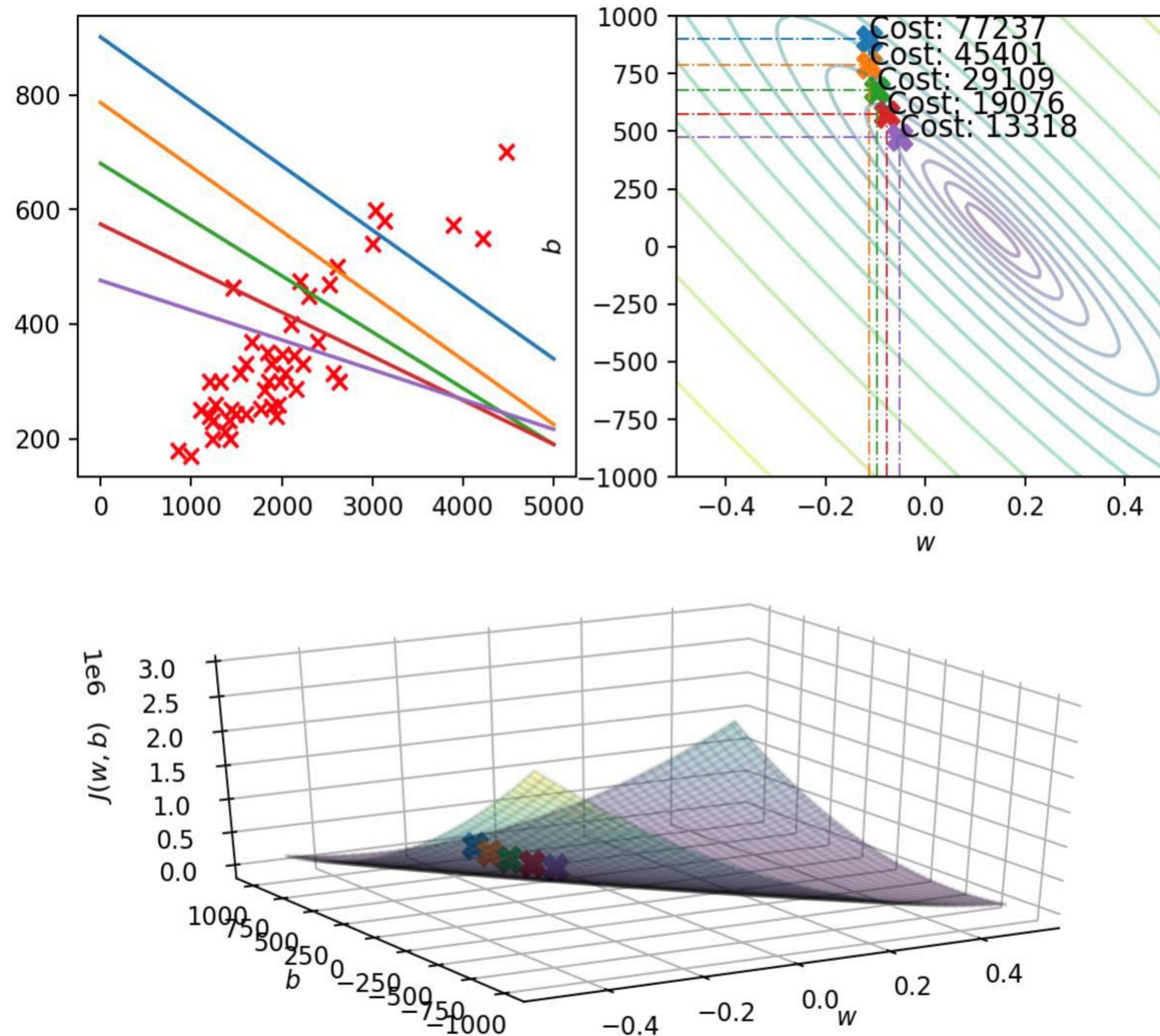
Gradient Descent for Linear Regression



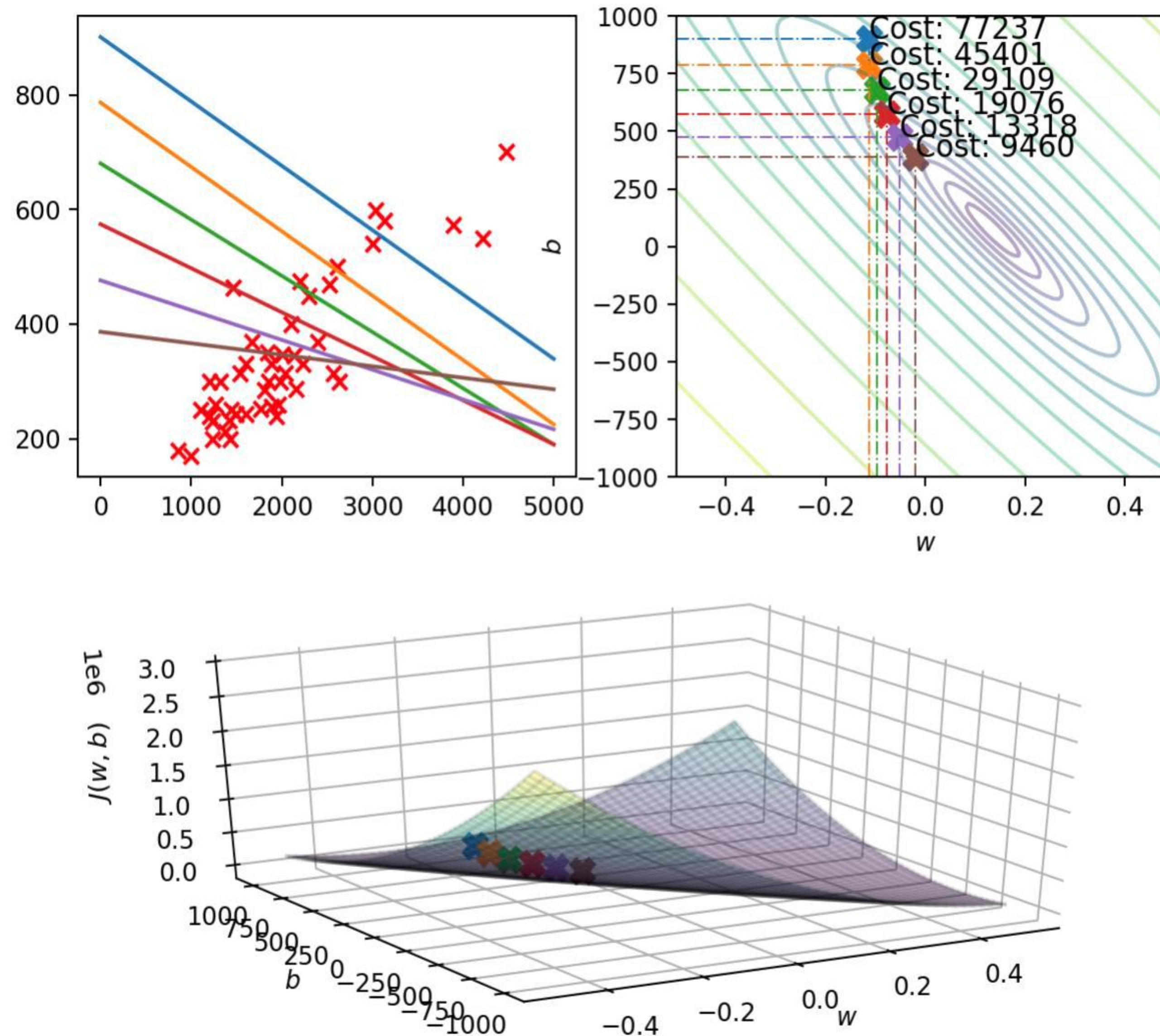
Gradient Descent for Linear Regression



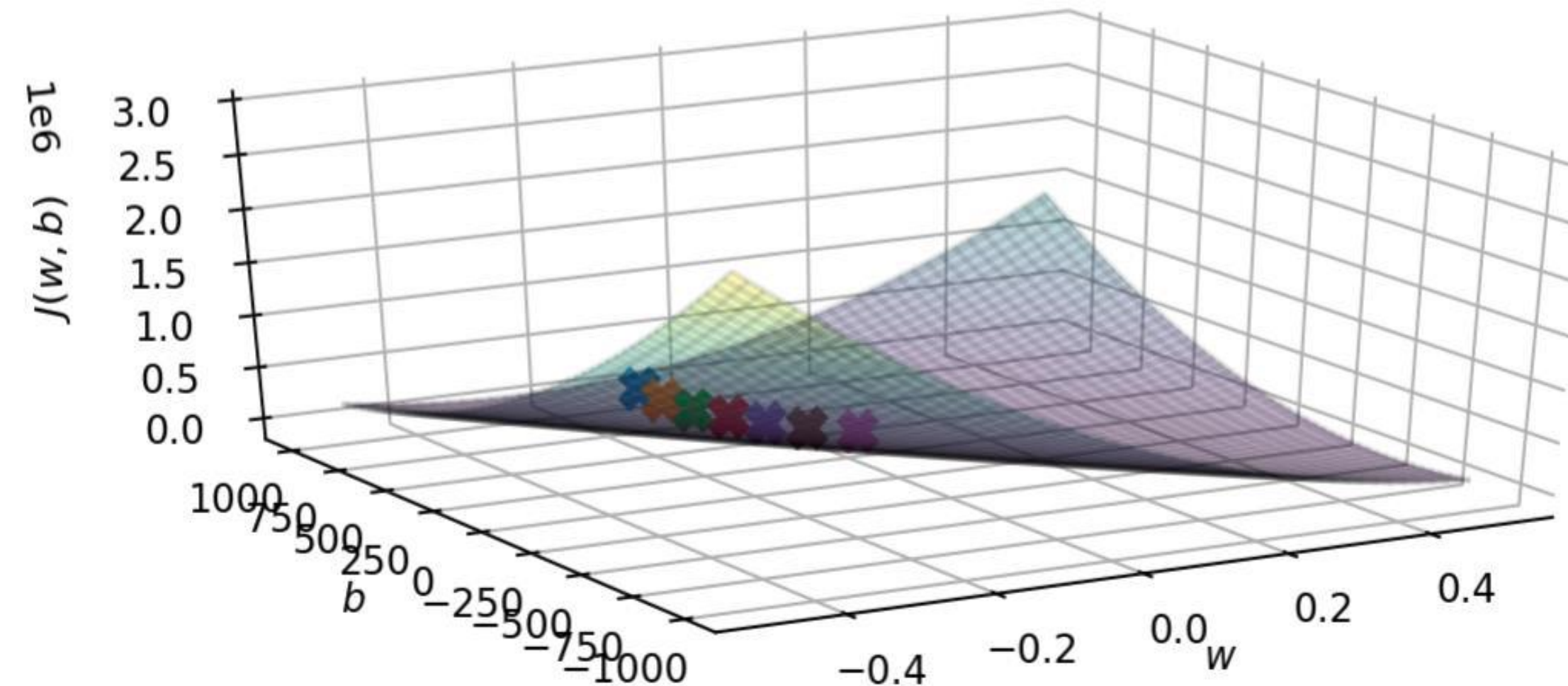
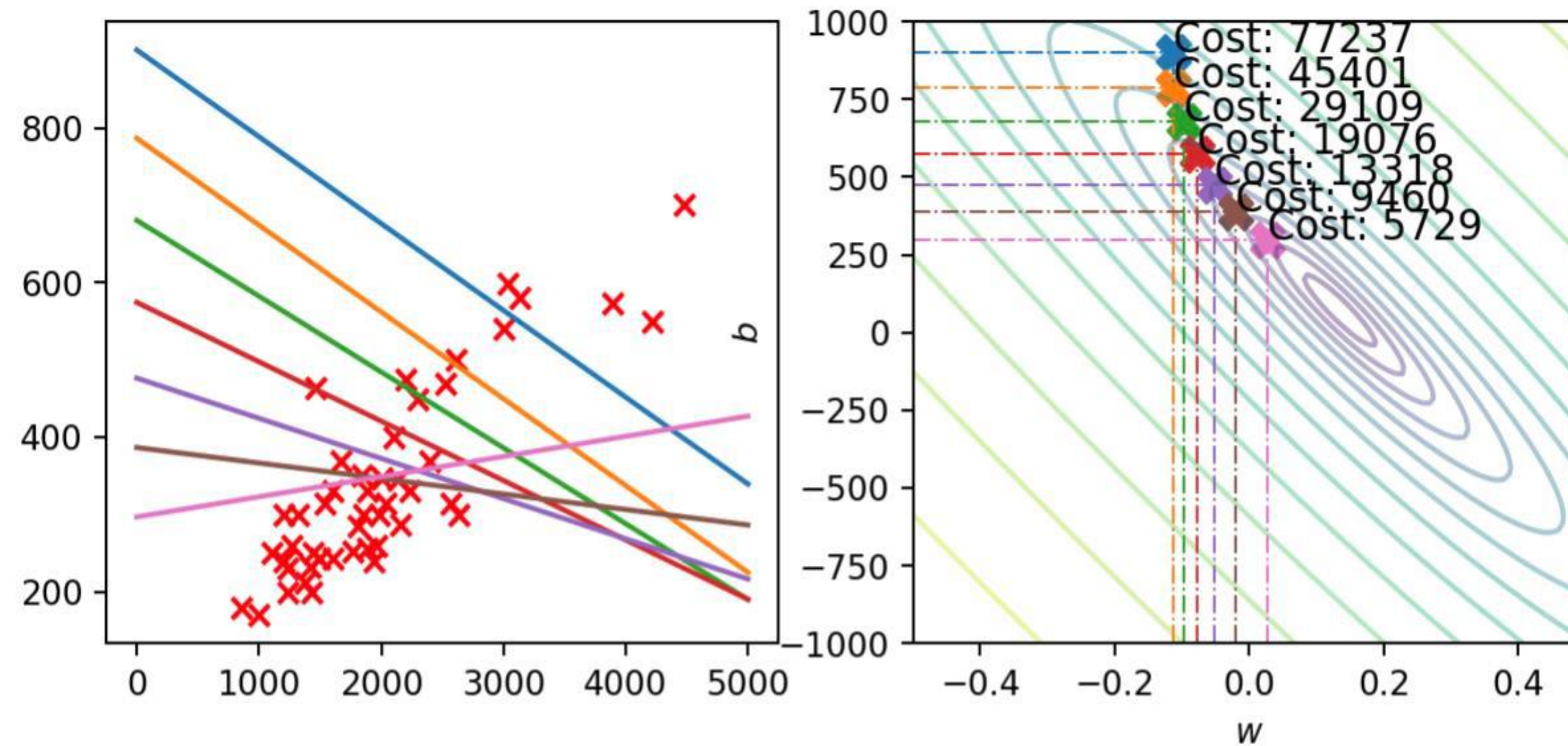
Gradient Descent for Linear Regression



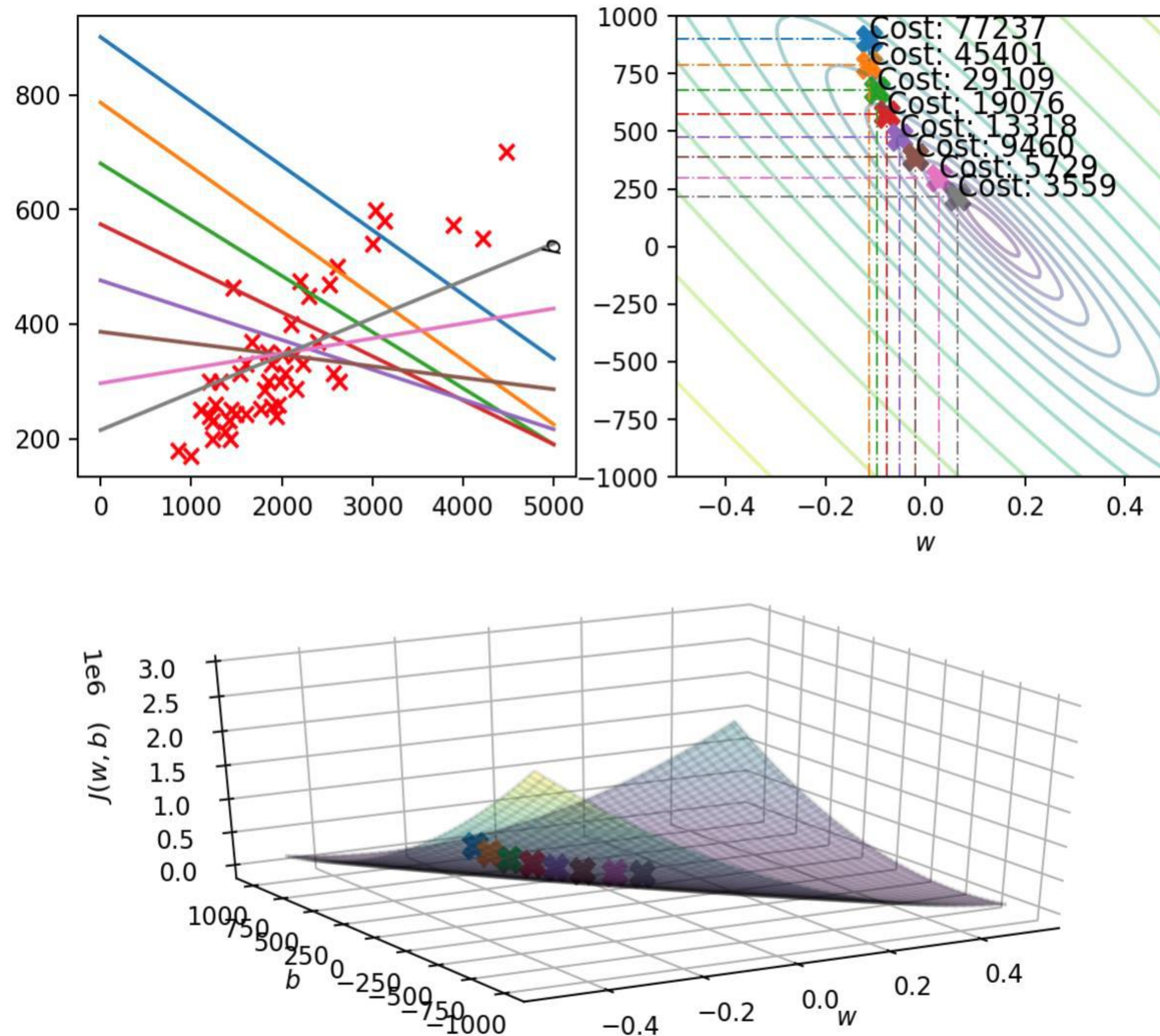
Gradient Descent for Linear Regression



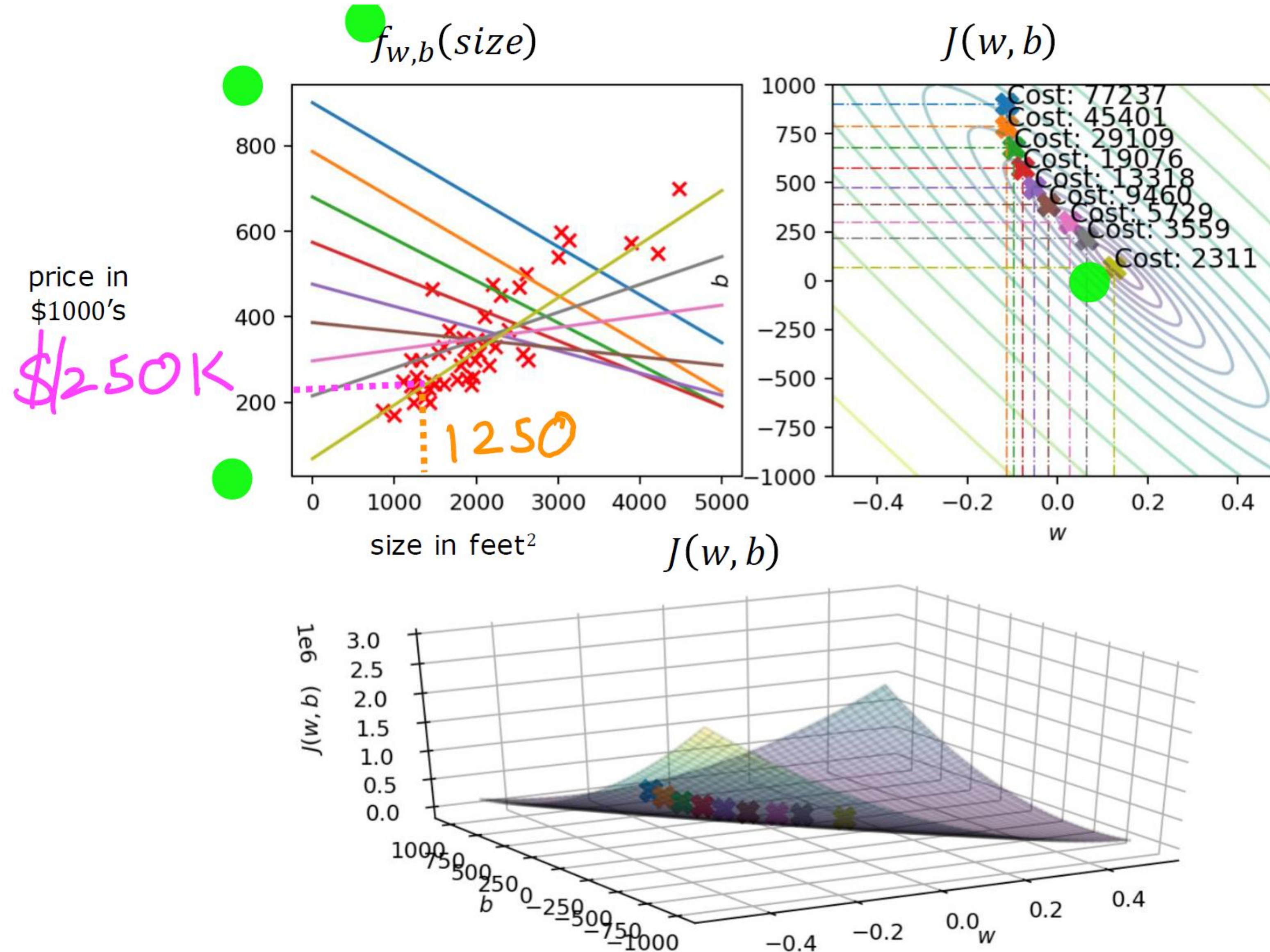
Gradient Descent for Linear Regression



Gradient Descent for Linear Regression



Gradient Descent for Linear Regression



Gradient Descent for Linear Regression

"Batch" gradient descent



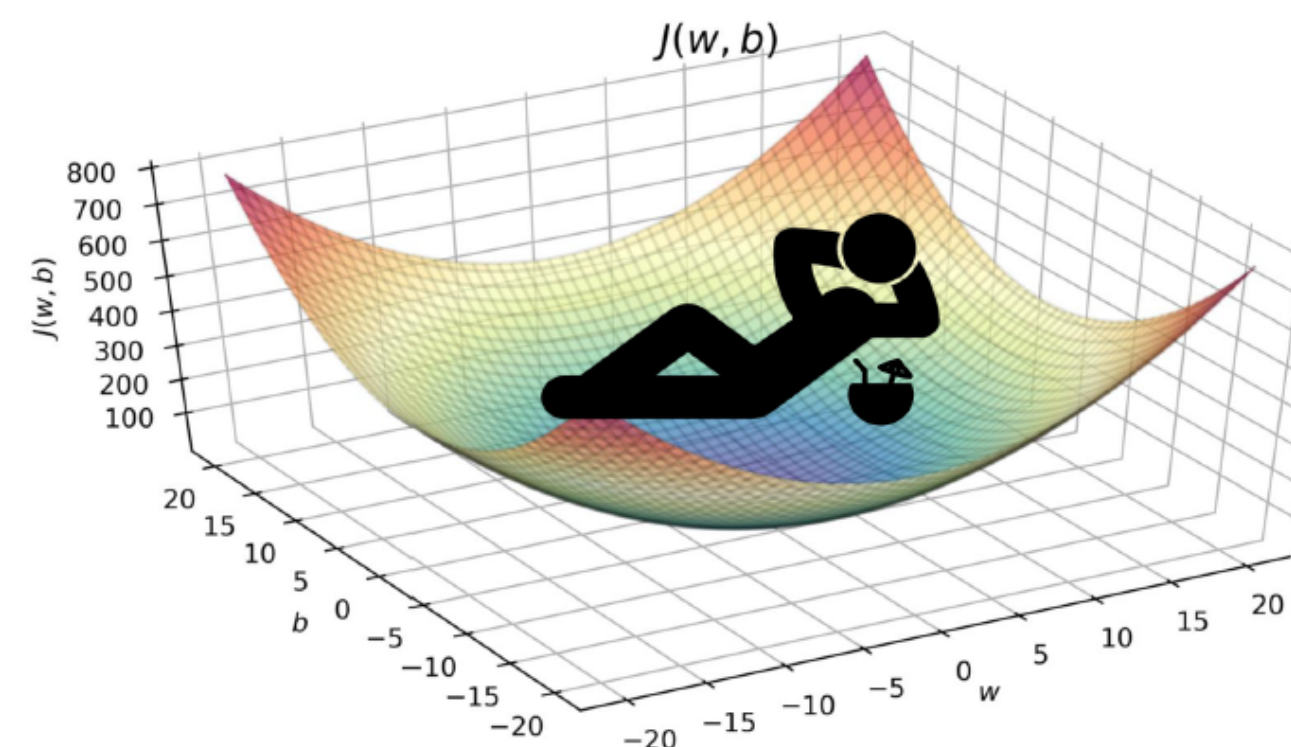
"Batch": Each step of gradient descent uses all the training examples.

other gradient descent: subsets

	x size in feet ²	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

$$\sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$



Gradient Descent for Linear Regression

Gradient

- Full-batch gradient descent

$$J = \frac{1}{M} \sum_{i=1}^M J^{(i)}$$

- Mini-batch gradient descent

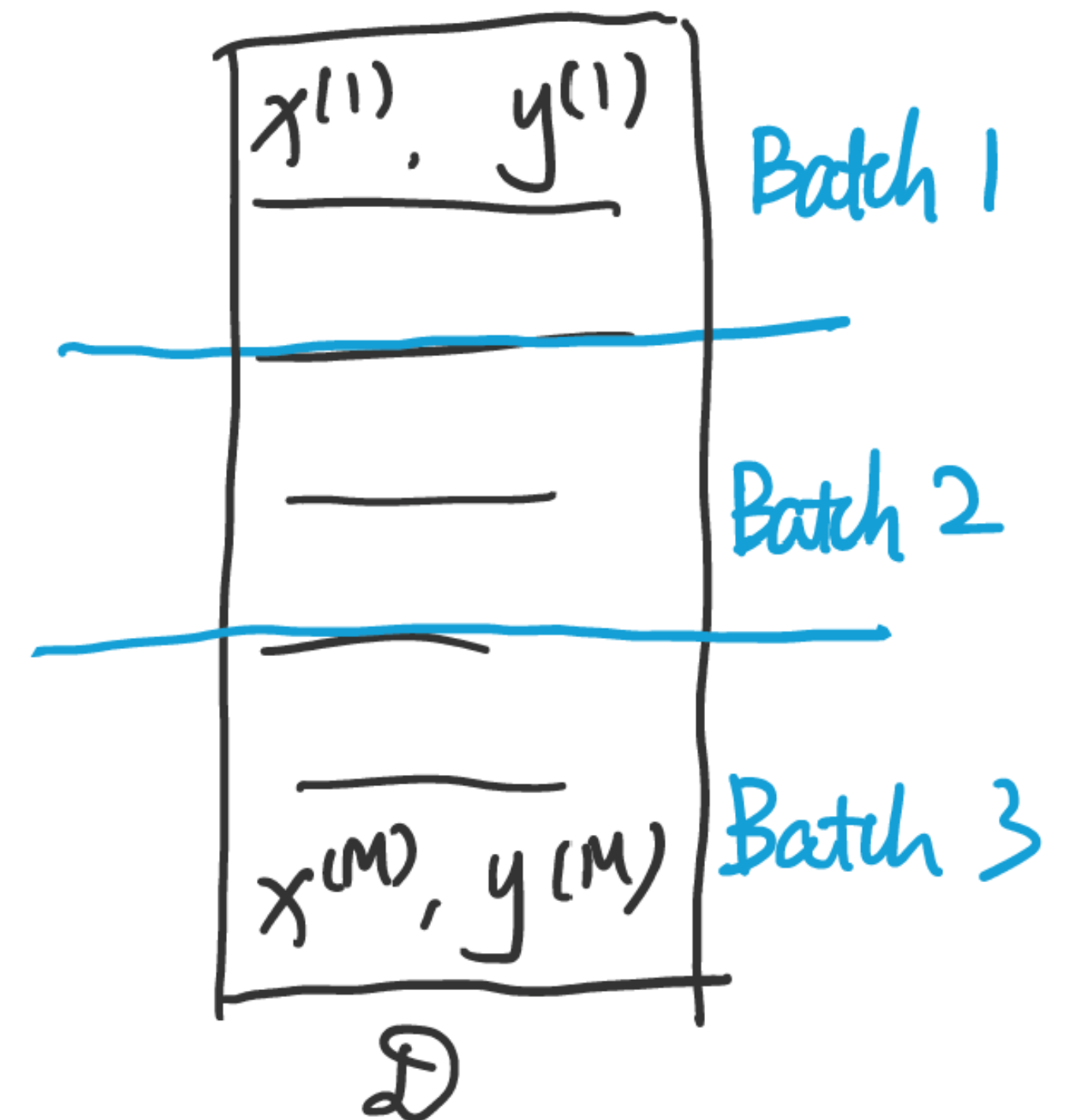
In each iteration in the loop,

$$J = \frac{1}{B} \sum_{b=1}^B J^{(n_b)}$$

where $B < N$, and $n_1, n_2, \dots, n_b \in \{x_1, \dots, x_M\}$

- Stochastic gradient descent (SGD)

Mini-batch with $B = 1$



Thank You

Any questions?

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Source & Special Thanks to Andrew Ng (Coursera)
Machine Learning Course

<https://www.youtube.com/@Deeplearningai>

